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# DETERMINING BEST DIET AND TEMPERATURE REGIME OF MONKEYFACE PRICKLEBACKS (CEBIDICHTHYS VIOLACEUS) FOR AQUACULTURE

A Thesis

Presented to

The Faculty of the Moss Landing Marine Laboratories

San José State University

In Partial Fulfillment

of the Requirements for the Degree

Master of Science

by

Luke C. Townsend

December 2024

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The Designated Thesis Committee Approves the Thesis Titled

# DETERMINING BEST DIET AND TEMPERATURE REGIME OF MONKEYFACE PRICKLEBACKS (CEBIDICHTHYS VIOLACEUS) FOR AQUACULTURE

by

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ABSTRACT

DETERMINING BEST DIET AND TEMPERATURE REGIME OF MONKEYFACE PRICKLEBACKS (CEBIDICHTHYS VIOLACEUS) FOR AQUACULTURE

by Luke C. Townsend

Wild freshwater eel populations are threatened due to overfishing and habitat degradation. Attempts to use aquaculture to close the eel life cycle have yet to be commercially adapted, owing largely to their complex lifecycle. Monkeyface pricklebacks (Cebidichthys violaceus) are intertidal fish that may serve as an alternative species for freshwater eels in aquaculture, as they possess several beneficial characteristics. This study aimed to optimize diet and culture temperatures to maximize monkeyface prickleback cultures by measuring growth, condition, metabolic rates, fillet composition, and critical thermal maximum (CTmax) of juvenile fish that were reared in two separate experiments: (1) on four different diets (fresh seaweed, seaweed pellet, mixed pellet, and soy pellet) and (2) across five different temperatures (14–26°C) when fed a single diet. Monkeyface pricklebacks exhibited the highest growth rates and body condition when fed the mixed and soy pellets and in the 20–23°C treatment. Metabolism did not differ between diets but was influenced by temperature. Macronutrient and fatty acid content in fillets were influenced by diet but not temperature. Fish could also increase their CTmax by 2°C after acclimating to 12°C warmer waters. If raised on soy-based pellets at 20°C, monkeyface pricklebacks may be adopted as an alternative, which could reduce pressure on freshwater eel populations. However, the fish is still an emerging aquaculture species and faces challenges such as diseases, parasites, aggressive intraspecies behavior, and challenges stimulating reproduction in captivity.

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LIST OF ABBREVIATIONS

AICc - Akaike Information Criterion

ANF - anti-nutritional factor

ANOVA - analysis of variance

ARR - acclimation response ratio

AS - aerobic scope

CBASS - Coral Bleaching Automated Stress System

CTmax - critical thermal maximum

CTmin - critical thermal minimum

DHA - C22:6n3 docosahexaenoic

EPA - C20:5n3 eicosapentaenoic/C24:0

FCR - feed conversion ratio

FK - Fulton's condition factor

GLMM - generalized linear mixed model

HSI - hepatosomatic index

IG - instantaneous growth

MLML - Moss Landing Marine Laboratories

MMR - maximum metabolic rate

MR - metabolic rate

NMDS – non-metric multidimensional scaling

NPAL - NP Analytical Laboratories

PERMANOVA - permutational analysis of variance

SDA - specific dynamic action

SGR - specific growth rate

SMR - standard metabolic rate

TGC - thermal growth coefficient

Tukey’s HSD - Honest Significant Difference

USA - United States of America

# Introduction

## Aquaculture as a Sustainable Practice

The steady increase in human population size and Earth's limited resources have led to a global food security crisis. Earth's population is expected to reach ~9 billion people by 2050, requiring an approximate 70% increase in future food production (Mc Carthy et al., 2018; Tyczewska et al., 2018). Current food production practices may account for this needed increase in the short term, but they may not be able to support the human population in the long term due to concerns with productivity and sustainability. This is especially apparent in wild fish landings, which increased for decades but peaked in the 1990s, indicating wild fisheries are at capacity (Delgado et al., 2003; Food and Agriculture Organization of the United Nations, 2022). For instance, bottom trawling is used to harvest wild fish in large quantities, but it damages marine habitats on the ocean floor, reducing the biomass that can be supported (Delgado et al., 2003; Hiddink et al., 2020). In effect, the fish supply may remain high in the short term, however, decreases in fishery productivity are likely to occur as marine habitats are damaged, resulting in fewer fish in the future. Seafood and marine fish in particular are vital sources of healthy food due to their high levels of omega-three fatty acids [C20:5n3 eicosapentaenoic/C24:0 (EPA) and C22:6n3 docosahexaenoic (DHA)] (Rexroad et al., 2021). Considering this, and the predicted shortfall of fish landings in the future, it is prudent to increase the fish supply to meet the recommended serving of 113 g/week in the United States of America (USA) (Rexroad et al., 2021). Thus, alternative methods are needed to meet the demand for increased fish production.

Aquaculture is one such method that has succeeded in increasing the fish supply. According to the Food and Agriculture Organization, approximately 90% of wild fish stocks are overfished or fished to capacity, which leaves ~10% of stocks that are underfished (Chu et al., 2020; Food and Agriculture Organization of the United Nations, 2022). This suggests that an increase in the fish supply will not likely occur from expanding wild fishery production as most stocks are fished to sustainable capacity. Despite the plateau in landings from wild fisheries, fish production has continued to rise by ~6% per year as freshwater and marine species are increasingly reared in captive settings through aquaculture (Delgado et al., 2003; Bostock et al., 2010; Tacon, 2020). Aquaculture currently accounts for more than half of global seafood production, with fish farming comprising ~66% of global aquaculture production (*The State of World Fisheries and Aquaculture*, 2022; Fong et al., 2022). At present, more than 208 species of fish are grown in aquaculture settings, which are valued at approximately US$139.7 billion annually (Tacon, 2020). Most of these species, however, are freshwater or diadromous with less than 6% of fish aquaculture being composed of marine species (Tacon, 2020). Currently, freshwater aquaculture is restricted by land availability and government policies implemented to reduce the environmental impacts of inland aquaculture (Costello et al., 2020; Costa-Pierce et al., 2022). In contrast, marine fish aquaculture has potential to grow and increase the fish supply, while freshwater aquaculture is limited in resources and wild fishery landings remain stagnant for the foreseeable future (Fong et al., 2022).

Although aquaculture is the fastest-growing food production industry globally, the USA lags in aquaculture development, ranking 18$^{th}$ in aquaculture production (National Marine

Fisheries Service, 2022). This is especially important as the USA imports ~2.77 million metric tons of seafood annually ($21.4 billion), making it the world's largest net importer of seafood (Fong et al., 2022; National Marine Fisheries Service, 2022). Meanwhile, the USA only exports ~1.09 million metric tons of seafood ($4.4 billion) and lands ~3.81 million metric tons ($4.8 billion), which equates to a seafood deficit (of 45,359 metric tons) and cost (valued at $12.2 billion) (National Marine Fisheries Service, 2022). Domestically, a net import system has innate issues such as reduced food security and public distrust of the product's safety from the possible use of banned additives; therefore, it is important to increase domestic seafood production in the USA by increasing aquaculture production, especially in states with high consumption rates (Fong et al., 2022).

**Aquaculture in California**

California is the USA's largest seafood consumer and importer, comprising ~25% of the nation's international imports (Fong et al., 2022). As previously mentioned, there are drawbacks associated with this dependence, as the state has little control over the application of banned substances as well as unethical and environmentally unsustainable fishing and culture practices (Fong et al., 2022). Additionally, it is predicted that exporting countries will likely decrease their exports as their domestic demand increases with time, leading to higher import prices (Fong et al., 2022). Increasing the state's aquaculture production would begin to reduce these drawbacks, while still implementing socially and environmentally safe practices. Fish are typically consumed at higher rates than shellfish in the USA; therefore, initial efforts should be focused on increasing fish supply (Rexroad et al., 2021). While culturing an established species would increase fish supply, it is advantageous to increase

species diversity to augment resilience in the aquaculture sector (Metian et al., 2020). As previously mentioned, marine fish are increasingly considered for culture because they are not intrinsically limited by land and government policies, depending on location and state policies (Costello et al., 2020; Costa-Pierce et al., 2022). Therefore, the development of a previously uncultured marine fish in the USA and California especially could be desirable to increase supply and species diversity.

Nutritional requirements are major considerations for selecting and developing an emerging species for aquaculture. Fishmeal and fish oil are typically major components of cultured fish diets due to their digestibility and complete nutrient profiles, and they come from concentrated forms of reduced forage fish (e.g., anchovies) (Delgado et al., 2003; Huntington & Hasan, 2009; Food and Agriculture Organization of the United Nations, 2022). This has led to aquaculture becoming one of the largest industry users of fishmeal and fish oil; ~69% and 75% of the global supply of wild capture forage fish were used by the fish and crustacean aquaculture sectors, respectively, in 2016 (Boyd et al., 2020). As fish aquaculture continues to grow at an average annual rate of 5.7% since 2000, the demand and consumption of fishmeal and fish oil will continue to increase, which may strain forage fish populations (Delgado et al., 2003; Boyd et al., 2020). In addition, the cost of fishmeal and fish oil has and will continue to rise since wild capture fisheries have remained stagnant while aquaculture production increases (Huntington & Hasan, 2009; Food and Agriculture Organization of the United Nations, 2022). This is exemplified by the cost of fishmeal rising from ~U.S.$452.1 per metric ton in 2000 to ~U.S.$1,415.2 per metric ton in 2022 due to increased demand (International Monetary Fund, 2024). Furthermore, the inclusion level of

fishmeal and fish oil within an aquafeed varies based on the dietary needs of the farmed species. Carnivorous fish like salmonids often perform better with diets higher in fishmeal (20–40%) than herbivorous/omnivorous fish like carp (~5%), which is likely due to adaptations associated with their feeding ecology (Huntington & Hasan, 2009; Zhang et al., 2022). Terrestrial ingredients have been implemented in the test diets of carnivorous fish as an alternative to fishmeal and fish oil, but many have been found to reduce growth and cause enteritis like soybean meal (Kumar et al., 2020). If a suitable terrestrial alternative is found, there could be other unintended economic costs such as increased demand and ultimately higher costs for other agriculture industries like poultry, which are already significant users of ingredients like soy for their feed (Bostock et al., 2010). Thus, marine aquaculture should prioritize herbivores over carnivores to lower feed costs and reduce their reliance on fishmeal and fish oil. However, carnivorous fish are typically reared in marine aquaculture settings because there is a higher demand and price paid for them despite the costs, which accounts for the expenses of mariculture infrastructure and feed (Zhang et al., 2022). Based on these potential issues, it would be ideal to consider developing new species for marine aquaculture that are already adapted for low or no animal protein diets, negating some of the economic and environmental impacts as mentioned above.

**Freshwater Eel Status and Sustainability**

Freshwater eels (*Anguillidiae* spp.) are a popular protein source used in many seafood dishes, but they are primarily sourced from a fishery currently considered unsustainable (Muthmainnah et al., 2016). In 2017, the USA imported 283 metric tons ($1,744,000) of freshwater eels, with California being the largest importer in the nation (Liddel, 2019; HS

Code 030326, n.d.). As a result of their seafood popularity, freshwater eels have substantially declined in the wild due to overfishing because the market supply is based on wild-caught adults and juveniles (Arai, 2014; Honda et al., 2016; Muthmainnah et al., 2016; Drouineau et al., 2018). Specifically, Japanese, European, and American eels have decreased in abundance since the 1980s due to fisheries targeting the eels for partial life cycle aquaculture (Arai, 2014). Habitat degradation, changes in environmental conditions, pollution, and parasites have also led to the decline in freshwater eels (Arai, 2014; Drouineau et al., 2018). Recovery of the species has been poor because the overfishing of juvenile eels has continued to impede natural recruitment (e.g., ~90–99% decline in Europe since the 1980s) (Drouineau et al., 2018). In addition, freshwater eels are catadromous species that reproduce in the open ocean, migrate to the coast as larvae/leptocephali, undergo a metamorphosis from larvae to juveniles, traverse freshwater streams as elvers, and remain in freshwater environments as adult eels (Muthmainnah et al., 2016; Drouineau et al., 2018). This complex lifecycle and the fact that larval eel recruitment depends on multiple environmental factors, such as trophic condition, currents, and salinity, make recruitment more difficult and complex than many other species (Drouineau et al., 2018). As climate change intensifies, larval recruitment will likely decrease further as temperature increases may reduce larval food sources (Drouineau et al., 2018). The combination of overfishing and reduced natural recruitment has thus led to the low supply of freshwater eels, which has created significantly high prices (~$35,000 per kilogram in Japan and ~$14,000 per pound in the USA for juvenile eels), making it more expensive than the endangered bluefin tuna (Specialty Food Association, 2019; Morgan, 2021).

While methods have been developed to culture freshwater eels, many challenges remain to close the species' life cycle in a commercially viable manner. As previously mentioned, the freshwater eel supply for the market is based on harvesting juvenile eels from the wild and rearing them to market size (50–250 g) (Ringuet et al., 2002; Arai, 2014; Honda et al., 2016; Muthmainnah et al., 2016; Drouineau et al., 2018). Research is being conducted to reduce the strain on the wild populations to rear all life stages of freshwater eels in a captive setting (Drouineau et al., 2018). Despite having success in artificially reproducing freshwater eels, the inability to rear American and European glass eels, along with the Japanese glass eels having no commercial adoption of culture technology, has prevented aquaculture from reducing pressures on wild stocks and facilitating their recovery (Drouineau et al., 2018).

Sustainable feed is also a concern with freshwater eels. Like other carnivorous fish, freshwater eels require fishmeal and fish oil in their diets (Ringuet et al., 2002; Glencross et al., 2007). As previously mentioned, this is unsustainable with current practices, and it is expensive. Like other carnivorous fish, Degani (1987) reported that including soybean meal reduced growth in juvenile European eels, making it less likely that alternative ingredients are a possible solution (Kumar et al., 2020). These outcomes from previous eel research suggest that an alternative species that thrives on fish-free feed may be a promising substitute, until an alternative ingredient without adverse health effects can be identified.

**Monkeyface Prickleback as a Potential Alternative to Freshwater Eels**

Introducing an alternative eel substitute species that could be reared in an aquaculture setting may aid in alleviating the commercial pressure on wild freshwater eel populations. Beyond market demand, the development of new species for aquaculture generally benefits

from characteristics that suit culture conditions, such as fast growth, efficient conversion of feed to mass, tolerance of fluctuating environmental conditions, and an ability to be reared at high stocking density (United Nations Development Program, 1979; Toomey et al., 2020). The monkeyface prickleback (*Cebidichthys violaceus*) warrants potential development as a new aquaculture species because it could serve as an alternative species to the imperiled wild-caught freshwater eels. It possesses several life history traits that are advantageous for aquaculture and is an herbivorous marine fish.

Though not technically an eel, *C. violaceus* is a temperate eel-shaped fish in the family Stichaeidae that lives in the intertidal zone from Baja California to Oregon (Marshall & Echeverria, 1991; Leet et al., 2001). In the wild, monkeyface pricklebacks have been found to live up to 18 years and reach a length of 76 cm, inhabiting crevices in rocky intertidal and shallow subtidal environments (Leet et al., 2001; Love, 2001). Monkeyface pricklebacks undergo an ontogenetic shift and become herbivores that consume a variety of macroalgal species found within the intertidal region, such as the green algae *Ulva* and red algae *Pyropia perforata* (Montgomery, 1977; Horn et al., 1982; Horn & Gibson, 1988; German et al., 2004; Gawlicka & Horn, 2005; German & Horn, 2006; German et al., 2014). Thus, the species have evolved to consume a diet that is inherently more sustainable for farming than carnivorous freshwater eels and most cultured marine fish generally, as they are not reliant on wild sources of fishmeal and fish oil or likely affected by the morbidities associated with feeding carnivorous fish a plant-based diet (Estruch et al., 2018). The similar morphology and reportedly similar taste to freshwater eels may allow the farmer to market monkeyface pricklebacks to the same markets that utilize eels, justifying a correspondingly high profit.

Juvenile monkeyface pricklebacks are reported to have a growth rate at ~5 mm $month^{-1}$ after the ontogenetic shift to herbivory (Helm, 1992). In comparison, wild American eels under fifteen years old (*Anguilla rostrata*) have an average growth rate of ~4 mm $month^{-1}$ in freshwater environments (Lamson et al., 2009). The species is adapted to the intertidal zone, where monkeyface pricklebacks can tolerate a wide environmental temperature range (9–24°C), making them highly eurythermal (Love, 2001). They can also aerially respire where they can remain out of water for up to ~15–37 hours (Horn & Riegle, 1981; Love, 2001). Thus, they can survive changes in temperature more easily than other farmed fish and may be amenable to aerial transport. Based on telemetry data, *C. violaceus* traverses ~2 $m^2$ or less within 40-minute periods, and multiple individuals can be found within the same crevice, suggesting that monkeyface pricklebacks may tolerate high stocking densities as they live sedentary lifestyles with multiple fish occupying crevices in the intertidal region (Ralston & Horn, 1986; Leet et al., 2001). This would allow a farmer to potentially grow them in a smaller space than other fish that require a lower stocking density due to competition and space requirements for swimming. Notably, the monkeyface prickleback is a benthic species that does not occupy the water column like other cultured fish, reducing their potential stocking density vertically. *C. violaceus* moving <1% of the time would also indicate that the species likely has a low metabolic rate and does not expend energy with constant movement, unlike many other cultured fish that live in the water column (Ralston & Horn, 1986). Through static respirometry, Edwards & Cech Jr. (1990) reported that juvenile monkeyface pricklebacks have an aerobic metabolic rate (MR) at ~0.54 mg $O_2$ $hr^{-1}$ at 15°C, which is relatively low in comparison to the similarly sized bald sculpin, an intertidal fish, that has a

measured MR nearly three times higher at ~1.43 mg $O_2$ $hr^{-1}$ at 15°C. This suggests that monkeyface pricklebacks are comparatively efficient in their energy usage, which may contribute to a low feed conversion ratio (FCR), reducing the feed cost as the farmer can feed less to attain market size. Since 1991, commercial landings for monkeyface pricklebacks have ranged from 5.4 to 424.6 kg in California, due to the challenge of fishing for them individually at low tide (Leet et al., 2001). These landings are low compared to the commercial fishery for freshwater eels at 16,911.5 kg on average in North Carolina from 2011–2020 (Atlantic States Marine Fisheries Commission, 2022). The lack of a commercial scale fishery to compete with and difficulty exploiting the wild population due to their rough habitat may allow monkeyface prickleback cultures to be more readily and socially accepted as a new species for aquaculture, compared to fish like salmon for which there is a robust wild fishery and residence to aquaculture—making the development of the species ideal from a social perspective. In summary, monkeyface pricklebacks have multiple biological characteristics that are beneficial for aquaculture (e.g., herbivory, eel-like morphology, environmental tolerance, and amenable to high densities) and there may be less controversy over the potential to compete with wild capture fisheries as they lack a large-scale commercial fishery. Together these factors warrant an investigation into whether this species may be a cultured and sustainable alternative to freshwater eels.

## Investigating the Optimal Diet for Monkeyface Pricklebacks

### Introduction

The herbivorous nature of monkeyface pricklebacks preconditions the likelihood that the species will be able to efficiently grow using sustainable, fish-free diets. As previously mentioned, *C. violaceus* undergoes an ontogenetic shift in diet from carnivory to herbivory at ~44 mm in length, unlike most fish (Montgomery, 1977; Horn et al., 1982; Horn & Gibson, 1988; German et al., 2004; Gawlicka & Horn, 2005; German & Horn, 2006; German et al., 2014). During this genetically predetermined shift, the prickleback's gut extends in length and increases production of key enzymes (e.g., α-amylase and lipase) that allow it to hydrolyze the high carbohydrate (e.g., starch, glycogen, and polysaccharides) and low lipid content of algae (Montgomery, 1977; Edwards & Horn, 1982; Horn et al., 1982; German et al., 2004; German & Horn, 2006; German et al., 2014, Kim et al., 2014). After transitioning to herbivory, monkeyface pricklebacks primarily consume chlorophyta (green) and rhodophyta (red) algae, with their preference changing seasonally as they consume carbohydrate-rich algae in the colder months and protein-rich algae in the warmer months (Horn et al., 1982; Horn et al., 1986). This seasonal selectivity, combined with their stomach's low pH of 3, comparatively lower than their intestinal tract, allows monkeyface pricklebacks to effectively digest macroalgae at an assimilation efficiency of ~42.65 % (Edwards & Horn, 1982). These adaptions likely predispose monkeyface pricklebacks as an emerging fish species for culture as it mitigates the reliance on the relatively unsustainable use of fishmeal and fish oil for their nutrition by being able to digest algae matter more readily.

An important component of the fed aquaculture industry is a diet that maximizes growth efficiently. Feed is a significant cost in aquaculture production that may exceed 50% of the farms' costs (De Silva & Turchini, 2009). In effect, a producer will seek to utilize a diet that rapidly and efficiently grows fish biomass while balancing feed costs. In contrast, a fish that grows quickly but has a poor feed conversion ratio (FCR) will be more expensive to produce because it will require more feed to grow at a similar rate. Thus, a diet that is both efficient and achieves rapid growth is necessary for monkeyface pricklebacks to be successfully cultured as an alternative to freshwater eels.

While whole macroalgae are the monkeyface prickleback's natural diet, formulated pellets are typically utilized in intensive aquaculture systems because they are designed to supply a complete diet that enhances growth and feed efficiency while maintaining water quality by reducing waste (Behnke, 2001; Sim et al., 2005). In addition, pellets are often formed via a heat extrusion process, which improves digestibility, palatability, and stability (Rahman et al., 2016). For plant ingredients that would be used for monkeyface prickleback pellets, the extrusion process likely improves digestibility by physically breaking the material down into a granular form, inactivating anti-nutritional factors (ANFs) like phytic acid that reduce carbohydrate assimilation, denaturing the plants' proteins, and increasing starch availability for amylase breakdown via starch gelatinization (Srihara & Alexander, 1984; Allan & Booth, 2004; Khater et al., 2014; Salim et al., 2023). Since monkeyface pricklebacks are marine herbivores, algae-based pellets that have undergone extrusion may be more digestible than whole seaweed due to fewer ANFs present and more starch being available (Wan et al., 2019). Formulated pellets are also nutrient-dense and readily bioavailable, which

is expected to reduce a species' metabolic cost of digestion or specific dynamic action (SDA), as opposed to other meal types (Allan & Booth, 2004; Secor, 2009; Khater et al., 2014; Rahman et al., 2016). For instance, the California yellowtail (*Seriola dorsalis*) had a lower SDA on a pelleted feed in comparison to a whole-fish meal (Parish et al., 2020). In effect, juvenile monkeyface pricklebacks fed pellets will likely have lower SDAs than individuals fed whole macroalgae. Furthermore, monkeyface pricklebacks fed pellets (Fris & Horn, 1993) had a slightly higher assimilation efficiency than whole seaweed (Edwards & Horn, 1982) on average (44% > 42%, respectively), suggesting that pellets improve assimilation efficiency. Considering these observations, monkeyface pricklebacks are likely to grow faster on a pelleted feed than whole macroalgae, assuming feed intake is equal. Fris & Horn (1993) conducted a feed study with monkeyface pricklebacks and reported that algae-based pellets with high protein concentrations maximize growth. Therefore, formulated pellets for monkeyface pricklebacks likely need to be high in protein to rear the species to market-size in a short period of time.

Terrestrial-based ingredients (e.g., soybean meal, corn meal, etc.) are often cost-effective for use as alternative feed ingredients in aquaculture (Bórquez & Hernandez, 2009). However, the effect of these ingredients on monkeyface prickleback growth has not been examined. Based on their herbivorous ecology, monkeyface pricklebacks likely have the potential to consume terrestrial-based ingredients that would otherwise negatively affect the growth of carnivorous fish because their digestive system is high in α-amylase, a carbohydrase that breaks down the carbohydrate components of plants (German et al., 2014; Kim et al., 2014). For instance, soy protein has been found to reduce weight gain, inflame

organs, and decrease food consumption in carnivorous fish (e.g., Salmonids) because these feeds are rich in ANFs and carbohydrates (Cottrell et al., 2020; Kumar et al., 2020). Thus, a feed study is needed to determine which alternative ingredient (algae and soy protein), or a combination of the two, is most efficient at rearing monkeyface pricklebacks.

I conducted a feed study that quantified growth rates, condition indices, feed intake, metabolic costs of digestion, and fillet composition of *C. violaceus* that were fed four different algal and terrestrial-based diets. I hypothesized that monkeyface pricklebacks receiving pelleted diets would have a higher growth rate, better condition, lower FCR, higher macronutrient and fatty acid content, and lower metabolic cost of digestion than monkeyface pricklebacks consuming the whole macroalgae diet.

## Materials and Methodology

A feed study was conducted at Moss Landing Marine Laboratories (MLML) to examine the growth, condition, feed intake, metabolic costs, and fillet composition of monkeyface pricklebacks fed whole algae, algae pellets, and soy-based pellets. The study began on August 16th, 2022, and concluded on July 17-19th, 2023. The study had four different dietary treatments with three different algae species that were cultured at MLML and were used for feed and pellet ingredients [the green algae sea lettuce (*Ulva* spp.), the red algae dulse (*Devaleraea mollis*) and nori (*Pyropia perforata*)]. These algae were selected because they have high protein content, are representative red and green macroalgae found in the natural diet of monkeyface pricklebacks, and are relatively easy to culture in land-based tumble systems (Horn et al., 1982; Gadberry et al., 2018). Soy protein concentrate was selected as the terrestrial ingredient because it is commonly used as a feed in agriculture and aquaculture, and it has been used previously as an ingredient in fish feed (Allan & Booth, 2004). Once pellets were formulated, juvenile fish were collected, and the study commenced. Growth measurements, feed intake, condition, metabolic costs of digestion, and fillet composition were measured over the study's duration and at the end of the study (~11 months).

### Pellet Formulation

Pellets were formulated as iso-nitrogenous (31.6% protein) and iso-lipidic (3.4% lipid). This was accomplished by determining proximate nutritional content in the algae (*Ulva* spp., *D. mollis*, and *P. perforata*) and soy ingredients and balancing the pelleted diets based on their nutritional composition. Algae and pellet samples were sent to NP Analytical

Laboratories (NPAL) for proximate analysis. The control diet (Diet 1) consisted of the whole algae *Ulva* spp., *D. mollis*, and *P. perforata*. The algae pellet (Diet 2) consisted of dried algae and a vitamin mix. The mixed pellet (Diet 3) consisted of a mix of dried algae, vitamins, and soy protein concentrate. The soy pellet (Diet 4) was mainly composed of soy protein concentrate with ~13% of *P. perforata* to increase palatability. The pellets were produced using a cold extrusion process with a California Pellet Mill. Based on the proximate analyses, pellets were formulated under the supervision of Dr. Ronald Johnson at NOAA Northwest Fisheries Science Center, as shown in Table 1.

**Table 1**

*Pellet Formulation for Algae, Mixed, and Soy Pellets Used in the Monkeyface Prickleback Feed Study (Upper Portion). Ingredients that Differ in Percent Content are Bolded. Protein, Lipid, and Nitrogen Content Indicates that the Algae, Mixed, and Soy Pellets are Isolipidic and Isonitrogenous (Lower Portion)*

| Ingredients (dry weight %) | Algae Pellet | Mixed Pellet | Soy Pellet |
|---|---|---|---|
| **Soy protein concentrate** | **0.00** | **15.00** | **30.00** |
| **Wheat flour** | **0.00** | **13.60** | **27.20** |
| **Microcrystalline cellulose** | **0.46** | **11.60** | **18.50** |
| **Pacific Dulse (*D. mollis*)** | **20.50** | **10.30** | **0.00** |
| **Sea lettuce (*Ulva* spp.)** | **30.50** | **15.30** | **0.00** |
| **Nori (*P. perforata*)** | **35.00** | **23.80** | **12.5** |
| **Soybean oil** | **0.00** | **1.20** | **2.40** |
| Trace minerals | 0.10 | 0.10 | 0.10 |
| Vitamin premix | 1.50 | 1.50 | 1.50 |
| Vitamin C | 0.50 | 0.50 | 0.50 |
| Choline | 0.50 | 0.50 | 0.50 |
| Betaine | 0.25 | 0.25 | 0.25 |
| L-methionine | 0.20 | 0.20 | 0.20 |
| L-lysine | 1.20 | 1.20 | 1.20 |
| L-threonine | 0.20 | 0.20 | 0.20 |
| Mono-Cal phosphate | 2.00 | 2.00 | 2.00 |
| Perma Pel | 2.00 | 2.00 | 2.00 |
| Taurine | 1.00 | 1.00 | 1.00 |
| Yttrium oxide | 0.01 | 0.01 | 0.01 |

(table continues)

**Table 1.** (continued)

| Calculated analysis (dry matter %) | Algae Pellet | Mixed Pellet | Soy Pellet |
|---|---|---|---|
| Crude Protein | 31.60 | 31.60 | 31.60 |
| Lipid | 3.40 | 3.40 | 3.40 |
| Nitrogen | 5.10 | 5.10 | 5.10 |

**Fish Collection, Husbandry, and Experimental Setup**

In the summer of 2022, juvenile monkeyface pricklebacks were collected from San Simeon, California, for the feed study (Institutional Animal Care and Use Committee protocol 1068 and California Department of Fish and Wildlife permits S-191050002-22189-001 and S-191050002-19126-001). On June 15–17$^{th}$, 206 juvenile monkeyface pricklebacks were collected in the early morning (~5am) at 35°39’12” N latitude and 121°15’32” W longitude, which is located near the Piedras Blancas State Marine Reserve (Figure 1). This was accomplished by walking into the intertidal at low tide, carefully overturning boulders, catching and identifying monkeyface pricklebacks, placing them in aerated coolers, and transporting them back to the Kenneth S. Norris Rancho Marino UC Ecological Reserve in Cambria, California and eventually MLML in aerated coolers (Figure 2; DeTolla et al., 1995). The 206 collected fish had an average total length of 14.3 cm (7.1–28.0 cm), and they had an average weight of 19.7 g (1.8–118.3 g) on July 7$^{th}$, 2022. Seventy-two monkeyface pricklebacks that were previously held at MLML were also added to the fish total, which had an average total length of 17.6 cm (13.9–24.0 cm) and an average weight of 25.7 g (12.7–67.0 g) on July 7$^{th}$, 2022.

**Figure 1.**

*Collection site of juvenile monkeyface pricklebacks at 35°39’12” N latitude and 121°15’32” W longitude.*

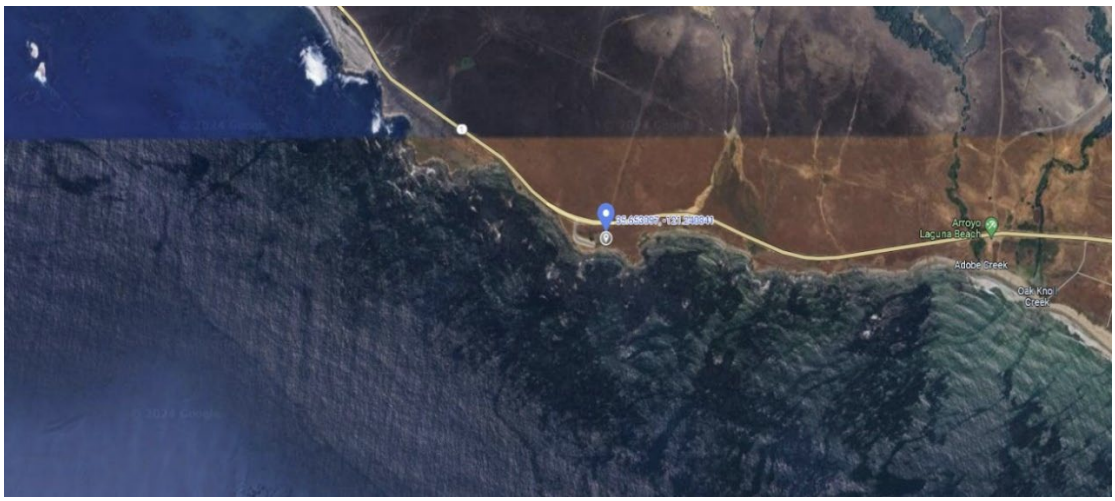



**Figure 2.**

*Monkeyface prickleback team collecting juvenile individuals at low tide in San Simeon, California.*



Fish were acclimated and quarantined in a 12-tank (350 L) flow-through system at 7.5 L/min with aeration for 14 days to ensure they were healthy with no predisposed health conditions (DeTolla et al., 1995). A Praziquantel bath was administered during this quarantine period to eliminate potential parasites. It was also administered monthly to remove potential parasites entering the flow-through system from Monterey Bay. Malachite green and melafix were used to prevent infections and heal injured fins, respectively. As the tanks are located outside at MLML's shore lab facility that sources its water from a single intake in Monterey Bay, the tanks were covered with UV-density cloth (90% knitted shade cloth) to maintain a dark shaded environment and prevent predation by birds. Tank temperatures were not controlled but varied with seawater intake.

The feed study commenced on August 16th, 2022. Prior to the beginning of the experiment, monkeyface pricklebacks were weighed, measured, and tagged with passive integrated transponders in the dorsal musculature to identify individuals. A total length range of 13.5–25.5 cm was used for the experiments, leading to 160 fish being used for the feed study. They had an average weight of 33.6 g (13.2–95.3g) and an average total length of 17.9 cm (13.6–25.5 cm). These fish were also sorted by size into three different size (in total length) subclasses: small (13.6–16.3 cm), medium (16.4–18.5 cm), and large (18.5–25.5 cm).

Dietary treatments and size subclasses were assigned to the 12 flow-through tanks at MLML's shore lab. To examine potential size effects on juvenile monkeyface prickleback growth, the feed treatments were blocked by size class with a small, medium, and large body size tank for each dietary treatment. This design was also used to limit the potential for larger individuals to outcompete or injure smaller individuals as observed in a pilot monkeyface

prickleback feed study, thereby reducing their growth and survivorship. Fish in the small subclass were randomly assigned and placed in small tanks. This process was also used for the medium and large subclasses. There were n = 13–14 fish per tank with 11 small polyvinyl chloride pipes in each tank for shelter.

Juvenile monkeyface pricklebacks consumed fresh whole macroalgae *Ulva* spp., *D. mollis*, and *P. perforata* before the feed study. To transition fish from macroalgae to pellets, individuals in the pelleted treatments were conditioned to eat pellets with Diet 2 (Table 1) for two weeks to encourage them to transition onto the base diet. These fish were fed at 2% of their total weight, while whole algae tanks were fed *ad libitum* for those two weeks. On August 16$^{th}$, 2022, fish were fed their treatment diets until the study's end on July 17-19$^{th}$, 2023. For the first four months of the study, pelleted treatments were fed incrementally at 0.5% of a tank's total fish weight until feeding ceased, while fish in the whole algae treatment were fed at 2% of body weight. Whole algae treatment was fed once a day, and the pelleted treatments were fed twice a day to maximize their growth with four hours separating each feeding. Tanks were siphoned daily during this four-month period, and a deep clean was performed once a week. For the remaining seven months of the study, pelleted and leafy algae tanks were fed at 0.75% body weight (everyday, twice a day) and 3% (three times a week) of their total tank weight, respectively. In addition, tanks were siphoned three times a week, and a deep clean was performed once a week. This switch in the feeding regimen was necessary to collect metabolic data in a reasonable timeframe.

**Sampling Protocols**

Before the study commenced, a subsample of pricklebacks from each treatment (seven fish total) was dissected for proximate analysis. This was accomplished by euthanizing the fish with a lethal dose of MS-222 (500 mg/L) and filleting them. Fillets from each subsample were kept frozen at -20°C, pooled together, mailed overnight, and analyzed by NPAL for proximate composition including moisture, protein, fat, ash, carbohydrate, calories, and fatty acid content. Nori (*P. perforata*) sheets, sea lettuce (*Ulva* spp.), dulse (*D. mollis*), and pellets were also sent to NPAL to examine the macronutrients and fatty acids listed above.

Total length and weight were measured throughout the feed study to track growth changes. This was accomplished by collecting individuals, anesthetizing them with MS-222 (0.96 g per 18.93 L, buffered with 1.92 g sodium bicarbonate), and then weighing and measuring each fish. Once weighed and measured, fish were placed in an aerated bucket with API Stress Coat Plus® to recover and then returned to their respective tank. For the first four months, juvenile monkeyface prickleback lengths and weights were measured bi-weekly. After FCR data was collected, lengths and weights were collected once a month for the remaining seven months.

To obtain a feed conversion ratio (FCR), the incremental feeding method was implemented for the pelleted tanks. A known weight of pellets (0.5% of tank weight) was introduced to each tank, and the remaining pellets were counted ten minutes after being added to each tank. Ten minutes was the chosen time limit due to the mixed pellets breaking down around that time. If five or more pellets were present after ten minutes, the tank was considered satiated, but if there were less than five pellets, another 0.5% tank weight

increment was added until satiation was achieved. Pellets were only counted for four months to obtain FCR values. Average pellet weight was obtained by counting and weighing out a handful of pellets, which was then multiplied by the number of pellets eaten to convert into grams for FCR calculations.

For the leafy algae tanks, FCR was calculated by drying out and weighing the amount fed along with the algae remaining after a feeding. This process first involved siphoning the tanks' leftover algae and fecal material. Water was then removed from fresh algae with a salad spinner. Two servings of daily algae were weighed out, with one serving being dried while the other was fed to the fish. The next morning, the tank leftovers were siphoned through a paint strainer while the uneaten dried algae was weighed. Lastly, the remaining food and fecal material were dried and weighed. The difference between the dry weight amount fed and leftover was the feed intake. This was performed once a week for four months to obtain feed intake. A feed intake for a sampled week was assumed to be the same every day until the following sampling occurred. Feed intakes were thus multiplied by seven and then summed into six feed intakes for each weigh-in period. A tank's total feed intake was then divided by its total weight gain to produce FCR values for each leafy algae tank.

Temperature was recorded throughout the study's duration. It was measured when pellets were fed and in one tank per treatment. It was found that the middle tanks shared nearly the same temperature as those in their three-tank set, allowing me to measure temperature in one tank per set to save time. A Bel-Art general-purpose liquid thermometer was used to measure temperature in each tank, which was checked with MLML's seawater intake temperature data.

At the study's endpoint, all monkeyface pricklebacks were euthanized with a lethal dose of MS-222, measured, weighed, dissected, and had their fillets frozen at -20°C. During the dissections, the liver weight of each fish was recorded for hepatosomatic index (HSI) calculations as a metric of energy storage and condition. Tanks were randomly selected for dissection, and the dissections lasted three days. Once dissections were complete, fillets were pooled by tank and shipped to NPAL for proximate analysis.

**Response Metrics**

A variety of metrics were quantified to examine potential differences among feeds. Specifically, fish growth (1–3), condition (4–5), survival (6), feed intake (7), and FCR (8) were calculated using the following equations:

$$Instantaneous\ Growth\ (IG) = \left(\frac{final\ weight}{initial\ weight}\right)^{\frac{1}{time}} x\ 100 \quad (1)$$

$$Specific\ Growth\ Rate\ (SGR) = (IG - 1)x\ 100 \quad (2)$$

$$Thermal\ Growth\ Coefficient\ (TGC) = \left(\frac{\left(final\ weight^{\frac{1}{3}} - initial\ weight^{\frac{1}{3}}\right)}{temperature\ x\ time}\right) x\ 1000 \quad (3)$$

$$Fulton's\ condition\ factor\ (K) = \left(\frac{fish\ weight\ (g)}{total\ length\ (cm)^3}\right) x\ 100 \quad (4)$$

$$Hepatosomatic\ index\ (HSI) = \left(\frac{liver\ weight\ (g)}{fish\ weight\ (g)}\right) x\ 100 \quad (5)$$

$$\%\ Survival = \left(\frac{final\ \#\ of\ fish}{initial\ \#\ of\ fish}\right) x\ 100 \quad (6)$$

$$Feed\ intake = feed\ offered\ (g) - feed\ waste\ (g) \quad (7)$$

$$Feed\ conversion\ ratio\ (FCR) = \frac{feed\ intake\ (g)}{weight\ gain\ (g)} \quad (8)$$

Growth was assessed by calculating individual instantaneous growth (IG) and then specific growth rates (SGR) from initial and final weights, along with the study's duration (Equations 1 & 2; Crane et al., 2020). Thermal growth coefficients (TGC) were also calculated to examine growth without the influence of temperature. A TGC for each individual fish was calculated for each month of the study by using the average temperature of each month and the time along with the weight between a month's measurements. Fulton's condition factor (FK) and HSI were calculated to examine fish condition among the dietary treatments at the end of the experiment. Mortalities were recorded upon occurrence and survivorship was calculated at the study's end. Feed intake and FCR were calculated to determine feed conversion efficiency. Calculations were computed in Excel and transferred to R for statistical analyses.

**Specific Dynamic Action**

The specific dynamic action (SDA) of juvenile monkeyface pricklebacks was measured to examine potential metabolic differences among diet treatments. SDA is likely to differ among pelleted treatments, as catfish that were fed pellets with high soy protein concentrations had a higher SDA than pellets with lower concentrations (Ai & Xie, 2006). To test this, oxygen consumption, $MO_2$ ($mgO_2\ g^{-1}\ h^{-1}$), was measured before and after a meal as the difference in $MO_2$ is a standard metric for calculating SDA (Tirsgaard et al., 2015).

Intermittent flow respirometry was implemented to measure the $MO_2$ of monkeyface pricklebacks (Figure 3). This was accomplished in a 75.71 L cooler at normoxic conditions using static chambers and AutoResp software (Loligo Systems). A chiller was used to ensure

the system was kept at a constant temperature of 12°C. Witrox-4 units with oxygen mini-sensors inserted into the outflow tubing of the chambers recorded changes in oxygen levels. Fish were randomly selected for SDA measurements, and eight fish were run per treatment (Edwards & Cech Jr., 1990; Chabot, Koenker, & Farrell, 2016). Individuals were fasted for 48-hours, based on gut evacuation rates upon consuming algae, in isolated flow-through tanks (Horn & Gibson, 1988; Chabot, Steffensen, and Farrell, 2016). During this time, the cooler was sanitized with a freshwater-bleach rinse that was neutralized with Kordon's Amquel Plus®. After 48 hours, fish were weighed, and the standard metabolic rate (SMR) was measured over 24 hours. SMR was calculated to determine a baseline metabolic rate and account for the stress of placing the fish in their designated chambers (Tirsgaard et al., 2015; Chabot, Koenker, & Farrell, 2016). Each cycle consisted of a five-minute flush, a one-minute waiting period, and a five-minute measuring period (Steffensen, 1989). After SMR was measured, fish were placed back into isolation for 24 hours before being fed. The feed amount was based on their tank's feed intake, calculated with Equation 7, assuming each fish had the same feed intake within a tank. Instead of anesthetizing and force-feeding, fish voluntarily consumed their meals, with remaining pellets being counted after 10 minutes of eating (Norin & Clark, 2017). After consuming their ration, fish were placed into the respirometry system no later than 30 minutes after eating, and SDA was measured over 48 hours. Once SDA measurements were finished, the fish were removed and placed back in their tanks.

The respR package was used to calculate the SMR and SDA of each fish (Harianto et al., 2019). First, the raw oxygen data for both SMR and SDA were organized into an Excel file

where the effective volume was calculated for each fish (Harianto et al., 2019). The data was then imported into R where the respR package was used to inspect the raw oxygen data, extract the lowest ranked rate from each cycle, and convert it by fish weight and the respirometer's effective volume (Harianto et al., 2019; R Core Team, 2020). For SMR, the calcSMR R function by Chabot, Koenker, & Farrell (2016) was used to calculate SMR by selecting rates after the initial two hours (to account for handling stress) with $R^2$ greater than 0.85 and running calcSMR with the selected rates. From these, SMR indices (mean of the lowest normal distribution, lowest 10% quantile, and quantile with $p = 0.2$) were calculated and the best was selected according to the recommendations by Chabot, Koenker, & Farrell (2016). To capture more points, two samples used $R^2$ of 0.70. For SDA, the calcSDA R script by Chabot, Koenker, & Farrell (2016) was used to calculate SDA magnitude ($mgO_2/g$) by selecting rates with an $R^2$ greater than 0.85 and running calcSDA (tau = 0.2, lambda = 24 hours, feed acclimation = 0.5 hours, and tolerance = 0.05) on the selected rates for each fish with their respective SMR as a baseline.

**Figure 3.**

*Intermittent flow respirometry setup to measure metabolic rates of monkeyface pricklebacks.*



**Statistical Analysis**

Potential treatment differences were analyzed using R with an $\alpha = 0.05$ for all statistical hypothesis tests (R Core Team, 2020). Generalized linear mixed models (GLMM) were used to examine differences among feed treatments for growth and condition with a sample size of 150 surviving fish. Multiple additive and interactive models were tested for each growth rate and condition index with restricted maximum likelihood. Models were selected based on the lowest Akaike Information Criterion (AICc) and evaluating a plot of the raw data. The selected additive GLMMs used response variables of SGR, FK, or HSI with independent predictors of diet treatment, body size (initial total length), and tank ID (random factor). For TGC, an additive GLMM containing TGC, size (initial total length), diet, month, and tank ID

(random factor) was selected. For models containing body size, a log transformation was applied to the size factor, and a Gaussian error distribution with a log link function was added to the model. Once a model was selected, a KS deviation test, dispersion test, and an outlier test were run to test assumptions. It should be noted that the GLMM for TGC had a significant KS deviation, but it is being presented since the QQ plot is nearly linear. A type II analysis of deviance was run with a chi-square test statistic to examine the model's significance. Estimated marginal means were calculated, then pairwise comparisons with a Tukey's Honest Significant Difference (HSD) adjustment were used to identify significant differences among treatments.

A one-way analysis of variance (ANOVA) was used to test significant differences in FCR, SDA coefficient, SMR, and fillet content (protein, fat, moisture, ash, carbohydrates, calories, and fatty acids) among dietary treatments. Normality and equal variance were tested by running a Shapiro-Wilk's normality test, conducting Bartlett and Levene's tests of equal variance, and examining the ANOVA's residuals. If the one-way ANOVA was significant, a Tukey's HSD multiple comparison of the mean was run to examine potential significant differences among treatments. Before a one-way ANOVA was applied to the fillet content data, a sample's protein, fat, ash, and carbohydrate content were converted from wet weight to dry weight via each sample's moisture content to evaluate differences in a dry-matter context. In addition, this conversion was also used for the pellet and fillet data collected before the study began. This was accomplished with the following equation (Equation 9):

$$Dry\ Weight\ Content\ (\%) = \frac{Wet\ Weight\ Content\ (\%)}{(100 - Moisture\ Content(\%))}\ x\ 100 \qquad (9)$$

Fatty acid content data was also standardized to examine each fatty acid in the context of each sample's total fatty acid content before running one-way ANOVAs. Each fatty acid value of a sample in wet weights was respectively divided by the total fat content of that sample in wet weight and multiplied by 100 (Equation 10).

$$Percent\ of\ Total\ Fat\ (\%) = \frac{Fatty\ Acid\ Content\ (\%)}{Total\ Fat\ Content\ (\%)}\ x\ 100 \quad (10)$$

Furthermore, the calculated SDA magnitude ($mgO_2/g$) was converted into an SDA coefficient for each fish to account for different amounts of feed consumed. This was accomplished by converting SDA magnitude into kjoules/g, multiplying it by the fish's weight, dividing it by the amount eaten in kjoules, and multiplying it by 100. FCR and fillet content had sample sizes of n = 12 (one value per tank). Meanwhile, SDA coefficient and SMR had sample sizes of n = 31. FCR, SMR, and fillet content met all the assumptions of a one-way ANOVA, but the SDA coefficient did not pass the Shapiro Wilk's normality test ($p = 0.0041$). Despite this, the SDA coefficient is still being reported for only three residuals were way off the QQ plot.

In addition to the one-way ANOVA, a non-metric multidimensional scaling (nMDS) test was run to compare treatment effects on fatty acids composition in a multivariate context, and a permutational analysis of variance (PERMANOVA) was used to test for treatment differences in fatty acid composition. C16:0 palmitic, C18:1n9C oleic, C18:2n6 linoleic/C19:1n10c, and C20:5n3 eicosapentaenoic/C24:0 (EPA) fatty acids were selected for the nMDS because they had significant differences among dietary treatments. Omega-3 and omega-6 were not included due to overlap with EPA and linoleic, respectively, and they are composed of multiple fatty acids. Despite not having significant differences among feed

treatments, C22:6n3 docosahexaenoic (DHA) was also included as it is an essential omega-three fatty acid. A low-stress score (< 0.2), an appropriate distance metric, and a high goodness of fit are required of an nMDS. The nMDS run with the selected fatty acids had a low-stress score of 0.000060, it had three dimensions, a Euclidean distance metric was applied, and the nMDS had a high goodness of fit with an $R^2$ of 1. A test of homogeneity of multivariate dispersion was conducted to assess the equal variance assumption of the PERMANOVA, which was met with a non-significant p-value at 0.96. The nMDS and PERMANOVA had the same sample size of n = 12.

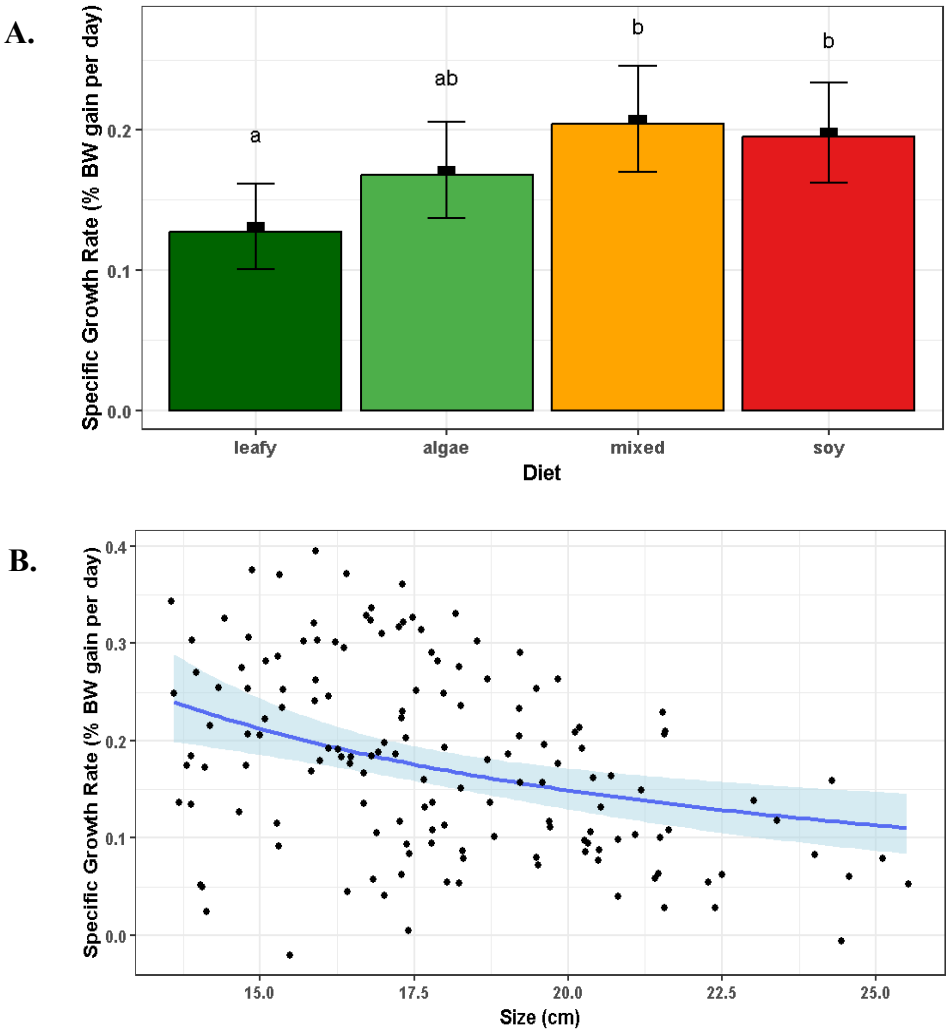
## Results

### Survivorship, Growth, and Condition

Juvenile monkeyface pricklebacks had a high survival rate (92.3% to 95.2%) across all treatments over the study's 11-month duration. Upon adjusting for body size and random tank factors, the specific growth rate (SGR) generalized linear mixed model (GLMM) indicated differences in SGR among diets for juvenile monkeyface pricklebacks (GLMM, $X^2 = 11.35$, p-value < 0.001; Figure 4A). Fish had significantly higher SGRs on the mixed and soy pellet diets than on the leafy algae diet, with estimated marginal means at 0.204% (mixed pellet), 0.195% (soy pellet), and 0.127% (leafy algae) bodyweight gain per day [Tukey's Honest Significant Difference (HSD), p-value < 0.05; Figure 4A]. The algae pellet did not significantly differ in SGR among any diets with an estimated marginal mean at 0.168% bodyweight gain per day, and the mixed pellet and soy pellet treatments did not differ between each other (Tukey's HSD, p-value > 0.05; Figure 4A). Furthermore, the GLMM indicated a negative correlation between size (total length) and SGR for juvenile monkeyface pricklebacks, with the estimated marginal means decreasing as size increased (GLMM, $X^2 = 13.97$, p-value < 0.001; Figure 4B).

**Figure 4.**

*Specific growth rate (SGR) of monkeyface pricklebacks as a function of diet and body size. (A) Bar plot depicting mean SGR across the dietary treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Black dots indicate estimated marginal means, error bars resemble the 95% confidence intervals, and letters are based on Tukey's HSD-adjusted comparisons. Bars that do not share the same letter are significantly different. (B) Non-linear regression between size (total length) and SGR with the blue line showing the estimated marginal mean, and the blue-shaded region depicting the 95% confidence interval around the mean. The black dots are the raw data points.*




The thermal growth coefficient (TGC) GLMM indicated a similar pattern with differences in TGC occurring among diets when adjusting for body size and random tank

variability (GLMM, $X^2 = 11.28$, p-value = 0.01; Figure 5A). Juvenile monkeyface pricklebacks had significantly higher TGCs on the mixed and soy pellet diets than the leafy algae diet, with estimated marginal means at 0.193 (mixed), 0.186 (soy), and 0.111 (leafy algae) (Tukey's HSD, p-value < 0.05; Figure 5A). The algae pellet did not significantly differ in TGC among any diets with an estimated marginal mean at 0.157, and TGC did not differ between the mixed and soy pellets (Tukey's HSD, p-value > 0.05; Figure 5A). TGC also differed over time in months (GLMM, $X^2 = 518.49$, p-value < 0.0001; Figure 5B), with the November and June months having significantly higher TGCs than the other months (Tukey's HSD, p-value < 0.05; Figure 5B). Temperature initially increased from August to September, decreased from September to the April/May month grouping, and then increased from April/May to the study's end in July for all treatments, varying from ~11–15°C (Figure 5C).

Juvenile monkeyface pricklebacks consuming the leafy algae (control) diet had a mean feed conversion ratio (FCR) of 5.37 (Figure 6). The algae, mixed, and soy diet treatments had mean FCRs of 2.92, 3.02, and 2.16, respectively (Figure 6). The one-way analysis of variance (ANOVA) for FCR differed significantly among diet treatments (ANOVA, $F_{3,8} = 14.46$, p-value = 0.0014), with the pelleted treatments having significantly lower FCRs (i.e., more efficient conversion) than the leafy algae treatment (Tukey's HSD, p-value < 0.05; Figure 6). The pellet treatments did not differ significantly in FCR among each other (Tukey's HSD, p-value > 0.05; Figure 6).

**Figure 5.**

*Thermal growth coefficient (TGC) of monkeyface pricklebacks as a function of diet and month in addition to temperature as a function of month. (A) Bar plot depicting thermal growth coefficient (TGC) of monkeyface pricklebacks across the dietary treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Black dots indicate estimated marginal means, error bars depict the 95% confidence intervals, and letters are based on Tukey's HSD-adjusted comparisons. Bars that do not share the same letter are significantly different. (B) Box plot depicting thermal growth coefficient (TGC) of monkeyface pricklebacks across the dietary treatments. Black boxes indicate estimated marginal means, error bars depict the 95% confidence intervals, and letters are based on Tukey's HSD-adjusted comparisons. Boxes that do not share the same letter are significantly different. (C) Line plot depicting temperature over time in months among the four diet treatments. The dots represent the average temperature at each time of measurement (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). The error bars represent standard error.*

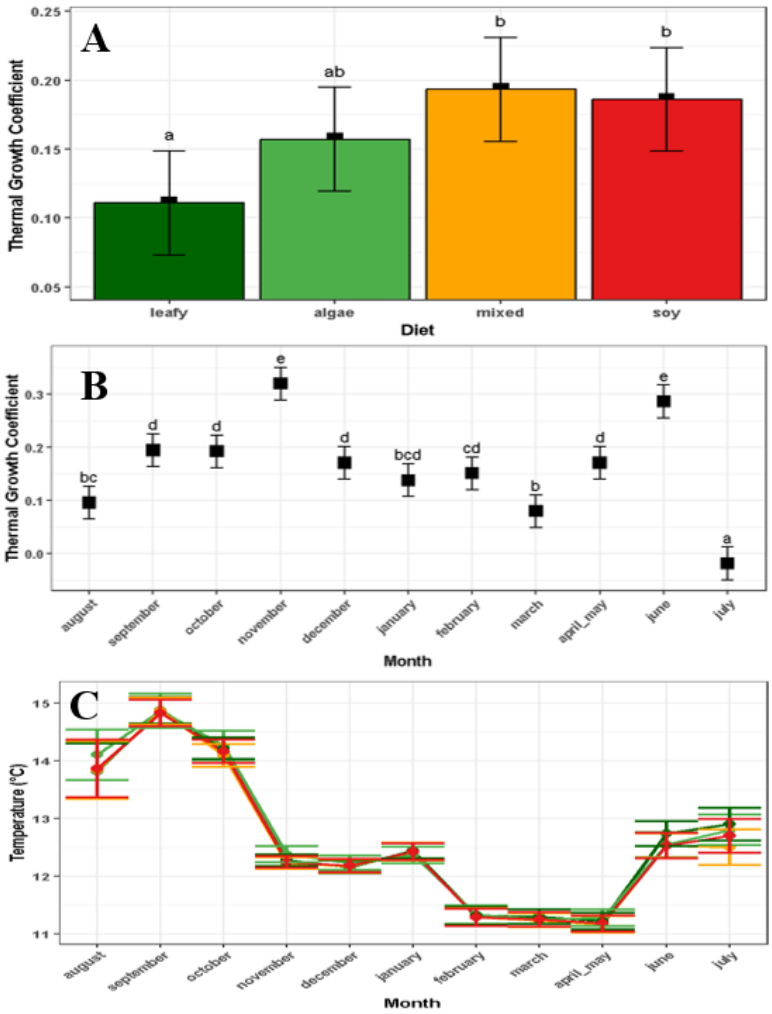
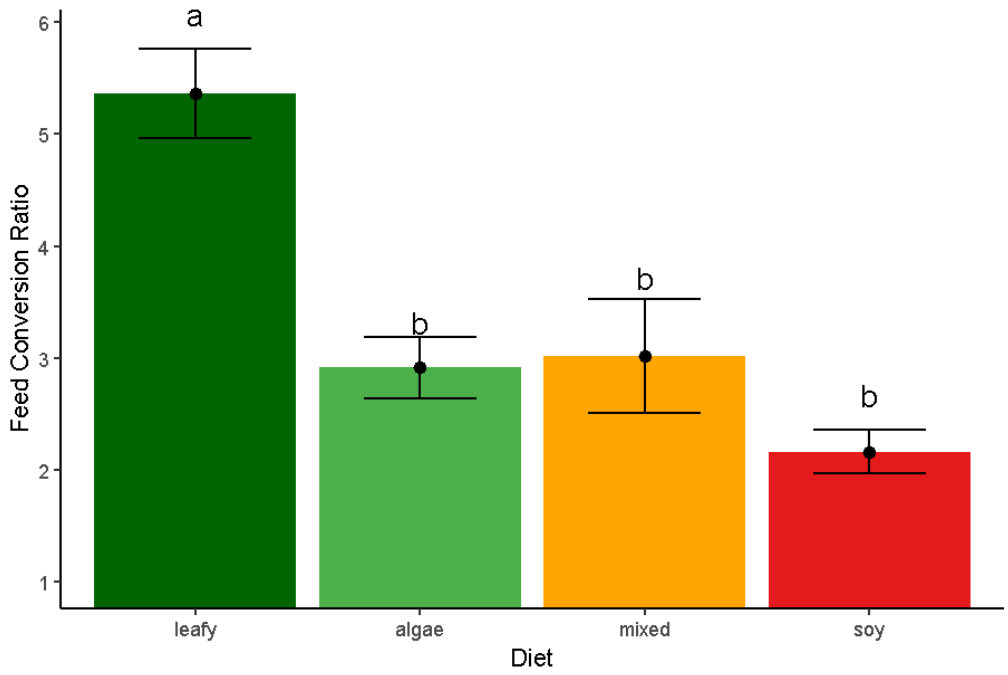


**Figure 6.**

*Bar plots illustrating the feed conversion ratio (high values = poor FCR) of monkeyface pricklebacks fed the four diet treatments. Black dots indicate the mean, error bars show standard error, letters are based on Tukey's multiple comparisons (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Bars that do not share the same letter are significantly different.*

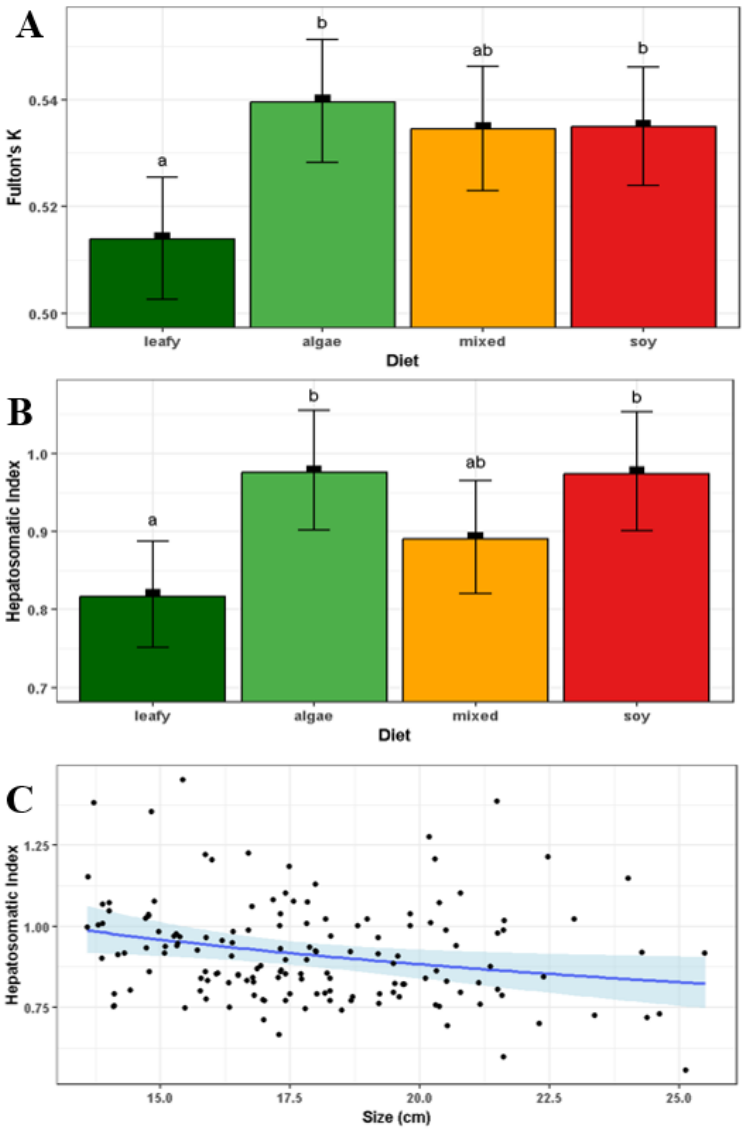



Juvenile monkeyface pricklebacks differed in Fulton's condition factor (FK) among diet treatments based on the GLMM that adjusted for fish body size and random tank variability (GLMM, $X^2 = 11.49$, p-value = 0.0093; Figure 7A). Fish fed the algae and soy pellet diets had a significantly higher FK than those eating the leafy algae diet with estimated marginal means at 0.540 (algae), 0.535 (soy), and 0.514 (leafy algae) (Tukey's HSD, p-value < 0.05; Figure 7A). The mixed pellet diet, however, did not differ from the other diets with an estimated marginal mean at 0.534, and the algae pellet and soy pellet diets did not significantly differ in FK between each other (Tukey's HSD, p-value > 0.05; Figure 7A). Similarly, hepatosomatic index (HSI) differed among diet treatments (GLMM, $X^2 = 12.84$, p-

value = 0.005; Figure 7B), with the algae and soy pellet diets having a significantly higher relative liver weight, reflecting more energy storage, compared to the leafy algae diet; estimated marginal means were 0.975 (algae), 0.974 (soy), and 0.817 (leafy algae) (Tukey's HSD, p-value < 0.05; Figure 7B). The mixed pellet diet did not differ in HSI from the other diets with an estimated marginal mean at 0.890, and the algae pellet and soy pellet treatments did not significantly differ in HSI between each other (Tukey's HSD, p-value > 0.05; Figure 7B). The HSI GLMM also detected a negative correlation between total length and HSI for juvenile monkeyface pricklebacks with liver condition decreasing as body size increased (GLMM, $X^2 = 5.9$, p-value = 0.015; Figure 7C).

**Figure 7.**

*Monkeyface prickleback body condition and energy storage in the feed study. (A) Bar plot depicting FK (higher FK = better condition) of monkeyface pricklebacks fed the different dietary treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Black dots indicate estimated marginal means, error bars resemble the 95% confidence intervals, and letters are based on Tukey's HSD-adjusted comparisons. Bars that do not share the same letter are significantly different. (B) Bar plot depicting mean HSI (higher HSI = higher energy reserves) across the dietary treatments. (C) Non-linear regression between size (total length) and HSI, with the blue line showing the estimated marginal mean, and the blue-shaded region showing the 95% confidence interval around the mean. The black dots are the raw data points.*



**Pellet and Fillet Composition**

Macronutrient content differed among the diets with protein, carbohydrates, and calories increasing while ash and moisture decreased from the algae to the soy pellet (Table 2). Nori (*P. perforata*) had lower protein and moisture content than pellets, and it had lower ash content, except for the soy pellet diet (Table 2). In addition, *P. perforata* had higher carbohydrate content than the pellets and higher calories, except for the soy pellet diet (Table 2). Since moisture values were not analyzed for *D. mollis* and *Ulva* spp., their wet weight content cannot be compared to the dry weight content values (Table 2).

The macronutrient content of juvenile monkeyface pricklebacks from the study's start was roughly similar between the leafy algae and pelleted diets (Table 3). Protein, however, was higher in the leafy algae diet than in the pelleted diets. In addition, the macronutrient and fatty acid content of monkeyface pricklebacks from the study's start are similar to reported values from the study's end (Tables 3 & 4).

Protein (ANOVA, $F_{3,8} = 3.96$, p-value = 0.053), ash (ANOVA, $F_{3,8} = 5.16$, p-value = 0.028), moisture (ANOVA, $F_{3,8} = 13.23$, p-value = 0.0018), carbohydrates (ANOVA, $F_{3,8} = 14.77$, p-value = 0.0013), and caloric (ANOVA, $F_{3,8} = 4.10$, p-value = 0.049) content in juvenile monkeyface pricklebacks differed among diet treatments (Figures 8A, 8C, 8D, 8E, & 8F; Table 4). However, there was no difference in fat content among the diet treatments (ANOVA, $F_{3,8} = 0.50$, p-value = 0.69; Figure 8B; Table 4). Although protein was marginally non-significant, differences were found among the treatments with the Tukey's HSD, so it is being reported as significant (Table 4). The leafy algae treatment had significantly higher protein content than the algae pellet treatment (Tukey's HSD, p-value < 0.05; Figure 8A;

Table 4), but the mixed and soy pellet treatments did not differ in protein content from the other treatments and each other (Tukey's HSD, p-value > 0.05; Figure 8A; Table 4). The leafy algae treatment had significantly higher ash content than the algae and mixed pellet treatments (Tukey's HSD, p-value < 0.05; Figure 8C; Table 4), but the soy pellet treatment did not differ in ash content from the other treatments (Tukey's HSD, p-value > 0.05; Figure 8C; Table 4). There was also no difference in ash content between the algae and mixed pellet diets (Tukey's HSD, p-value > 0.05; Figure 8C; Table 4). Moisture content was significantly higher in the leafy algae treatment than in the other treatments (Tukey's HSD, p-value < 0.05; Figure 8D; Table 4). In addition, there was no difference in moisture content among the pellet diets (Tukey's HSD, p-value > 0.05; Figure 8D). Carbohydrates were significantly higher in the leafy algae treatment than in the other diet treatments (Tukey's HSD, p-value < 0.05; Figure 8E; Table 4). There was also no difference in carbohydrate content among the pellet diets (Tukey's HSD, p-value > 0.05; Figure 8E). Unlike the other fillet content variables, caloric content was significantly higher in the algae pellet treatment than the leafy algae treatment (Tukey's HSD, p-value < 0.05; Figure 8F; Table 4), with the mixed and soy pellet treatments not differing from the other treatments and each other (Tukey's HSD, p-value > 0.05; Figure 8F; Table 4).

**Figure 8.**

*Bar plots depicting patterns of proximate composition of monkeyface prickleback fillets across the different diet treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Shown is the percent of dry matter protein (A), fat (B), ash (C), and carbohydrate (E) content in fillets among dietary treatments. Moisture (D) and calories (F) are displayed on a wet weight basis in fillets among dietary treatments. Black dots indicate the mean, error bars show standard error, and the letters are based on Tukey's multiple comparisons. Bars that do not share the same letter are significantly different.*

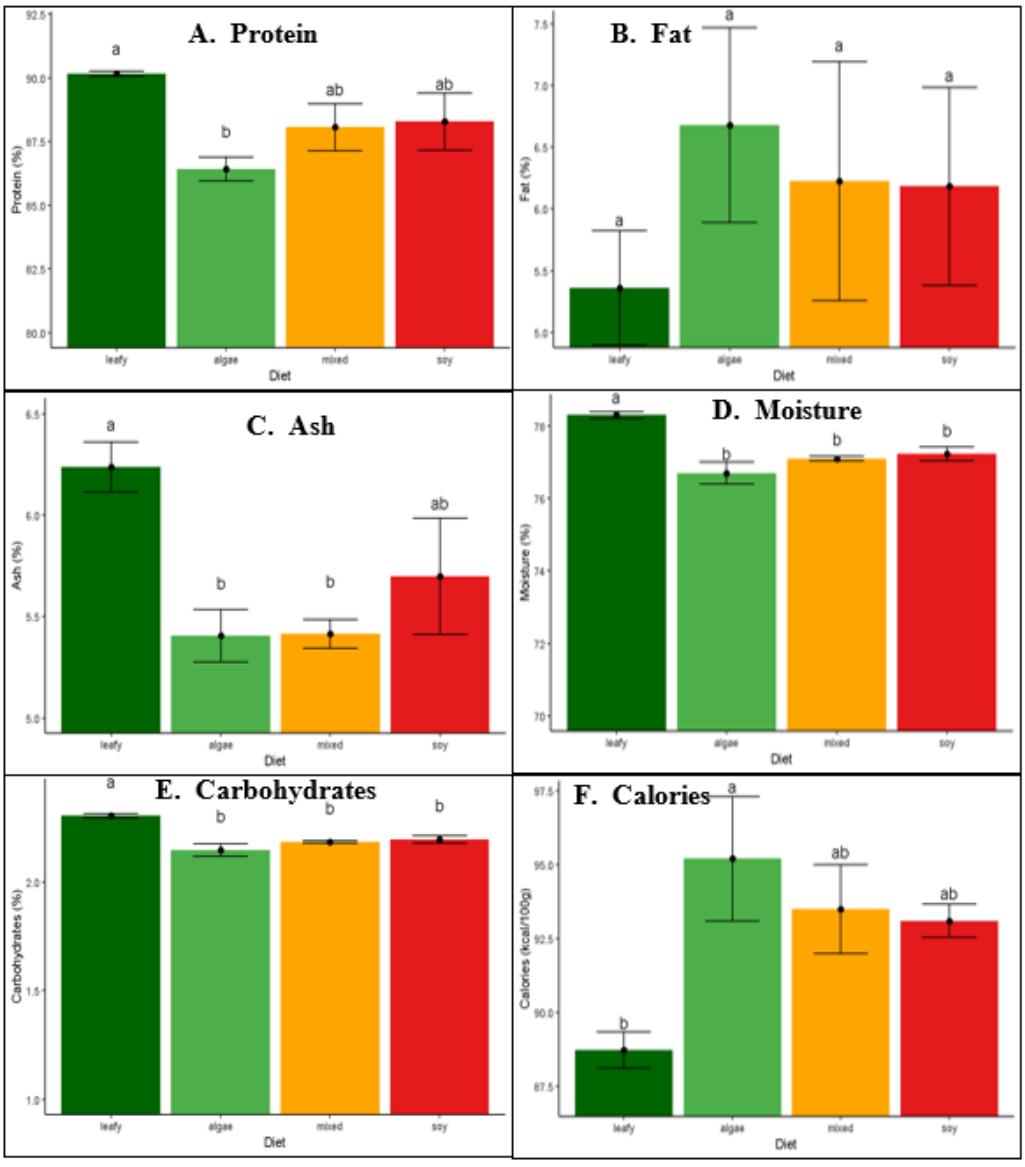


**Table 2**

*Macronutrient Content of Diets (Protein, Fat, Ash, Moisture, Carbohydrate, and Caloric Content) (Upper Portion). Fatty Acid Content of Diets (Lower Portion)*

| Macronutrient | Algae Pellet | Mixed Pellet | Soy Pellet | *P. perforata* Sheets | *Ulva* spp. | *D. mollis* |
|---|---|---|---|---|---|---|
| Protein (%) | 32.99 | 33.79 | 34.94 | 29.10 | 23.10* | 21.50* |
| Ash (%) | 23.03 | 14.65 | 6.17 | 10.69 | 26.20* | 30.90* |
| Moisture (%) | 13.60 | 13.30 | 11.00 | 6.88 | | |
| Carbohydrates (%) | 41.44 | 48.44 | 55.17 | 56.70 | | |
| Calories (kcal/100g) | 277.00 | 310.00 | 351.00 | 349.00 | | |

| Fatty Acids (% of Total Fat) | Algae Pellet | Mixed Pellet | Soy Pellet | *P. perforata* Sheets | *Ulva spp.* | *D. mollis* |
|---|---|---|---|---|---|---|
| C18:3n3 Linolenic | 5.53 | 5.07 | 4.75 | 1.79 | | |
| C20:4n6 Arachidonic/C23:0 | 2.16 | 1.47 | 1.19 | 4.34 | | |
| C20:5n3 Eicosapentaenoic/C24:0 (EPA) | 21.89 | 12.76 | 4.60 | 28.43 | | |
| C22:6n3 Docosahexaenoic (DHA) | 1.75 | 1.47 | 1.19 | 1.79 | | |
| Monounsaturated Fatty Acids | 9.82 | 15.15 | 19.25 | 7.40 | | |
| Omega 3 Fatty Acids | 33.29 | 20.26 | 9.49 | 29.55 | | |
| Omega 6 Fatty Acids | 8.86 | 30.26 | 44.48 | 13.72 | | |
| Polyunsaturated Fatty Acids | 42.46 | 50.74 | 54.03 | 43.27 | | |
| Saturated Fatty Acids | 28.90 | 22.79 | 19.73 | 27.49 | | |
| Total Fat (g/100g) | 2.28 | 2.72 | 3.35 | 2.23 | | |
| Trans Fatty Acids | 3.54 | 1.99 | 1.19 | 1.79 | | |

* *Ulva* spp. and *D. mollis* moisture values were not analyzed and reported as wet weights.

**Table 3**

*Macronutrient Content of Juvenile Monkeyface Prickleback Fillets Before the Feed Study Started (Protein, Fat, Ash, Moisture, Carbohydrate, and Caloric Content) (Upper Portion). Fatty Acid Content of Juvenile Monkeyface Prickleback Fillets Before the Feed Study Started (Lower Portion)*

| Macronutrient | Leafy Algae | Pelleted Diets |
|---|---|---|
| Protein (%) | 85.11 | 47.49 |
| Fat (%) | 4.03 | 3.66 |

(table continues)

**Table 3.** (continued)

| | | |
|---|---|---|
| Ash (%) | 4.74 | 5.96 |
| Moisture (%) | 81.20 | 79.70 |
| Carbohydrates (%) | 6.12 | 42.90 |
| Calories (kcal/100g) | 75.40 | 80.10 |
| | | |
| Fatty Acids (% of Total Fat) | Leafy Algae* | Pelleted Diets |
| C18:3n3 Linolenic | | 11.76 |
| C20:4n6 Arachidonic/C23:0 | | 11.76 |
| C20:5n3 Eicosapentaenoic/C24:0 (EPA) | | 17.97 |
| C22:6n3 Docosahexaenoic (DHA) | | 11.76 |
| Monounsaturated Fatty Acids | | 17.59 |
| Omega 3 Fatty Acids | | 33.24 |
| Omega 6 Fatty Acids | | 11.76 |
| Polyunsaturated Fatty Acids | | 43.24 |
| Saturated Fatty Acids | | 30.00 |
| Total Fat (g/100g) | | 0.34 |
| Trans Fatty Acids | | 11.76 |

* There was insufficient sample for the leafy algae fatty acid analysis.

## Table 4

*Macronutrient Content in Means and Standard Errors of Monkeyface Prickleback Fillets Across the Diet Treatments After the Feed Study (Protein, Fat, Ash, Moisture, Carbohydrate, and Caloric Content) (Upper Portion). Fatty Acid Content in Means and Standard Errors of Monkeyface Prickleback Fillets Among Dietary Treatments (Lower Portion). Rows that are Bolded were Significantly Different in Composition Among the Diet Treatments*

| Macronutrient | Leafy Algae | Algae Pellet | Mixed Pellet | Soy Pellet | F-Value (3,8) | P-Value |
|---|---|---|---|---|---|---|
| **Protein (%) *** | **90.17 ± 0.11** | **86.42 ± 0.47** | **88.07 ± 0.93** | **88.29 ± 1.13** | **3.96** | **0.053*** |
| Fat (%) | 5.36 ± 0.46 | 6.68 ± 0.79 | 6.23 ± 0.97 | 6.18 ± 0.80 | 0.50 | 0.69 |
| **Ash (%)** | **6.24 ± 0.12** | **5.41 ± 0.13** | **5.41 ± 0.70** | **5.70 ± 0.29** | **5.16** | **0.028** |
| **Moisture (%)** | **78.30 ± 0.10** | **76.70 ± 0.31** | **77.10 ± 0.06** | **77.23 ± 0.19** | **13.23** | **0.0018** |
| **Carbohydrates (%)** | **2.30 ± 0.01** | **2.15 ± 0.03** | **2.18 ± 0.01** | **2.20 ± 0.02** | **14.77** | **0.0013** |

(table continues)

**Table 4.** (continued)

| **Calories (kcal/100g)** | **88.73 ± 0.61** | **95.20 ± 2.11** | **93.50 ± 1.50** | **93.10 ± 0.57** | **4.10** | **0.049** |
|---|---|---|---|---|---|---|
| | | | | | | |
| Fatty Acids (% of Total Fat) | Leafy Algae | Algae Pellet | Mixed Pellet | Soy Pellet | F-Value (3,8) | P-Value |
| **C16:0 Palmitic** | **19.50 ± 0.22** | **21.05 ± 0.18** | **17.62 ± 0.08** | **15.56 ± 0.07** | **247.40** | **3.17E-08** |
| **C18:1n9C Oleic** | **11.05 ± 0.21** | **12.88 ± 0.60** | **15.80 ± 0.57** | **17.32 ± 1.09** | **16.66** | **0.00084** |
| **C18:2n6 Linoleic/C19:1n10c** | **6.23 ± 0.83** | **4.29 ± 0.69** | **18.48 ± 1.12** | **30.36 ± 0.71** | **201.10** | **7.18E-08** |
| C18:3n3 Linolenic | 6.23 ± 0.83 | 4.52 ± 0.58 | 5.26 ± 1.13 | 4.48 ± 1.14 | 0.75 | 0.56 |
| C20:4n6 Arachidonic/C23:0 | 6.35 ± 0.72 | 4.29 ± 0.69 | 5.26 ± 1.13 | 4.48 ± 1.14 | 0.98 | 0.45 |
| **C20:5n3 Eicosapentaenoic/C24:0 (EPA)** | **23.59 ± 0.21** | **21.54 ± 0.98** | **14.63 ± 0.77** | **8.15 ± 1.20** | **64.65** | **5.96E-06** |
| C22:5n3 Docosapentaenoic | 6.42 ± 0.84 | 4.29 ± 0.69 | 5.26 ± 1.13 | 4.48 ± 1.14 | 1.00 | 0.44 |
| C22:5n6 Docosapentaenoic | 6.23 ± 0.83 | 4.29 ± 0.69 | 5.26 ± 1.13 | 4.48 ± 1.14 | 0.83 | 0.51 |
| C22:6n3 Docosahexaenoic (DHA) | 6.23 ± 0.83 | 4.29 ± 0.69 | 5.26 ± 1.13 | 4.48 ± 1.14 | 0.83 | 0.51 |
| Monounsaturated Fatty Acids | 19.18 ± 0.59 | 21.72 ± 1.14 | 22.43 ± 0.79 | 23.56 ± 1.73 | 2.62 | 0.12 |
| **Omega 3 Fatty Acids** | **35.94 ± 0.78** | **33.70 ± 1.20** | **23.28 ± 1.12** | **14.63 ± 1.99** | **53.44** | **1.23E-05** |
| **Omega 6 Fatty Acids** | **10.26 ± 0.22** | **7.96 ± 0.39** | **22.77 ± 0.80** | **34.28 ± 0.24** | **669.30** | **6.09E-10** |
| **Polyunsaturated Fatty Acids** | **46.28 ± 0.98** | **41.68 ± 1.49** | **45.99 ± 0.29** | **48.90 ± 2.07** | **4.73** | **0.035** |
| **Saturated Fatty Acids** | **27.23 ± 0.30** | **28.76 ± 0.16** | **24.04 ± 0.21** | **20.95 ± 0.04** | **302.60** | **1.43E-08** |
| Total Fat (g/100g) | 0.67 ± 0.10 | 0.98 ± 0.14 | 0.83 ± 0.16 | 1.00 ± 0.21 | 0.98 | 0.45 |
| Trans Fatty Acids | 6.23 ± 0.83 | 4.29 ± 0.69 | 5.26 ± 1.13 | 4.48 ± 1.14 | 0.83 | 0.51 |

* Protein was marginally non-significant at $\alpha$=0.05.

Fatty acid content differed among diets (Table 2). Linolenic, arachidonic, C20:5n3 eicosapentaenoic/C24:0 (EPA), C22:6n3 docosahexaenoic (DHA), omega 3, saturated, and

trans-saturated fatty acids decreased in content from algae to the soy pellet (Table 2). Monounsaturated, omega 6, polyunsaturated, and total fat increased in content from algae to the soy pellet (Table 2). P. perforata had lower linolenic and monounsaturated fatty acid content than the pellets but higher EPA and DHA fatty acid content (Table 2).

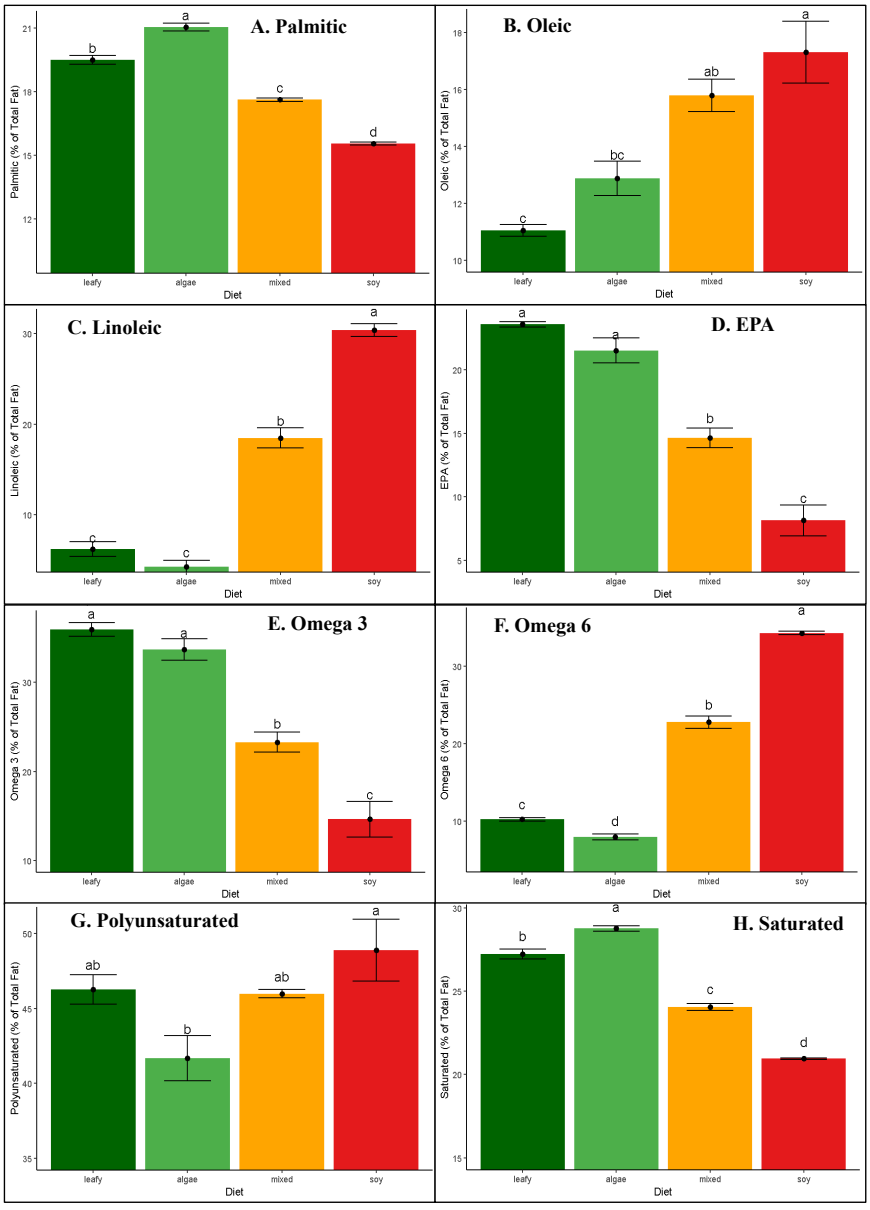
Palmitic (ANOVA, $F_{3,8}$ = 247.40, p-value = 3.17E-08), oleic (ANOVA, $F_{3,8}$ = 16.66, p-value = 0.00084), linoleic (ANOVA, $F_{3,8}$ = 201.10, p-value = 7.18E-08), EPA (ANOVA, $F_{3,8}$ = 64.65, p-value = 5.96E-06), omega-3 (ANOVA, $F_{3,8}$ = 53.44, p-value = 1.23E-05), omega-6 (ANOVA, $F_{3,8}$ = 669.30, p-value = 6.09E-10), polyunsaturated (ANOVA, $F_{3,8}$ = 4.73, p-value = 0.035), and saturated (ANOVA, $F_{3,8}$ = 302.60, p-value = 1.43E-08) fatty acids in juvenile monkeyface pricklebacks differed among diet treatments (Figures 9A, 9B, 9C, 9D, 9E, & 9F; Table 4). Fish fed the algae pellet had significantly higher palmitic fatty acid content than the other diets (Tukey's HSD, p-value < 0.05; Figure 9A). Additionally, the fish fed the leafy algae diet had significantly higher palmitic fatty acid content than the mixed and soy pellet treatments (Tukey's HSD, p-value < 0.05; Figure 9A), and the mixed pellet diet was significantly higher in palmitic content than the soy pellet treatment (Tukey's HSD, p-value < 0.05; Figure 9A). Oleic fatty acid content was significantly higher in fish fed the soy pellet diet than the algae pellet and leafy algae diets (Tukey's HSD, p-value < 0.05; Figure 9B), while those on the mixed pellet diet did not differ in oleic content between the soy and algae pellet treatments (Tukey's HSD, p-value > 0.05; Figure 9B), but it was significantly higher than those on the leafy algae diet (Tukey's HSD, p-value < 0.05; Figure 9B). Fish fed the algae pellet, and leafy algae diets did not differ from each other in oleic fatty acid content (Tukey's HSD, p-value > 0.05; Figure 9B). Fish fed the soy pellet diet had significantly

higher content for linoleic fatty acid than the other diets (Tukey's HSD, p-value < 0.05; Figure 9C). In addition, the mixed pellet treatment had significantly higher linoleic content than the algae pellet and leafy algae diets (Tukey's HSD, p-value < 0.05; Figure 9C). The algae pellet and leafy algae treatments did not differ in linoleic fatty acid content (Tukey's HSD, p-value > 0.05; Figure 9C). Fish eating the leafy algae and algae pellet diets did not differ in EPA content between each other (Tukey's HSD, p-value > 0.05; Figure 9D), but they were significantly higher than the mixed and soy pellet diets (Tukey's HSD, p-value < 0.05; Figure 9D). The mixed pellet treatment was also significantly higher than the soy pellet diet in EPA fatty acid content (Tukey's HSD, p-value < 0.05; Figure 9D). Similarly, omega-3 fatty acid content did not differ between fish fed the leafy algae and algae pellet treatments (Tukey's HSD, p-value > 0.05; Figure 9E), but the diets were significantly greater in omega-3 content than the mixed and soy pellet diets (Tukey's HSD, p-value < 0.05; Figure 9E). Additionally, fish on the mixed pellet diet were significantly higher than the soy pellet treatment in omega-3 fatty acid content (Tukey's HSD, p-value < 0.05; Figure 9E). The soy pellet had significantly higher omega-6 fatty acid content than the other diets (Tukey's HSD, p-value < 0.05; Figure 9F). In addition, fish fed the mixed pellet diet had significantly higher omega-6 fatty acid content than the algae pellet and leafy algae treatments and the leafy algae diet was significantly higher in omega-6 content than the algae pellet treatment (Tukey's HSD, p-value < 0.05; Figure 9F). Fish eating the soy pellet diet had significantly higher polyunsaturated fatty acid content than the algae pellet treatment (Tukey's HSD, p-value < 0.05; Figure 9G), but the mixed pellet and the leafy algae diets did not differ in polyunsaturated fatty acid content from the algae pellet and soy pellet treatments and each

other (Tukey's HSD post-hoc comparison, p-value > 0.05; Figure 9G). Fish fed the algae pellet had significantly higher saturated fatty acid content than the other diets (Tukey's HSD, p-value < 0.05; Figure 9H). Additionally, fish fed the leafy algae diet had significantly higher saturated fatty acid content than the mixed pellet and soy pellet treatments (Tukey's HSD, p-value < 0.05; Figure 9H), and the mixed pellet diet was significantly higher in saturated fatty acid content than the soy pellet treatment (Tukey's HSD, p-value < 0.05; Figure 9H). The remaining fatty acids did not differ in content among diet treatments for juvenile monkeyface pricklebacks (ANOVA, p-value > 0.05; Table 4).

**Figure 8.**

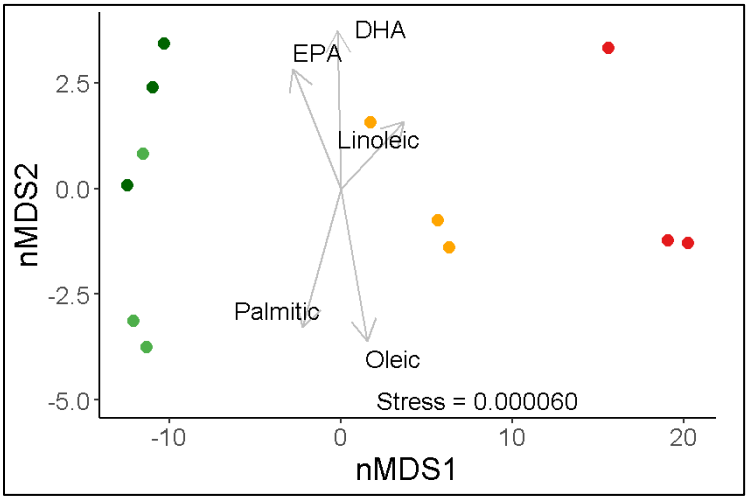
*Bar plots (A-F) depicting the fatty acid composition of monkeyface prickleback fillets across the different diet treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Shown are fatty acids of palmitic (A), oleic (B), linoleic (C), EPA (D), omega-3 (E), omega-6 (F), polyunsaturated (G), and saturated (H) content. Black dots indicate the mean, error bars show standard error, and the letters are based on Tukey's multiple comparisons of the mean. Bars that do not share the same letter are significantly different.*



In a multivariate context, the permutational analysis of variance (PERMANOVA) detected significant effects of dietary treatments on fillet content for palmitic, oleic, linoleic, EPA, and DHA fatty acids (PERMANOVA, $F_{3,8} = 71.86$, p-value = 0.001, $R^2 = 0.96$). The non-metric multidimensional scaling (nMDS) test visually supports the PERMANOVA result, with the nMDS axis 1 displaying distinct separation among the diet treatments (Figure 10). Linoleic, oleic, EPA, and palmitic fatty acids distinguish the groups, with linoleic being the main driver for differences between mixed and soy pellet diets. EPA and palmitic acid were higher in the two algae diets. Oleic also explains the distinction between the mixed and soy pellet groups, but to a lesser extent. The nMDS axis 2 shows separation between the leafy algae and algae pellet diets, which are mainly driven by palmitic and oleic fatty acids being higher in fish fed the algae pellets, while EPA, DHA, and linoleic were higher in fish fed the leafy algae.

**Figure 9.**

*A non-metric multidimensional scaling (nMDS) of palmitic, oleic, linoleic, EPA, and DHA in fillets among dietary treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Arrows indicate strength (arrow length) and direction of fatty acids in relation to dietary treatments.*

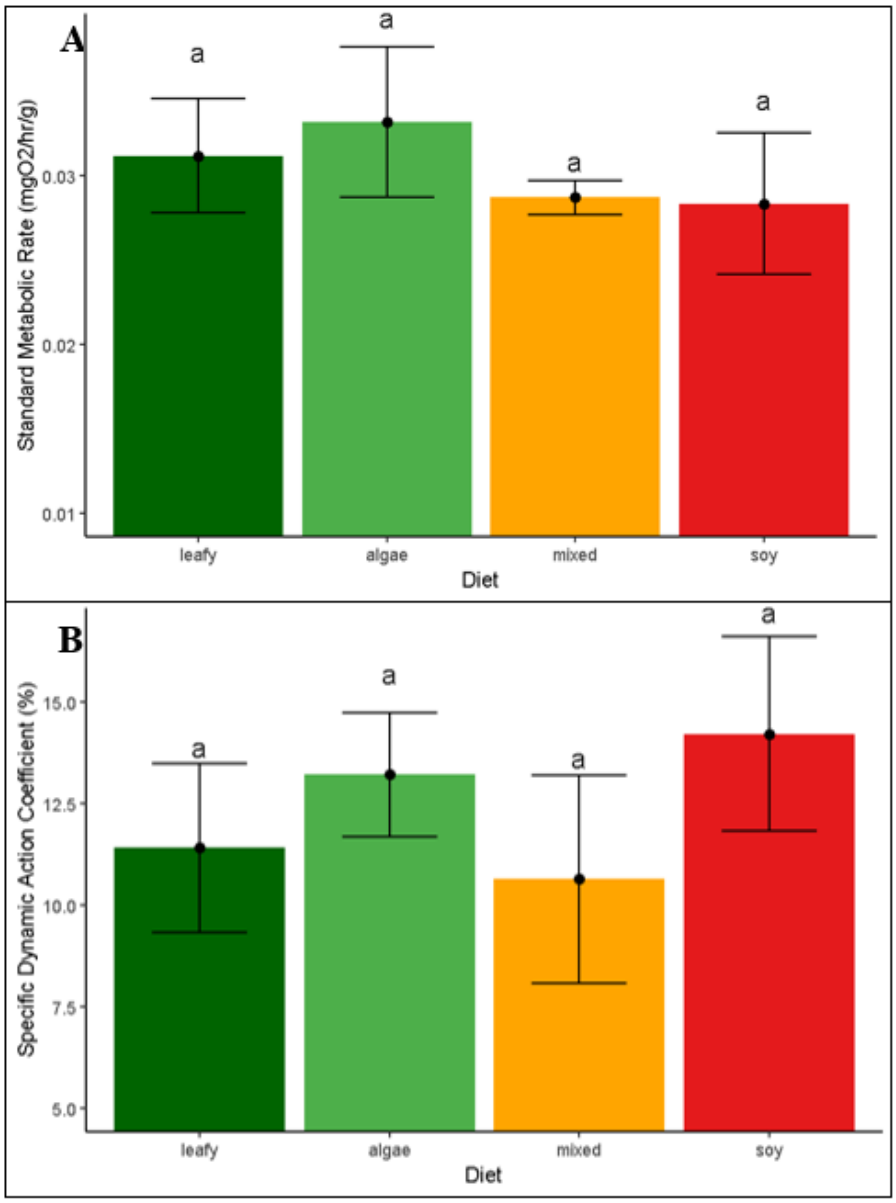


**Metabolic Rates**

Across all diet treatments, juvenile monkeyface pricklebacks had mean standard metabolic rates (SMRs) of 0.03 $mgO_2/hr/g$ (Figure 11A). SMR did not differ among diet treatments (ANOVA, $F_{3,27} = 0.445$, p-value = 0.72; Figure 11A). The leafy algae, algae pellet, mixed pellet, and soy pellet treatments had mean specific dynamic action (SDA) coefficients of 11.4%, 13.2%, 10.6%, and 14.2%, respectively (Figure 11B). This resulted in the SDA coefficients not differing among diet treatments (ANOVA, $F_{3,27} = 0.53$, p-value = 0.66; Figure 11B), indicating that the costs of digestion were similar for all diets in this study.

**Figure 9.**

*Standard metabolic rates (SMRs) and specific dynamic action (SDA) coefficient of monkeyface pricklebacks consuming the different diets. (A) Bar plot illustrating the standard metabolic rate (SMR) of monkeyface pricklebacks across the dietary treatments (leafy algae = dark green, algae pellet = light green, mixed pellet = yellow, soy pellet = red). Black dots indicate the mean, error bars show standard error, and the letters are based on Tukey's multiple comparisons consisting of the mean. Bars that do not share the same letter are significantly different. (B) Bar plot depicting specific dynamic action (SDA) coefficient as a function of the different dietary treatments.*



## Discussion

The feed study was conducted to determine the optimal diet to rear monkeyface pricklebacks for aquaculture. Monkeyface pricklebacks grew faster and more efficiently and were in better condition on the pelleted diets than their natural whole algae diet, supporting the study's hypothesis that the fish consuming the pelleted feeds would have higher growth rates, be in better condition, and lower feed conversion ratios (FCR). However, not all the macronutrients were higher in content for the pelleted feeds, with fish eating the leafy algae generally having higher macronutrient content. Similarly, fatty acids were higher in pelleted diets for some but not all fatty acids. Furthermore, monkeyface pricklebacks did not have lower metabolic costs of digestion on the pelleted diets, refuting the study's hypothesis that specific dynamic action (SDA) would be lower for fish consuming pelleted feeds. Based on biological and economic factors, pellets containing soy protein perform comparatively better in monkeyface prickleback growth trials, as they enable fish to reach market-size quickly, efficiently, and healthily, thereby reducing farm costs. However, soy-based feeds are less environmentally sustainable, and their use would likely raise the cost of soy, impacting other agrofeed industries that are reliant on soy. In contrast, leafy algae have more environmental benefits and a higher quality macronutrient profile, despite yielding lower growth, condition, and efficiency in the fish that would ultimately cost farmers more. Therefore, farmers need to weigh these benefits and downfalls when considering diets for monkeyface pricklebacks as an aquaculture alternative to freshwater eels.

**Survivorship, Growth, and Condition**

Juvenile monkeyface pricklebacks were found to have higher growth rates on pelleted diets than the leafy algae control diet (Figures 4A & 5). Smaller monkeyface pricklebacks also had higher specific growth rates (SGRs) than larger-sized fish, which has been observed in other fish as size is inversely related to fish growth (Sigourney et al., 2008; Figure 4B). Thus, the observed correlation between size and SGR for monkeyface pricklebacks can be supported. The superior growth of the pelleted diets may be attributed to a number of factors including efficient feed conversion, digestibility, stability, and complete nutritional profiles which are often associated with pelleted feeds and their extrusion (Behnke, 2001; Allan & Booth, 2004; Sim et al., 2005; Khater et al., 2014; Rahman et al., 2016). This has been observed in the cichlid fish *Oreochromis niloticus* which had more weight gain on a commercial cichlid feed than four different species of leafy algae, as the pellets were easier to detect and nutritionally balanced (Tadesse et al., 2003). However, it may be due to differences in diet composition. The pelleted diets had higher protein content than the leafy algae diet (Table 2). Since protein plays a vital role in fish growth, the observed significant differences in SGR and thermal growth coefficient (TGC) may be due to the pelleted diets' higher protein content than the leafy algae (Manam, 2023). This is likely given Fris & Horn (1993) reported monkeyface pricklebacks had higher growth consuming pellets with higher protein content than diets with less protein. The pelleted diets also had higher total fat content than *P. perforata*, which may have led to the fish consuming pelleted diets having higher growth rates since fatty acids are essential for fish growth (Turchini et al., 2009; Manam, 2023; Table 2). However, C20:5n3 eicosapentaenoic/C24:0 (EPA), an important fatty acid

for fish growth, was higher in *P. perforata*, but fish consuming leafy algae had lower growth rates (Tadesse et al., 2003; Table 2). Nevertheless, juvenile monkeyface pricklebacks reared on pellets instead of leafy algae will reach market size more quickly.

A positive trend between SGR and soy protein inclusion was present. As soy protein increased in the pelleted diets, SGR also increased; fish consuming the algae pellet had a lower mean growth rate than those consuming the mixed and soy pellets (Figures 4A & 5). While soy protein has been found to hamper fish growth at levels greater than 40% inclusion as it inflames the fish's intestines, it has also been found to not impact growth below 40% inclusion like in the Asian seabass (*Lates calcarifer*) and gilthead seabream (*Sparus aurata*) (Krogdahl et al., 2000; Sales, 2009; Kumar et al., 2020). The highest inclusion rate of soy protein in the soy pellet was at ~30%, below this threshold. The mixed and soy pellet diets had higher protein content than the leafy algae and algae pellet diets, possibly explaining the higher growth rates for mixed and soy pellet treatments (Manam, 2023; Table 2). Monounsaturated, omega-6, polyunsaturated, and total fatty acid content were also higher in the mixed and soy pellet diets than in the leafy algae and algae pellet diets (Table 2). Since omega-6 has been found to positively impact fish growth, mixed and soy pellet treatments having significantly higher SGR and TGC values may be due to their respective feeds having higher omega-6 content (Wu et al., 2024; Table 2). While omega-3s also influence growth, and they decreased in fatty acid content from algae to soy pellets, their decrease in magnitude may not have been large enough in comparison to omega-6's increase, resulting in the mixed and soy pellets having higher growth values (Wu et al., 2024; Table 2). Because the mixed and soy pellets resulted in higher growth rates than the leafy algae and algae pellet diets,

either one of those diets would allow juvenile monkeyface pricklebacks to grow more quickly.

Despite accounting for temperature effects on growth, SGR and TGC revealed similar diet treatment growth effects in the fish among diet treatments (Figures 4A & 5A). As previously mentioned, the feed study was conducted in a flow-through seawater system that sourced its seawater from a single intake in Monterey Bay. As a result, incoming seawater was relatively similar in temperature among tanks and treatments because their inflows were ultimately connected to the same seawater intake (Figure 5C). Since UV density covers were attached to every tank to reduce sun exposure, any observed temperature differences would be due to water inflow, which varied from ~11–15°C (Figure 5C). SGR calculations were solely influenced by the temperature of the inflowing water, which TGC accounted for. Upon examining TGC over time, no discernable patterns appear, ruling out the possibility of growth rates differing by season (Figure 5B). Therefore, the observed differences in growth between treatments are due to diet because TGC accounts for temperature effects on growth, and it showed similar growth rates in addition to the same pattern observed with SGR.

Juvenile monkeyface pricklebacks on pelleted diets had lower FCRs (Figure 6). Similar to the growth rates, this may be attributed to pellets likely being more digestible than the leafy algae due to the extrusion process (Khater et al., 2014; Rahman et al., 2016). In broiler chickens, it has been found that pellets improve FCR as the nutrients are more digestible and feed intake rates are higher (Muramatsu et al., 2015). Thus, it can be inferred that monkeyface pricklebacks fed pellets could digest nutrients more efficiently than fish consuming leafy algae, equating to a higher weight gain. In addition, the fish fed the pelleted

treatments could consume more feed, attributing to the higher weight gain that likely increased more than the increase in feed intake. However, there are uncertainties associated with the measured FCR values. For the leafy algae, a single feed intake was measured each week and was assumed to remain constant for that week, which is not likely. Fecal material was also difficult to separate from the leftover feed, which could underestimate the amount of algae consumed by the monkeyface pricklebacks. Regarding the pellets, their poor stability likely led to pellets being counted improperly, with one pellet dividing into multiple pieces, resulting in an underestimation of pellets eaten and, thus, lower FCR. Additionally, monkeyface pricklebacks were assumed to be satiated if they did not eat five pellets, but pellets were not removed once satiation was determined. In effect, the fish may have eaten more pellets than recorded, underestimating FCR values for the pelleted treatments. Because both leafy algae and pellet FCR methods likely underestimated the actual FCR values, the observed pattern may still apply, depending on the magnitude of the underestimation. Monkeyface pricklebacks fed pellets would reduce feed costs for farmers since pellets have lower FCRs than the leafy algae diet.

For both Fulton's condition factor (FK) and hepatosomatic index (HSI), a positive trend between pelleted diets and condition was observed with the fish from the pelleted treatments being in better body condition and having more energy storage compared to those fed the leafy algae diet (Figures 7A & 7B). This may be due to the pellets possibly being more digestible than the leafy algae, allowing for macronutrients and fatty acids to be absorbed and incorporated more efficiently, which would result in improved energy storage and body condition (Khater et al., 2014; Rahman et al., 2016). In combination with the higher protein

and total fat content in the pellets, the pelleted treatments would likely have a more complete diet as higher amounts of macronutrients and fatty acids can be utilized by fish fed these diets as opposed to those on the leafy algae diet (Table 2). Since fish that meet protein and fatty acid requirements are healthier and have more energy, it can be supported that they have higher FKs and HSIs (Oliva-Teles, 2012). Smaller monkeyface pricklebacks had higher HSI than larger-sized fish (Figure 7C). Similar patterns have been observed in other fish. For example, smaller salmonids require less dietary energy than larger individuals (Cowey & Cho, 1993). This likely results in the smaller fish having higher energy reserves because the larger individuals would use more energy to account for their larger size. Since HSI estimates a fish's energy reserves, the observed correlation between size and HSI for monkeyface pricklebacks can be supported (Chellappa et al., 1995).

At the end of the feed study, juvenile monkeyface pricklebacks had a high total survivorship, displaying greater than 90% for all diet treatments. Six mortalities were recorded within the first month of the experiment, which may be attributed to stress (Schreck & Tort, 2016). Three more monkeyface pricklebacks passed away in March 2023, seven months into the study. These mortalities were likely due to repeated MS-222 use potentially impacting gill function (Chance et al., 2018). One mortality also occurred in March 2023 after a storm clouded the incoming ocean water, potentially suffocating the juvenile monkeyface prickleback. In total, ten monkeyface pricklebacks passed away throughout the 11-month feed study that were likely due to stress, repeated MS-222 use, or a storm event; monkeyface pricklebacks likely did not pass away from the diet treatments. Therefore,

monkeyface pricklebacks can be reared on any of the four diets and maintain high survivorship.

Based on survivorship, growth, FCR, and condition data, monkeyface pricklebacks are healthier, more efficient, and grow faster on pelleted diets than the whole algae diet (Figures 4A, 5, 6, 7A, & 7B). They also grow slightly faster on pellets containing soy protein (Figure 4A). Based on biological and economic factors, monkeyface pricklebacks should be reared on pelleted feeds containing soy protein to reach market-size more quickly, efficiently, and in better condition than the other diets tested here, ultimately reducing costs for farmers. However, this would likely raise the cost of soy and place pressure on other agrofeed industries that rely on soy (Bostock et al., 2010). Soybean production also has negative environmental impacts. For instance, genetically modified soybean farming has led to water contamination, cancer-related pesticide use, deforestation that reduces carbon-uptake potential, and soil health (Phélinas & Choumert, 2017). In contrast, marine algae require less resources (e.g., land space, freshwater, and fertilizers) to produce more biomass compared to most land-based crops while combating climate change and eutrophication by removing carbon dioxide, phosphorus, and nitrogen from the surrounding water column (Zhang et al., 2024). However, the lower growth, efficiency, and condition of monkeyface pricklebacks consuming leafy algae would likely extend the time needed to reach market-size, increasing costs for farmers. Therefore, a farmer needs to consider which diet suits their farm best to rear monkeyface pricklebacks.

Regardless of the diet selected, the highest calculated SGR was 0.2% bodyweight gain per day for fish consuming the soy pellet, which is relatively low in comparison to European

eels (*Anguilla anguilla*) that have SGRs ranging from 0.48% (fish < 10 g) to 0.6% (fish > 30 g) bodyweight gained per day (Karipoglou & Nathanailides, 2009). The observed FCR values in monkeyface pricklebacks were also higher than those observed in the European eels, with monkeyface pricklebacks having their lowest FCR at 2.16 in the soy pellet diet and European eels having their lowest FCR at 1.80 in eels greater than 30 g that were fed dry pellets containing ~55% protein and ~16% lipids (Karipoglou & Nathanailides, 2009). However, this may be due to the eels being reared at warmer temperatures (~25°C) than the monkeyface pricklebacks (average of ~12.5°C) (Karipoglou & Nathanailides, 2009). Pellets containing soy protein may be comparatively better to rear monkeyface pricklebacks at a fast rate, but it may not be fast and efficient enough to supplement the freshwater eel supply given low SGR and high FCR values relative to the European eel, unless higher temperatures improve their SGR and FCRs.

**Fillet Composition**

The leafy algae diet generally had higher protein, ash, moisture, and carbohydrate content than the other diet treatments in the fillets of juvenile monkeyface pricklebacks (Figures 8A, 8C, 8D, & 8E). However, it had the lowest caloric content (Figure 8F). These observations may be due to differences in the macronutrient content among the diets, but their respective macronutrient patterns do not match (Tables 2 & 4). For example, the leafy algae feed had the lowest protein content compared to the pelleted diets, but the juvenile monkeyface pricklebacks consuming the leafy algae had the highest protein content in their fillets (Tables 2 & 4). In effect, differences in the diets' macronutrient content did not likely influence the observed differences in the fillet content. Alternatively, the differences in assimilation

efficiencies of diets and their ingredients may have led to the observed differences in fillet content. As previously mentioned, monkeyface pricklebacks had a slightly higher assimilation efficiency on pellets than leafy algae (Edwards & Horn, 1982; Fris & Horn, 1993). If this was the case, then monkeyface pricklebacks fed pelleted diets would likely have higher macronutrient content in their fillets than the leafy algae, but monkeyface pricklebacks had generally lower macronutrient content from consuming pelleted diets, implying that the leafy algae may have had a higher assimilation efficiency. Regarding the ingredients, fish consuming leafy algae could assimilate more macronutrients than the pelleted diets because monkeyface pricklebacks have adapted a specialized digestive system to consume algae (Montgomery, 1977; Edwards & Horn, 1982; Horn et al., 1982; German et al., 2004; German & Horn, 2006; German et al., 2014, Kim et al., 2014). However, despite having similar ingredient compositions, the algae pellet diet had generally lower macronutrient content than the leafy algae. This infers that differences in macronutrient content are not due to differences in ingredient assimilation efficiencies. Nevertheless, the leafy algae treatment generally had higher macronutrient content than the pelleted diets, resulting in their fillets being more nutritious for human consumption from a macronutrient perspective. Additionally, their fillets would likely taste more savory than the others because they had a higher protein content, making them more desirable for consumers (Liem & Russell, 2019).

Palmitic, oleic, linoleic, EPA, omega-3, omega-6, polyunsaturated, and saturated fatty acids in monkeyface prickleback fillets differed among dietary treatments (Table 4). For linolenic, arachidonic, EPA, C22:6n3 docosahexaenoic (DHA), monounsaturated, omega-3,

omega-6, polyunsaturated, saturated, and trans-unsaturated fatty acids along with total fat, a roughly similar pattern was observed between fillets and diets (Tables 2 & 4). For instance, EPA content decreased in monkeyface prickleback fillets from the leafy algae diet to the soy pellet treatment, which also occurred in the diets (Tables 2 & 4). The content of other fatty acids not measured in the diets (e.g., palmitic fatty acid) can be inferred from each diet's composition. Macroalgae of the phylum chlorophyta (green algae) have mean palmitic, oleic, and linoleic fatty acid content (% of total fatty acid methyl esters) in *Ulva* spp. at 50.1%, 5.5%, and 5.7%, respectively (Pereira et al., 2012). C22:5n3 docosapentaenoic and C22:5n6 docosapentaenoic were not detected in *Ulva* spp. (Premarathna et al., 2022). For rhodophyta (red algae), palmitic, oleic, and linoleic fatty acids have means of 53.2%, 2.8%, and not detected in *Asparagopsis armata*, respectively (Pereira et al., 2012). C22:5n3 docosapentaenoic was not detected in any of the rhodophyta *Gracilaria corticata*, *Acanthophora spicifera*, *Gelidiopsis variabilis,* and *Jania adhaereus*, but C22:5n6 docosapentaenoic was measured at 0.67% (mean value of fatty acid composition) in *Acanthophora spicifera* (Premarathna et al., 2022). Normal soybeans have lower palmitic fatty acid content (% crude oil) than both macroalgal phylum at 12% but higher oleic and linoleic content at 21% and 55%, respectively (Medic et al., 2014). Docosapentaenoic fatty acids are also not detected in soy protein concentrate (*Eat This Much*, n.d.). In effect, monkeyface pricklebacks consuming the leafy algae diet should have higher palmitic fatty acid content but lower oleic and linoleic fatty acids in their fillets than the mixed and soy pellet diets, which is supported by this study's results (Table 4). Thus, the

differences in the fatty acid content of monkeyface prickleback fillets among dietary treatments are likely due to their diet's fatty acid content.

Linoleic and oleic fatty acids were the main drivers of differences between the mixed and soy pellet treatments (Figure 10). This is likely due to the higher linoleic and oleic content in the fillets of those diets than the leafy algae and algae pellet diets (Figures 9B & 9C). Meanwhile, palmitic and EPA fatty acids drove the distinction of the leafy algae and algae pellet groups, likely due to higher fatty acid content in their fillets than the mixed and soy pellet treatments (Figures 9A, 9D, & 10). In addition, slight differences in DHA may have also driven the distinction of the leafy algae and algae pellet groups, despite DHA not significantly differing amongst the dietary treatments (Table 4). Palmitic and oleic fatty acids drove the distinction between the leafy algae and algae pellet diets, for the algae pellet treatment had higher palmitic and oleic content in their fillets than the leafy algae treatment (Figures 9A, 9B, & 10).

The statistical analysis of the macronutrient and fatty acid content in the fillets of juvenile monkeyface pricklebacks has associated uncertainties. NPAL recommended at least 10 g of a sample per test, which resulted in monkeyface prickleback fillets being pooled to run all the desired tests. This led to only three samples per treatment, which does not consider the size differences among tanks within a treatment. In effect, macronutrient and fatty acid data should be interpreted cautiously. The macronutrients and fatty acids within the fillets of monkeyface pricklebacks that may be consumed by humans depend on the diet being fed to the fish. Based on the growth, condition, and FCR data, either the mixed or the soy pellet is optimal for rearing the monkeyface pricklebacks. In effect, monkeyface pricklebacks fed

either of these diets would likely reflect the macronutrient and fatty acid content found in fillets of monkeyface pricklebacks fed those diets.

**Metabolic Rates**

Standard metabolic rate (SMR) and SDA did not differ among dietary treatments for juvenile monkeyface pricklebacks (Figures 11A & 11B). In European eels, SMR has been reported to differ among diets with differing fatty acid content; SMR was higher in eels consuming a diet rich in saturated fatty acids as opposed to highly unsaturated fatty acids (McKenzie, 2001). However, SMR was not found to differ in rainbow trout for diets differing in protein and lipid content (Eliason et al., 2007). Despite differences in fatty acid content among treatments, SMR was not found to differ, inferring that monkeyface prickleback's SMR does not change regardless of diet, at least for the diets used in this study. Regarding SDA, it was initially hypothesized that pellets would have a lower metabolic cost of digestion (lower SDA) than the leafy algae due to pellets being more digestible and bioavailable (Allan & Booth, 2004; Secor, 2009; Khater et al., 2014; Rahman et al., 2016). This hypothesis could not be supported as SDA did not differ among the leafy algae and the pelleted diets. Similarities in SDA across diet treatments have also been reported for southern catfish *Silurus meridionalis*, where SDA coefficients did not differ despite changes in soy protein inclusion from 13 to 39% (Ai & Xie, 2006).

The limitations of the SMR and SDA methods may have led to the rejection of the hypothesis. Background respiration was not adjusted for SMR and SDA because it was not captured properly, likely resulting in SMR and SDA being overestimated. If metabolic rate was overestimated for a fish in a specific treatment more than the other treatments, then the

lack of observed differences may be due to the methods as opposed to treatment effects. Regardless of diet, monkeyface pricklebacks were found to have similar metabolic costs of digestion (Figure 11B). In effect, growth rates were likely not impacted by differences in metabolism. Therefore, the pelleted feeds, specifically the mixed and soy pellets, should be used to grow monkeyface pricklebacks for aquaculture because metabolism did not differ among dietary treatments.

In summary, monkeyface pricklebacks had the highest growth after consuming diets with soy protein. Based on biological and economic factors, monkeyface pricklebacks should be reared on a soy-based diet, but the leafy algae diet is more environmentally sustainable and produces a more nutritious and possibly savory fillet due to a higher macronutrient content. Farmers need to consider all factors before selecting a diet to rear monkeyface pricklebacks for commercial aquaculture. Despite being able to consume these diets, the growth rates were less than those observed in European eels. Although an untested diet may induce a faster growth rate in monkeyface pricklebacks, diet alone may not produce monkeyface pricklebacks quickly enough for aquaculture.

## Identifying the Optimal Temperatures to Rear Monkeyface Pricklebacks

### Introduction

Most fish species are ectotherms, and their physiological performance depends on the environmental temperature (Wang & Overgaard, 2007; Volkoff & Rønnestad, 2020). As temperatures rise, dissolved oxygen decreases while oxygen demands often increase, which leads to elevated metabolic rates (Jobling, 1997; Clarke & Johnston, 1999; Denderen et al., 2020; Volkoff & Rønnestad, 2020). In addition, increased temperatures may increase rates of digestion with the rise in enzyme activity, and the growth rate may increase in response to the elevated metabolism, depending on food consumption (Jobling, 1997; Daniel et al., 2008; Denderen et al., 2020; Volkoff & Rønnestad, 2020). Furthermore, juvenile fish typically exhibit enhanced growth in warmer waters but require additional food intake to account for the heightened metabolism, which often equates to a lower feed conversion ratio (FCR) (Morita et al., 2010; Cavrois-Rogacki et al., 2019; Csik et al., 2023). Past the ideal and pejus ("turning worse") temperatures, fish growth will slow down as the cardiovascular system is unable to provide ample oxygen for growth given other metabolic demands, such as cellular maintenance and heat shock protein production (Figure 12; Wang & Overgaard, 2007; Logan & Buckley, 2015; Volkoff & Rønnestad, 2020; Alfonso et al., 2021). All fish species typically display this downward parabolic or "bell-shaped" physiological performance curve as a result of temperature due to their ectothermic nature (Figure 12; Wang & Overgaard, 2007; Volkoff & Rønnestad, 2020). To develop monkeyface pricklebacks as an aquaculture species, it is therefore crucial to account for the above-mentioned growth factors of temperature, metabolism, and feed intake. While it is likely that higher temperatures will

increase monkeyface prickleback growth, as they are marine ectotherms, simultaneous changes in metabolic demand make determining an ideal culture temperature that is both fast and efficient for growth more nuanced. Their optimal growth temperature is likely different from other marine cultured fish since the monkeyface prickleback is an intertidal species and may have a more significant physiological tolerance for changes in temperature than subtidal species (Hilton et al., 2010; Logan & Somero, 2010; Logan & Buckley, 2015; Volkoff & Rønnestad, 2020; Willis et al., 2021). No studies have examined the relationship between monkeyface prickleback growth and temperature.

Beyond optimal temperatures, thermal effects on aquaculture species impact fish health (Debnath et al., 2006). With long-term exposure to higher temperatures, the frequency and intensity of infections and diseases are predicted to increase as susceptibility to parasites and the strength of immune responses are temperature and stress-dependent (Alfonso et al., 2021). Climate change models predict oceans will warm 1–4°C by 2100, likely leading to reduced fish health from thermal stress (Alfonso et al., 2021). Several studies have examined the effects of a warming ocean on aquaculture, making an examination of this factor important for the development of new aquaculture species (Gubbins, 2006; Barange et al., 2018; Reid et al., 2019). Thus, the thermal limits of monkeyface pricklebacks need to be determined to ensure sustainability in captive settings. Typically, as temperatures rise above the pejus temperature and approach a fish's thermal limit, unorganized movement will begin to occur as the fish reaches its critical thermal maximum (CTmax) (Figure 12; Becker & Genoway, 1979; Beitinger et al., 2000; Wang & Overgaard, 2007). CTmax is a commonly used measurement to determine an organism's upper thermal limit as it can be quantified in a

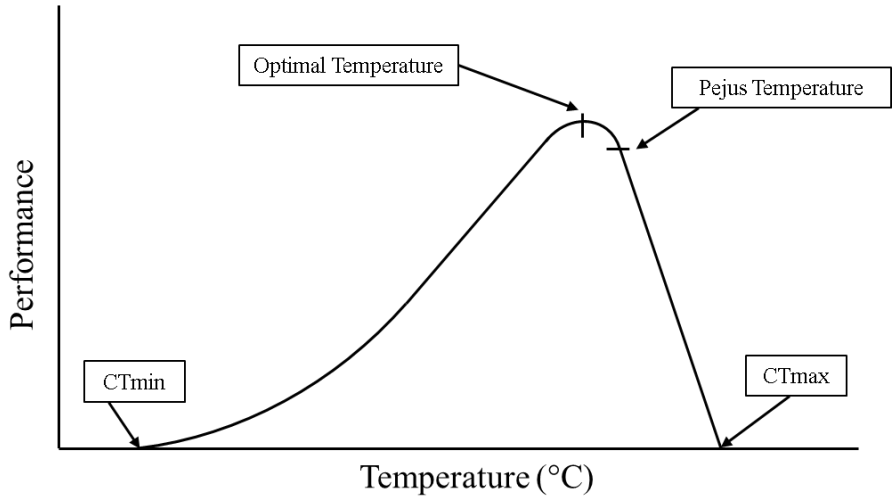
standardized way, and different species exhibit similar behavioral responses (Desforges et al., 2023). If a species has a high CTmax, it can withstand high temperatures for acute exposures (Becker & Genoway, 1979; Logan & Buckley, 2015; Xia et al., 2017). Additionally, fish that have acclimated to warmer temperatures have been shown to experience a shift in their CTmax, with those acclimated to the warmest temperature exhibiting the largest change in their CTmax–allowing them to physiologically withstand extreme temperatures more effectively than non-acclimated individuals (Becker & Genoway, 1979; Logan & Buckley, 2015; Xia et al., 2017). Furthermore, the thermal nature of a fish species influences their CTmax, with eurythermal (wide temperature range) fish generally having a higher CTmax than stenothermal (narrow temperature range) fish (Becker & Genoway, 1979). The monkeyface pricklebacks' aerial respiration capabilities and eurythermal nature suggest that it can withstand high temperatures, and it may have a relatively high CTmax, like other air-breathing fish, such as the mudskipper (Love, 2001; Dabruzzi et al., 2019). Since the closely related high cockscomb *Anoplarchus purpurescens* (within the family Stichaeidae) has a measured CTmax between 29.5–30°C, monkeyface pricklebacks likely share a similar CTmax (Barton, 1985). However, individuals acclimated to warmer-than-ambient temperatures will likely have a CTmax higher than 30°C, potentially allowing the acclimated fish to be reared in warmer settings. Thus, a study is needed to quantify the monkeyface prickleback's ideal rearing temperature and establish their thermal limits for aquaculture.

I conducted a temperature study that quantified growth rates, condition, feed intake, metabolic rates, fillet composition, and the critical thermal maximum of *C. violaceus* following acclimation to five different temperatures. I hypothesized that monkeyface

pricklebacks would have a higher growth rate, better condition, lower FCR, higher macronutrient and fatty acid content, and higher metabolic rates at warmer temperatures below the pejus temperature than ambient conditions. Additionally, the CTmax of monkeyface pricklebacks was hypothesized to increase upon acclimating to warmer temperatures.

**Figure 12.**

*A typical thermal performance curve for marine ectotherms with a critical thermal minimum (CTmin), an ideal temperature at peak performance, a pejus temperature whereby performance decreases, and critical thermal maximum (CTmax). Based on Volkoff & Rønnestad (2020) along with Wang & Overgaard's (2007) performance figures for fish.*



## Materials and Methodology

After the feed study, a temperature study was initiated at Moss Landing Marine Laboratories (MLML) to examine the growth, condition, feed intake, metabolic costs, filet composition, and thermal limits of monkeyface pricklebacks reared at different temperatures. The study began on October 2nd, 2023, and concluded on December 26th, 2023. The study had five different temperature treatments: 14°C, 17°C, 20°C, 23°C, and 26°C. These treatments were determined based on the predicted and observed thermal range of monkeyface pricklebacks, with the AquaMaps model predicting a range of 9.0–17.0°C (Kaschner et al., 2016) while it has been observed that the fish occupy a range of 16–26°C (Pomeroy, n.d.) and 9–24°C (Love, 2001). The treatments were also determined from observations of the elevated thermal tolerance typical of intertidal species (Madeira et al., 2012). Soy-based pellets were formulated, and juvenile fish were collected and acclimated to temperature treatments before the start of this experiment. Growth measurements, feed intake, condition, metabolic rates, critical thermal maximum, and fillet composition were measured over the study's duration and at its end (~3 months).

### Pellet Formulation

Pellets were formulated based on the prior feed study results, and the soy-based feed was chosen since it had the highest specific growth rate (SGR) and maximizing growth was the goal of this study. The pellet's protein content was increased to 36.9%, and the lipid content was increased to 5.8% to stimulate growth rates further. In addition, C22:6n3 docosahexaenoic (DHA) gold microalgae powder® from DSM Nutrition was added to increase omega-3 fatty acid content. Diet formulation was corrected upon conducting

proximate analysis on the algal ingredients used. The pellets were produced using a cold extrusion process with a California Pellet Mill. Midway through the experiment, diet reserves were running low, so more were made. The second diet batch produced had slightly different protein, lipid, and nitrogen content, resulting in an average being reported for the calculated analysis of the pellets in Table 5. However, all temperature groups received the same type of slightly modified pellet beginning ~6 weeks into the study. These pellets were also formulated under the supervision of Dr. Ronald Johnson at NOAA Northwest Fisheries Science Center.

**Temperature Treatment Tank Design**

A flow-through temperature design was designed to rear juvenile monkeyface pricklebacks at different temperatures in MLML's aquarium room. This was accomplished by creating a heated water jacket around triplicate tanks within a water table. The tanks were placed on top of cement blocks so that displaced tank water could flow out of the water table's outflow via gravity. One to two 500W heaters were placed near each water table's inflow and connected to Inkbird temperature controllers with a temperature sensor inside the middle tank to maintain temperature treatments. The 14°C treatment used a chiller instead of a heater with a built-in temperature sensor. A pump was placed within the water table to transport heated/chilled water from the water table into each tank, with flow rates controlled by airline valves at a rate of 17.98 L $hr^{-1}$. A float valve was attached to the inflow of each water table to maintain the water level. Each tank had eight polyvinyl chloride pipes for monkeyface pricklebacks to shelter in (Figure 13).

**Table 5**

*Pellet Formulation for the Temperature Study Showing the Percentage of Ingredients (Above) and the Average Protein, Lipid, and Nitrogen Composition (Below)*

| Ingredients (dry weight %) | |
|---|---|
| Soy protein concentrate | 30.00 |
| Wheat flour | 27.30 |
| Nori (*P. perforata*) | 27.00 |
| DHA Gold | 4.00 |
| Soybean oil | 2.40 |
| Trace minerals | 0.10 |
| Vitamin premix | 1.50 |
| Vitamin C | 0.50 |
| Choline | 0.50 |
| Betaine | 0.25 |
| L-methionine | 0.20 |
| L-lysine | 1.20 |
| L-threonine | 0.20 |
| Mono-Cal phosphate | 1.90 |
| Perma Pel | 2.00 |
| Taurine | 1.00 |
| | |
| Calculated analysis (dry matter %) | |
| Crude Protein | 37.75 |
| Lipid | 5.85 |
| Nitrogen | 6.00 |

**Figure 13.**

*Juvenile monkeyface pricklebacks occupying one of their polyvinyl chloride tube shelters.*



**Fish Collection, Husbandry, and Experimental Design**

In the summer of 2023, juvenile monkeyface pricklebacks were collected from San Simeon, California, for the temperature study (Institutional Animal Care and Use Committee protocol 1068 and California Department of Fish and Wildlife permits S-191050002-22189-001 and S-191050002-19126-001). On June 5–6$^{th}$, 148 juvenile monkeyface pricklebacks were collected in the early morning at 35°39’12” N latitude and 121°15’32” W longitude (same location for the feed study fish collection; Figure 1). The collection method implemented for the feed study was also used for the temperature study: overturning boulders, catching juvenile monkeyface pricklebacks with hand nets, and eventually transporting them back in aerated coolers to MLML. The 148 collected fish had an average total length of 15.1 cm (9.8–30.8 cm) and an average weight of 21.0 g (5.3–143.7 g) on June 14$^{th}$, 2023. Fish were acclimated and quarantined at MLML’s shore lab in two tanks (1,440 L) with flow-through seawater at 7.5 L/min with aeration for 14 days to ensure they were healthy with no predisposed health conditions (DeTolla et al., 1995). A Praziquantel bath was

administered during this quarantine period to eliminate potential parasites. A praziquantel bath was also administered at the beginning of the experiment once the fish had acclimated to their temperature treatments. Melafix was used to heal any injured fins observed throughout the study.

The temperature study began on October 2nd, 2023, when pellets were processed and delivered to MLML. Before pellets arrived, juvenile monkeyface pricklebacks were weighed, measured, and tagged with passive integrated transponders in the dorsal musculature to identify individuals. Once quarantine had passed, the fish that fit the total length range of 13.4–21.5 cm were moved to MLML's aquarium room for the temperature study. To increase the study's sample size, 20 juvenile monkeyface pricklebacks previously held in MLML's aquarium room were added that had an average total length of 13.9 cm (13.2–14.9 cm) and an average weight of 14.9 g (11.9–19.3 g) on June 13th, 2023. In total, 116 fish were used for the temperature study with an average total length of 15.9 cm (13.4–21.5 cm) and weight of 23.7 g (12.3–50.6 g) on October 2nd, 2023.

Temperature treatments were randomly assigned to water tables and their triplicate set of tanks. Unlike the feed study, juvenile monkeyface pricklebacks were not separated by size for the temperature study. Individual fish were randomly assigned to their treatment tanks at n = 8 fish per tank (n = 24 per treatment).

Juvenile monkeyface pricklebacks were consuming the macroalgae sea lettuce (*Ulva* spp.), dulse (*D. mollis*), and nori (*P. perforata*) before initiating the temperature study. To condition monkeyface pricklebacks to consume the formulated pellets, fish were fed formulated pellets (Table 2) for two weeks at 0.75% of their total tank weight. At the same

time, the fish were acclimated to their respective temperature treatments by increasing or decreasing the temperature of each tank by 1°C per day until treatment conditions were attained (Tirsgaard et al., 2015; Xia et al., 2017). For the study's duration (12 weeks), fish were fed incrementally at ~1 g until four or more pellets remained at each feeding, and they were fed twice a day (at least four hours apart). Tanks were siphoned daily, with a deep clean occurring every other week.

Antibiotics were administered roughly a month into the temperature study. After acclimating the fish and starting the temperature study, skin lesions began to appear on some of the fish. Infected fish passed away within a couple of days. The mortalities occurred in every treatment, indicating that the cause was not temperature-dependent but opportunistic. The veterinary staff at Monterey Bay Aquarium were contacted to examine post-mortem samples, and they concluded that a systemic bacterial infection (*Aeromonas* spp.) was afflicting the fish. Ceftazidime was injected into the dorsal musculature at 20 mg/kg every five days. Three injections were administered for each fish, which reduced mortalities and allowed the study to continue.

**Sampling Protocols**

Total length and weight were measured biweekly throughout the temperature study. Anesthesia was achieved by placing 9.46 L tanks in each trough, filling the 9.46 L tanks with the trough's heated/chilled water, mixing MS-222 (0.48 g per 9.46 L) along with baking soda (0.96 g per 9.46 L) in each tank, and anesthetizing fish one tank at a time. Once fish were netted and anesthetized, their tank was cleaned with a vinegar-water mixture to remove potential bacteria, and fish were weighed on a calibrated scale and measured with a fish

board. After weighing and measuring a fish, juvenile monkeyface pricklebacks were placed in an aerated bucket with API Stress Coat Plus® to recover. Upon regaining movement, the fish were returned to their respective tank filled with water at their treatment temperature. This process was repeated for each tank.

To obtain a feed intake rate and feed conversion ratio (FCR), juvenile monkeyface pricklebacks were initially fed at 4 g increments until satiation was achieved. However, on October 13, 2023, the increments were switched to 1 g due to the limited supply of formulated pellets. Before switching to 1 g increments, the remaining pellets were not counted but recorded as either full, three-fourths, half, one-fourth, or all the pellets eaten. To get a better estimate of feed intake and extend the formulated pellet supply, pellets were fed at 1 g increments that were counted as they were fed. If the fish no longer expressed interest in the food, a 30-second timer was started. If fish ate during that period, the timer was restarted. This was repeated until the timer reached 30 seconds without a fish eating a pellet. If three or fewer pellets were left after 30 seconds, another 1 g increment was added, and the counting process was repeated. Satiation was deemed to have occurred when four or more pellets remained after 30 seconds. The average pellet weight was obtained by counting and weighing out a handful of pellets, which was then multiplied by the number of pellets eaten to convert it into grams for feed intake and FCR calculations.

At the study's endpoint, all juvenile monkeyface pricklebacks were euthanized with a lethal dose of MS-222, measured, weighed, dissected, and had their fillets frozen at -20°C. During the dissections, the liver weight of each fish was recorded for hepatosomatic index (HSI) calculations. However, HSI data is not reported since HSI could not be measured

before the systemic bacterial infection occurred, making comparisons difficult. Tanks were randomly selected for dissection, and the dissections lasted three days. Once dissections were complete, fillets were pooled by tank and overnighted to NP Analytical Laboratories (NPAL) for proximate analysis.

**Response Metrics**

Growth, condition, survivorship, feed intake, and FCR were used as the primary metrics for the temperature study (Equations 1–2, 4, & 6–8). Fish growth was analyzed by calculating instantaneous growth (IG) and then specific growth rates (SGR) from initial and final weights along with the study's duration (Equations 1 & 2). For each measurement period, percent weight gain was also calculated to examine cumulative percent growth (Equation 11).

$$Percent\ Weight\ Gain = \left(\frac{Final\ weight - Initial\ weight}{Initial\ Weight}\right) x\ 100 \quad (11)$$

Since the systemic bacterial infection and antibiotic injections hampered fish growth, only the first four weeks of survivorship and growth data from living fish were used for data analysis. Like the growth rate calculation, Fulton's condition factor (FK) was computed using only the first four weeks of growth data from live fish. Mortalities were recorded upon occurrence, and survivorship was calculated at the study's end. Feed intake and FCR were calculated to determine feed conversion efficiency. However, negative FCR values from fish weight loss in the highest temperature treatment led to calculating the percent total tank weight consumed per day, which was produced using the following equation (Equation 12):

$$Total\ Tank\ Weight\ Consumed\ per\ Day\ (\%) = \left(\frac{Amount\ Eaten\ per\ Day}{Tank\ Weight}\right) x\ 100 \quad (12)$$

Calculations were computed in Excel and transferred to R for statistical analysis as described below for each metric.

**Metabolic Rates**

Intermittent flow respirometry was used to measure juvenile monkeyface pricklebacks' standard metabolic rate (SMR) and maximum metabolic rate (MMR) at different temperatures. Aerobic scope (AS), the maximum oxygen supply for aerobic activities above SMR, was calculated as the difference between MMR and SMR to provide insight in to how temperature changes impact monkeyface prickleback fitness (Clark et al., 2013; Farrell, 2013). The same Autoresp system and coolers from the feed study were used to conduct respirometry for the temperature study (e.g., 75.71 L coolers, Witrox-4 units, and oxygen mini sensors). Fish within the 10–20 ratio of wet weight volume to respirometry volume were selected. Eight fish were run for the end-range temperatures (14 & 26°C), while six were run for the remaining temperatures (17, 20, & 23°C) due to time constraints. Before respirometry was conducted, fish were acclimated to their treatment temperatures for eight (14°C), eight (17°C), eight (20°C), nine (23°C), and seven (26°C) weeks. A fasting period of 48 hours was used for the 14°C treatment, but it was reduced to 24 hours for the warmer temperature treatments (17, 20, 23, & 26°C) as gut retention typically decreases as temperature increases (Tirsgaard et al., 2015; Volkoff & Ronnestad, 2020). Once juvenile monkeyface pricklebacks of a temperature treatment were fasted, the coolers housing the respirometry system were filled with heated/chilled water that was kept constant using chillers/heaters. Before SMR was conducted, background respiration was measured for at least five flush-wait-measure cycles, and the fish were weighed with a calibrated scale. Each

cycle consisted of a five-minute flush, a one-minute waiting period, and a five-minute measuring period (Steffensen, 1989). SMR was measured over a 24-hour period under normoxic conditions. The 20°C treatment was extended by four hours due to an endcap malfunction in one of the respirometry chambers. Once SMR data had been collected, but before the second background respiration run, MMR was measured by removing the fish from its chamber, chasing it in a 5-gallon bucket with a net for five minutes, leaving it out of water for at least one minute to ensure exhaustion, placing it back in the same chamber, and running one five minute measure cycle (Clark et al., 2013; Sandblom et al., 2014; Rosewarne et al., 2016). After measuring MMR for each fish, juvenile monkeyface pricklebacks were returned to their respective tanks, and background respiration was measured for at least five flush-wait-measure cycles. AS was calculated by subtracting an individual's MMR from its SMR (Clark et al., 2013).

The respR package was used to calculate the SMR and MMR of each fish (Harianto et al., 2019). First, the raw oxygen data for background respiration, SMR, and MMR were organized into an Excel file where the effective volume was calculated for each fish. The data was then imported into R where the respR package was used to inspect the raw oxygen data, extract oxygen consumption rates, adjust for background respiration, convert units, and select rates (Harianto et al., 2019; R Core Team, 2020). Average background respiration was calculated before and after each respirometry run for each fish, and it was estimated at each time point of the experiment by assuming a linear relationship (Harianto et al., 2019). The highest-ranked rates from the one five-minute cycle were extracted for MMR and adjusted for background respiration at each time point (Harianto et al., 2019). MMR was then

converted by fish weight and effective volume, and the highest rate was selected (Harianto et al., 2019). For SMR, the lowest ranked rate from each cycle was extracted, adjusted, and converted (Harianto et al., 2019). These rates were sequentially selected by first removing the initial ten rates (to account for handling stress), then selecting rates with $R^2$ greater than 0.90. From these, the lowest $10^{th}$ percentile rates were selected, and a mean value was calculated in $mgO_2/hr/g$ (Harianto et al., 2019).

**Critical Thermal Maximum**

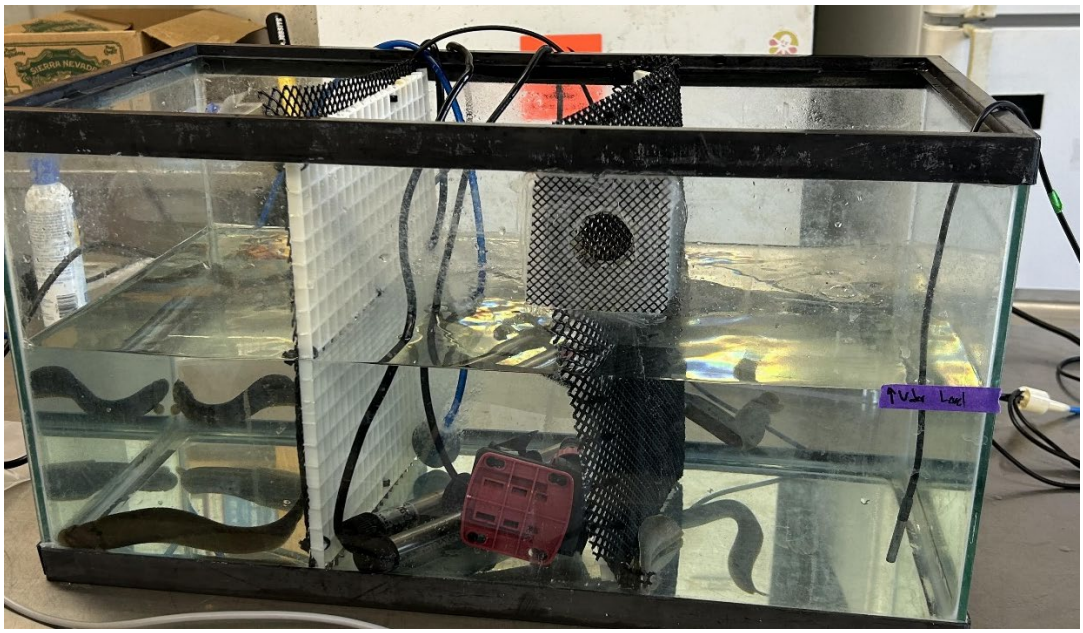
Nine to ten weeks into the study, the critical thermal maximum (CTmax) was quantified with an acute temperature ramp. Fish were fasted before CTmax trials by following the fasting durations used for the respirometry trials. Once fasted, fish were acclimated to the experimental tank (75.71 L aquaria) for 1 hour at their treatment temperature before exposing them to a constant, gradual increase at ~0.3°C/min (Becker & Genoway, 1979; Beitinger et al., 2000; Xia et al., 2017; Dabruzzi et al., 2019). A Coral Bleaching Automated Stress System (CBASS) was used to measure and ramp temperature in the experimental tank with two 500W heaters (Figure 5; Evensen et al., 2023). Dividers were placed inside tanks to prevent fish from receiving burns from heaters, a pump was added to circulate the water to ensure temperature was evenly mixed throughout the tank, and oxygen was bubbled into the tank to maintain normoxic conditions (Figure 14). Since monkeyface pricklebacks do not have swim bladders, the typical loss of equilibrium cue for a fish reaching the CTmax could not be used (Becker & Genoway, 1979; Beitinger et al., 2000; Xia et al., 2017; Dabruzzi et al., 2019). Thus, a loss of righting response was implemented with CTmax equating to the fish laying on its side for at least a minute (Dabruzzi et al., 2019). This was accomplished by

occasionally tipping the fish over and timing its return to an upright state. If a juvenile monkeyface prickleback was unable to right itself within a minute, then the temperature was recorded and determined as the fish's CTmax. Once CTmax was reached, fish were weighed and placed into a tank at their treatment temperature with API Stress Coat Plus® to recover from the acute thermal stress (Barton, 1978). This method was tested in pilot trials before the start of the temperature study to determine its validity. As previously mentioned, the CTmax experiment was conducted nine to ten weeks into the study. At this time, the selected fish from the 26°C treatment were in poor condition. Thus, CTmax for the 26°C treatment was determined when the fish could not right themselves within a minute and were not moving. The acclimation response ratio (ARR) was calculated with Equation 13 to determine the effect of temperature acclimation on the juveniles' CTmaxes, where delta Heat Tolerance is the delta CTmax (Xia et al., 2017).

$$ARR = \frac{\Delta\, Heat\ Tolerance}{\Delta\, Acclimation\ Temperature} \tag{13}$$

**Figure 14.**

*Experimental setup for critical thermal maximum experiment with the Coral Bleaching Automated Stress System (CBASS) temperature sensors, two heaters, an air stone, and a pump to circulate water inside the tank.*



**Statistical Analysis**

Temperature treatment effects were analyzed using R with an $\alpha = 0.05$ for all statistical hypothesis tests (R Core Team, 2020). Generalized linear mixed models (GLMMs) were used to examine differences among temperature treatments for growth and condition with a sample size of n = 103 fish before antibiotics were administered. Multiple additive and interactive models were tested for each growth rate and condition index with restricted maximum likelihood. Models were selected based on the lowest AICc and evaluating a plot of the raw data. The selected additive GLMMs containing SGR or FK with temperature as a fixed factor and tank ID as a random factor were chosen. Once a model was chosen, a KS deviation test, a dispersion test, and an outlier test were run to test assumptions. A type II

analysis of deviance was run with a chi-square test statistic to examine the model's significance. Estimated marginal means were then calculated and pairwise comparisons with a Tukey's Honest Significant Difference (HSD) adjustment were used to identify significant differences among treatments.

A repeated measures analysis of variance (ANOVA) with a mixed-effects model was used to test for significant differences in cumulative percent growth among temperature treatments over time. A sample size of n = 103 fish before antibiotics were administered was used. The assumptions of a repeated measures ANOVA are that residuals are normal, homoscedasticity, and variance-covariance matrix among within subjects' measures is circular. These assumptions were tested and met by examining the model residuals.

Linear regression examined potential associations between temperature and feed consumption. Non-linear regressions were used for SMR, MMR, AS, and CTmax. The assumptions of linear and non-linear regressions are independence, normality, and homoscedasticity. Normality and homoscedasticity were tested by running a Shapiro-Wilk's normality test and examining the regression's residuals. The weight of feed pellets consumed per day relative to the total weight of fish in a tank (i.e., percent total tank weight consumed) had a sample size of n = 15 (one value per tank), metabolic rates had samples sizes of n = 34, and CTmax had a sample size of n = 40. For the regressions with temperature, SMR, MMR, AS, and CTmax met all the assumptions. However, the linear regression between temperature and the total weight of pellets consumed per tank did not pass the Shapiro Wilk's normality test, but it is still being reported since a bootstrap linear regression also

produced a p-value below 0.05 and only one residual is outside the normal range in the ggqqplot.

A one-way ANOVA was used to test for significant differences in fillet content (protein, fat, moisture, ash, carbohydrates, calories, and fatty acids) among the temperature treatments. The assumptions of a one-way ANOVA are independence, normality, and equal variance among the treatments. Normality and equal variance were tested by running a Shapiro-Wilk's normality test, conducting Bartlett and Levene's tests of equal variance, and examining the ANOVA's residuals. If the one-way ANOVA was significant, a Tukey's HSD test was run to examine potential significant differences among treatments. Before a one-way ANOVA was applied to the fillet content data, protein, fat, ash, and carbohydrate content of a sample were converted from wet weight to dry weight via each sample's moisture content to evaluate differences in a dry-matter context (Equation 9). Fatty acid content data was also manipulated to examine each fatty acid in the context of each sample's total fatty acid content before running one-way ANOVAs. Each fatty acid value of a sample in wet weight was respectively divided by the total fat content of that sample in wet weight and multiplied by 100 (Equation 10). Fillet content had sample sizes of n = 14 (one value per tank but only two values for 20°C due to mortalities). Ash and protein did not pass the Shapiro-Wilk's normality test, but the others met all the assumptions of a one-way ANOVA. Ash and protein were still reported as their small sample size may have led to them not being normal.

A non-metric multidimensional scaling (nMDS) test was run to compare treatment effects on fatty acid composition in a multivariate context, and a permutational analysis of variance (PERMANOVA) was used to test for treatment differences in fatty acid composition.

Monounsaturated, polyunsaturated, saturated, and trans-unsaturated fatty acids were selected for the nMDS because they encompass the fatty acids into four fatty acid classifications. A low stress score (<0.2), an appropriate distance metric, and a high goodness of fit are required of an nMDS. The nMDS run with the selected fatty acids had a low stress score of 0.000086, it had three dimensions, a Euclidean distance metric was applied, and the nMDS had a high goodness of fit with an $R^2$ of 1. A test of homogeneity of multivariate dispersion was conducted to assess the equal variance assumption of the PERMANOVA, which was met with a non-significant p-value at 0.59. The nMDS and PERMANOVA had the same sample size of n = 15.

## Results

### Survivorship, Growth, and Condition

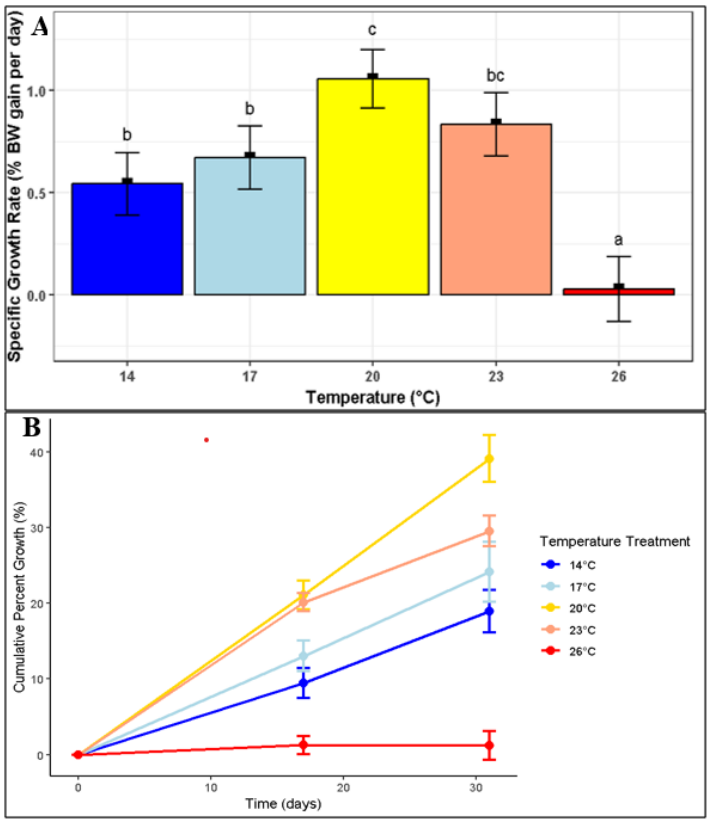
In total, juvenile monkeyface pricklebacks had a survival rate of 87.9% for the first four weeks of the study. The 20°C treatment had the highest survival rate at 95.8%, while the 17°C treatment had the lowest at 79.2%. The remaining temperature treatments (14°C, 23°C, & 26°C) had survival rates at 87.5%, 87.0%, and 90.5%, respectively.

Upon adjusting for tank variability, the specific growth rate (SGR) generalized linear mixed model (GLMM) indicated differences in SGR among temperature treatments for juvenile monkeyface pricklebacks (GLMM, $X^2 = 99.21$, p-value < 0.0001; Figure 15A). Fish reared at 20°C had significantly higher SGRs than fish acclimated to 14°C, 17°C, and 26°C with estimated marginal means at 1.05% (20°C), 0.54% (14°C), 0.67% (17°C), and 0.03% (26°C) bodyweight gain per day [Tukey's Honest Significant Difference (HSD), p-value < 0.05; Figure 15A]. The 23°C treatment did not significantly differ in SGR from the 14°C, 17°C, and 20°C with an estimated marginal mean at 0.83% bodyweight gain per day (Tukey's HSD, p-value > 0.05; Figure 15A), but it was significantly higher than the 26°C treatment (Tukey's HSD, p-value < 0.05; Figure 15A). Additionally, there was no difference in SGR between the 14°C and 17°C treatments (Tukey's HSD, p-value > 0.05; Figure 15A). The repeated measures analysis of variance (ANOVA) with a mixed effects model also detected differences in growth over time among temperature treatments, with the 20°C treatment having the highest cumulative percent growth at ~40% on day 31, while the fish in the 26°C had the lowest growth at ~0% (Figure 15B). Juvenile monkeyface prickleback cumulative percent growth increased during the study period (Time: $F_{2,196} = 277.27$, p-value

< 0.0001), the rate of growth was affected by both time and temperature (Time x Temperature: $F_{8,196} = 20.70$, p-value < 0.0001) indicating that the growth trajectories in the different temperature treatments were diverging from each other over time, and growth did differ among temperatures (Temperature: $F_{4,98} = 25.33$, p-value < 0.0001).

**Figure 15.**

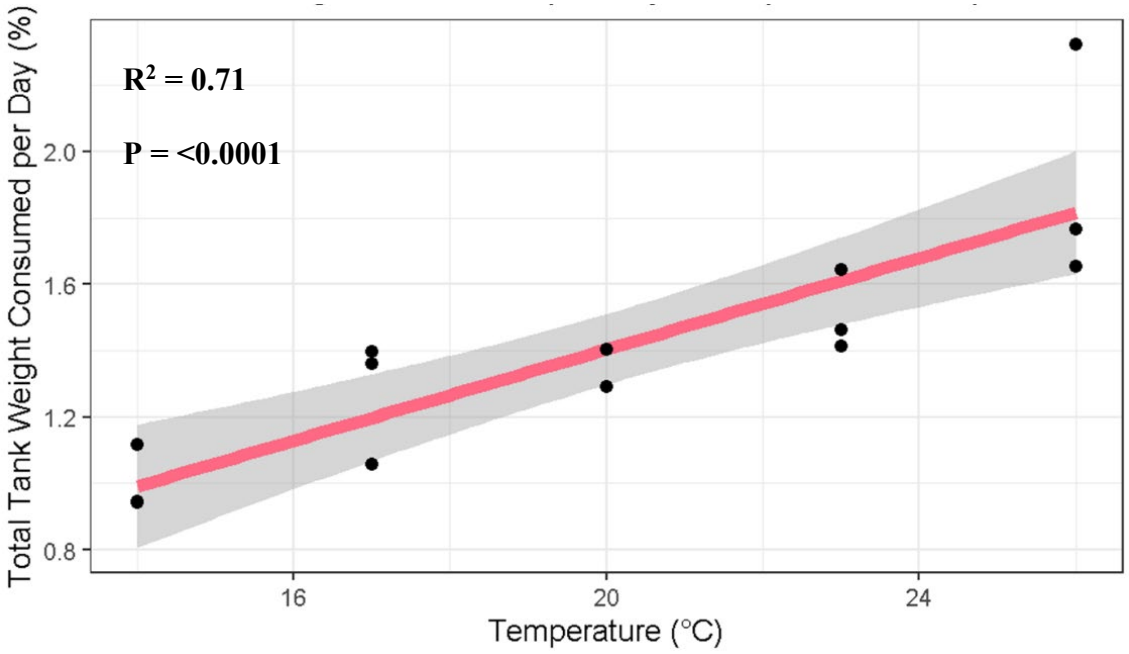
*Specific growth rate (SGR) and cumulative percent growth of monkeyface pricklebacks at different rearing temperatures. (A) Bar plot depicting the SGR of monkeyface pricklebacks in the temperature study (14°C = blue, 17°= light blue, 20°C = yellow, 23°C = light red, 26°C = red). Black dots indicate estimated marginal means, error bars resemble the 95% confidence intervals, and letters are based on Tukey's-adjusted comparisons. Bars that do not share the same letter are significantly different. (B) Cumulative percent growth over time as a line plot among temperature treatments. The dots represent the average cumulative percent growth at each time of measurement. The error bars represent standard error.*



The weight of feed pellets consumed, standardized by the weight of fish in each tank, increased linearly with temperature based on the linear regression (Figure 16). Thus, fish reared at warmer temperatures exhibited an elevated feed intake (Linear Regression, $F_{1,13}$ = 34.72, p-value < 0.0001, Adjusted $R^2$ = 0.71; Figure 16).

**Figure 16.**

*Linear regression depicting the weight of feed consumed per day standardized by the total fish weight in a tank as a function of the temperature treatments. The grey region represents the 95% confidence interval.*

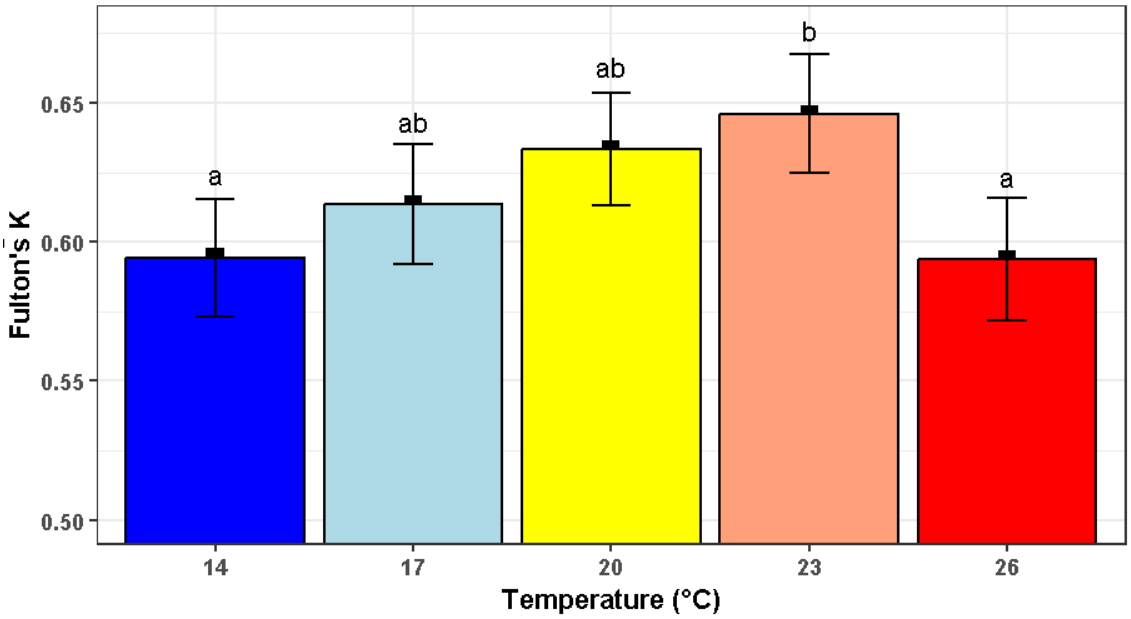



Juvenile monkeyface pricklebacks differed in Fulton's condition factor (FK) among the temperature treatments based on the GLMM adjusted for tank variability (GLMM, $X^2$ = 18.87, p-value = 0.00083; Figure 17). The 23°C treatment had a significantly higher FK than the 14°C and 26°C treatments with estimated marginal means at 0.646 (23°C), 0.594 (14°C), and 0.594 (26°C) (Tukey's HSD, p-value < 0.05; Figure 17). The 17°C and 20°C treatments, however, did not differ in FK from the 14°C, 23°C, and 26°C treatments and each other with

estimated marginal means at 0.613 (17°C) and 0.633 (20°C) (Tukey's HSD, p-value > 0.05; Figure 17). The 14°C and 26°C treatments also did not differ in FK between each other (Pairwise comparisons with Tukey's adjustment, p-value > 0.05; Figure 17).

**Figure 17.**

*Bar plots illustrating Fulton's condition factor (FK) of monkeyface pricklebacks in the temperature study (14°C = blue, 17°= light blue, 20°C = yellow, 23°C = light red, 26°C = red). Black dots indicate estimated marginal means, error bars resemble the 95% confidence intervals, and letters are based on Tukey's-adjusted comparisons. Bars that do not share the same letter are significantly different.*



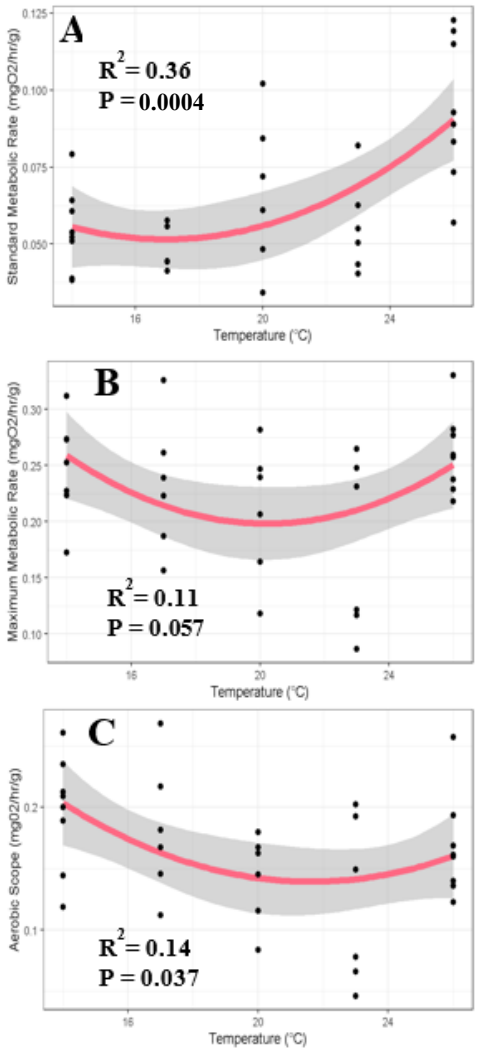

**Metabolic Rates**

Based on a non-linear regression, temperature influenced the standard metabolic rate (SMR) of juvenile monkeyface pricklebacks, with SMR increasing exponentially as temperature increased (Non-linear Regression, $F_{2,31} = 10.15$, p-value = 0.0004, Adjusted $R^2 = 0.36$; Figure 18A). Maximum metabolic rate (MMR), however, was marginally not significantly associated with temperature as it remained roughly constant despite temperature

increases (Non-linear Regression, $F_{2,31} = 3.138$, p-value = 0.057, Adjusted $R^2 = 0.11$; Figure 18B). Since SMR increased while MMR remained constant as temperature increased, aerobic scope (AS) decreased as temperature increased (Non-Linear Regression, $F_{2,31} = 3.678$, p-value = 0.037, Adjusted $R^2 = 0.14$; Figure 18C).

**Figure 18.**

*Monkeyface prickleback metabolic responses in the temperature study. (A) Standard metabolic rate as a function of temperature. (B) Maximum metabolic rate (MMR as a function of temperature. (C) Aerobic scope (AS) as a function of temperature. The grey region represents the 95% confidence interval.*



**Critical Thermal Maximum**

Based on a non-linear regression, critical thermal maximum (CTmax) was influenced by temperature for juvenile monkeyface pricklebacks (Non-linear Regression, $F_{2,37} = 35.19$, p-value < 0.0001, Adjusted $R^2 = 0.64$; Figure 19). As the temperature increased, the CTmax increased until it reached a peak of ~23°C (Figure 19). Past ~23°C, the CTmax slightly decreased as temperature increased (Figure 19). This can also be observed by examining the average CTmax at each temperature, with averages increasing from 14°C, peaking within the 20°C to 23°C range, and then declining past 23°C (Table 6). Furthermore, acclimation response ratios (ARRs) resemble this same pattern, for ARRs are positive at 14-17°C and 17-20°C, neutral at 20-23°C, and negative at 23-26°C (Table 6).

**Figure 19.**

*Non-linear regression between the critical thermal maximum (CTmax) of monkeyface pricklebacks and temperature. The grey region represents the 95% confidence interval.*

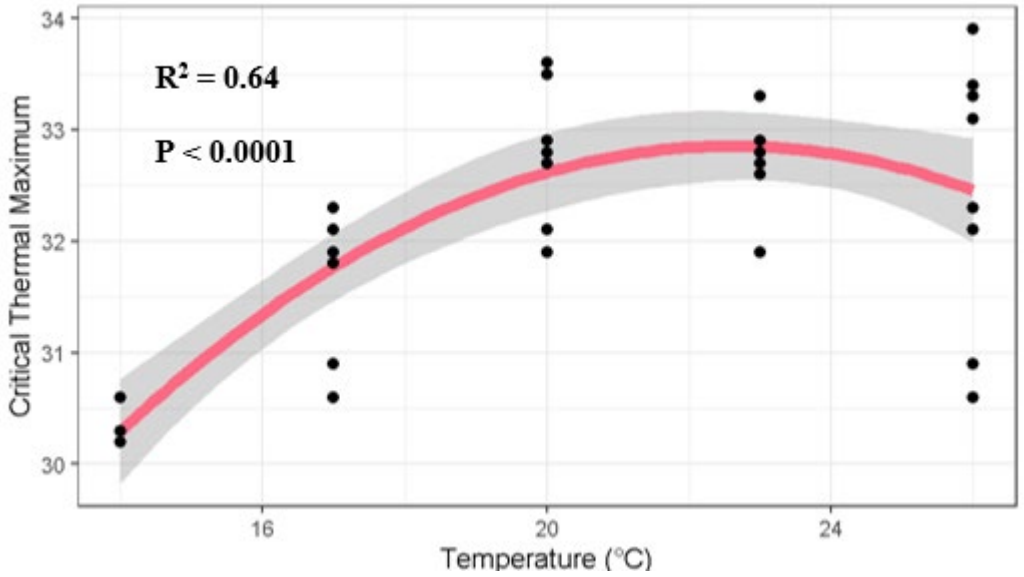


**Table 5**

*The Table Above the Split Shows the Average Critical Thermal Maximum (CTmax) per Temperature Treatment. The Table Below the Split Shows Acclimation Response Ratios (ARR) at Different Temperature Ranges*

| Temperature Treatment | Average CTmax |
|---|---|
| 14°C | 30.36 |
| 17°C | 31.58 |
| 20°C | 32.80 |
| 23°C | 32.80 |
| 26°C | 32.45 |
| | |
| Temperature Ranges | ARR (Δ CTmax/Δ Acclimation Temperature) |
| 14-17°C | 0.40 |
| 17-20°C | 0.41 |
| 20-23°C | 0.00 |
| 23-26°C | -0.12 |

**Pellet and Fillet Composition**

Macronutrient content was similar between the initial and second pellets used in the temperature study (Table 7). Fatty acids were also similar between the two pellets (Table 7). Protein (ANOVA, $F_{4,9} = 1.35$, p-value = 0.33), fat (ANOVA, $F_{4,9} = 1.51$, p-value = 0.28), moisture (ANOVA, $F_{4,9} = 3.37$, p-value = 0.06), and carbohydrate (ANOVA, $F_{4,9} = 1.77$, p-value = 0.22) content did not differ among temperature treatments in the fillets of juvenile monkeyface pricklebacks (Figures 20A, 20B, 20D, and 20E; Table 8). However, there was a difference in ash (ANOVA, $F_{4,9} = 5.01$, p-value = 0.021) and caloric (ANOVA, $F_{4,9} = 4.23$, p-value = 0.034) content among the temperature treatments (Figures 20C & 20F; Table 8). The 14°C and 26°C treatments had significantly higher ash content than 23°C treatment (Tukey's HSD, p-value < 0.05; Figure 20C). The 17°C and 20°C treatments did not differ in ash

content from the other treatments and each other (Tukey's HSD, p-value > 0.05; Figure 20C). Caloric content was significantly higher in the 23°C treatment than the 26°C treatment (Tukey's HSD, p-value < 0.05; Figure 20F), but it did not differ from the other treatments (Tukey's HSD, p-value > 0.05; Figure 20F). The 14°C, 17°C, and 20°C treatments did not differ amongst each other (Tukey's HSD, p-value > 0.05; Figure 20F).

**Table 6**

*Macronutrient Content of Diets (Protein, Fat, Ash, Moisture, Carbohydrate, and Caloric Content) (Upper Portion). Fatty Acid Content of Diets (Lower Portion)*

| Macronutrients | Initial Pellet | Second Pellet | Average |
|---|---|---|---|
| Protein (%) | 39.69 | 40.85 | 40.27 |
| Fat (%) | 6.24 | 6.74 | 6.49 |
| Ash (%) | 7.12 | 7.51 | 7.32 |
| Moisture (%) | 15.60 | 10.40 | 13.00 |
| Carbohydrates (%) | 47.04 | 44.87 | 45.95 |
| Calories (kcal/100g) | 340.00 | 362.00 | 351.00 |
| | | | |
| Fatty Acids (% of Total Fat) | Initial Pellet | Second Pellet | Average |
| C16:0 Palmitic | 19.33 | 20.47 | 19.90 |
| C18:1n9C Oleic | 10.13 | 10.12 | 10.12 |
| C18:2n6 Linoleic/C19:1n10c | 27.27 | 27.73 | 27.50 |
| C18:3n3 Linolenic | 3.12 | 2.89 | 3.00 |
| C20:4n6 Arachidonic/C23:0 | 0.87 | 0.95 | 0.91 |
| C20:5n3 Eicosapentaenoic/C24:0 (EPA) | 6.30 | 5.26 | 5.78 |
| C22:5n3 Docosapentaenoic | 0.87 | 0.95 | 0.91 |
| C22:5n6 Docosapentaenoic | 4.65 | 4.69 | 4.67 |
| C22:6n3 Docosahexaenoic (DHA) | 11.71 | 11.80 | 11.76 |
| Monounsaturated Fatty Acids | 12.51 | 12.63 | 12.57 |
| Omega 3 Fatty Acids | 21.65 | 20.50 | 21.07 |
| Omega 6 Fatty Acids | 33.33 | 33.65 | 33.49 |
| Polyunsaturated Fatty Acids | 55.19 | 54.03 | 54.61 |
| Saturated Fatty Acids | 25.11 | 26.30 | 25.71 |
| Total Fat (g/100g) | 4.62 | 4.22 | 4.42 |
| Trans Fatty Acids | 0.87 | 0.95 | 0.91 |

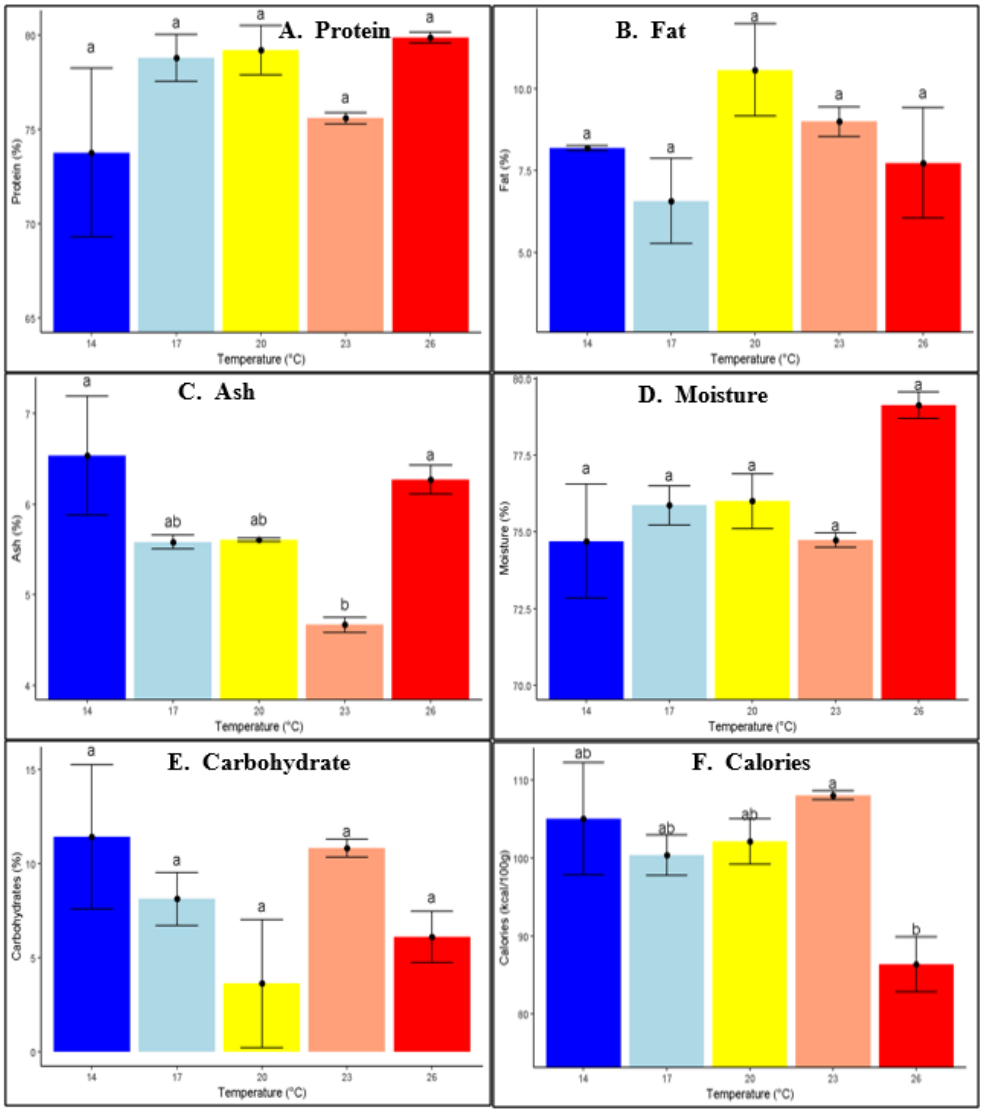
## Table 7

*Macronutrient Content (Mean ± Standard Error) of Monkeyface Prickleback Fillets Across the Temperature Treatments at the Conclusion of the Study (Upper Portion). Fatty Acid Content (Mean ± Standard Error) of Fillets Among Temperature Treatments (Lower Portion). Rows that are Bolded were Significantly Different in Composition Among the Temperature Treatments*

| Macronutrients | 14°C | 17°C | 20°C | 23°C | 26°C | F-Value (4,9) | P-Value |
|---|---|---|---|---|---|---|---|
| Protein (%) | 73.77 ± 4.47 | 78.79 ± 1.23 | 79.22 ± 1.30 | 75.59 ± 0.30 | 79.87 ± 0.28 | 1.35 | 0.33 |
| Fat (%) | 8.18 ± 0.081 | 6.56 ± 1.30 | 10.57 ± 1.42 | 8.99 ± 0.45 | 7.73 ± 1.69 | 1.51 | 0.28 |
| **Ash (%)** | **6.53 ± 0.66** | **5.58 ± 0.073** | **5.6 ± 0.019** | **4.67 ± 0.084** | **6.27 ± 0.16** | **5.01** | **0.021** |
| Moisture (%) | 74.70 ± 1.86 | 75.87 ± 0.64 | 76.00 ± 0.90 | 74.73 ± 0.24 | 79.13 ± 0.43 | 3.37 | 0.06 |
| Carbohydrates (%) | 11.42 ± 3.83 | 8.12 ± 1.41 | 3.62 ± 3.41 | 10.82 ± 0.48 | 6.11 ± 1.35 | 1.77 | 0.22 |
| **Calories (kcal/100g)** | **105.03 ± 7.18** | **100.37 ± 2.58** | **102.10 ± 2.90** | **108.00 ± 0.58** | **86.40 ± 3.52** | **4.23** | **0.034** |
| | | | | | | | |
| Fatty Acids (% of Total Fat) | 14°C | 17°C | 20°C | 23°C | 26°C | F-Value (4,9) | P-Value |
| **C16:0 Palmitic** | **17.28 ± 0.35** | **17.73 ± 0.15** | **18.74 ± 0.16** | **18.79 ± 0.21** | **18.61 ± 0.32** | **6.68** | **0.0088** |
| C18:1n9C Oleic | 14.96 ± 0.08 | 14.35 ± 0.21 | 15.06 ± 0.02 | 14.80 ± 0.06 | 14.41 ± 0.74 | 0.69 | 0.62 |
| C18:2n6 Linoleic/C19:1n10c | 19.28 ± 1.06 | 17.19 ± 0.35 | 17.04 ± 0.30 | 17.04 ± 0.36 | 15.39 ± 1.74 | 1.94 | 0.19 |
| C18:3n3 Linolenic | 2.80 ± 0.19 | 3.75 ± 0.51 | 3.42 ± 0.25 | 3.48 ± 0.54 | 4.6 ± 1.67 | 0.56 | 0.70 |
| C20:4n6 Arachidonic/C23:0 | 2.47 ± 0.50 | 3.75 ± 0.51 | 3.42 ± 0.25 | 3.48 ± 0.54 | 4.60 ± 1.67 | 0.73 | 0.60 |
| C20:5n3 Eicosapentaenoic/C24:0 (EPA) | 5.68 ± 0.15 | 6.12 ± 0.043 | 5.52 ± 0.11 | 5.76 ± 0.27 | 6.79 ± 0.56 | 2.62 | 0.11 |
| C22:5n3 Docosapentaenoic | 2.49 ± 0.48 | 3.75 ± 0.51 | 3.42 ± 0.25 | 3.48 ± 0.54 | 4.60 ± 1.67 | 0.72 | 0.60 |
| C22:5n6 Docosapentaenoic | 3.57 ± 0.051 | 4.03 ± 0.33 | 3.59 ± 0.081 | 3.81 ± 0.29 | 4.87 ± 1.52 | 0.49 | 0.74 |
| C22:6n3 Docosahexaenoic (DHA) | 9.76 ± 0.12 | 10.26 ± 0.24 | 10.24 ± 0.15 | 10.10 ± 0.29 | 10.77 ± 1.00 | 0.51 | 0.73 |
| Monounsaturated Fatty Acids | 23.14 ± 0.51 | 21.80 ± 0.48 | 23.92 ± 0.03 | 23.73 ± 0.29 | 21.76 ± 1.61 | 1.38 | 0.32 |
| Omega 3 Fatty Acids | 19.81 ± 0.10 | 20.68 ± 0.30 | 19.65 ± 0.20 | 19.91 ± 0.58 | 21.92 ± 1.69 | 1.14 | 0.40 |
| Omega 6 Fatty Acids | 25.60 ± 1.03 | 25.18 ± 0.58 | 23.38 ± 0.20 | 23.49 ± 0.37 | 23.37 ± 0.77 | 2.39 | 0.13 |
| Polyunsaturated Fatty Acids | 45.40 ± 1.01 | 45.89 ± 0.31 | 43.02 ± 0.01 | 43.38 ± 0.76 | 45.29 ± 1.04 | 2.40 | 0.13 |
| **Saturated Fatty Acids** | **24.53 ± 0.50** | **25.14 ± 0.20** | **26.25 ± 0.02** | **26.26 ± 0.25** | **26.43 ± 0.57** | **4.53** | **0.028** |
| Total Fat (g/100g) | 1.80 ± 0.45 | 1.10 ± 0.15 | 1.18 ±0.085 | 1.21 ± 0.21 | 1.09 ± 0.31 | 1.08 | 0.42 |
| Trans Fatty Acids | 2.47 ± 0.50 | 3.75 ± 0.51 | 3.42 ± 0.25 | 3.48 ± 0.54 | 4.60 ± 1.67 | 0.73 | 0.60 |

**Figure 20.**

*Bar plots illustrating patterns of proximate composition of monkeyface prickleback fillets at the conclusion of the temperature study (14°C = blue, 17°= light blue, 20°C = yellow, 23°C = light red, 26°C = red). Shown are values as a percent of dry matter for protein (A), fat (B), ash (C), and carbohydrate (E) content. Moisture (D) and calories (F) are displayed on a wet weight basis in fillets between temperature treatments. Black dots indicate the mean, error bars show standard error, and letters are based on the Tukey's HSD post-hoc test. Bars that do not share the same letter are significantly different.*




Palmitic and saturated fatty acids in juvenile monkeyface pricklebacks differed among temperature treatments (ANOVA, $F_{4,9} = 6.68$ and 4.53, p-value < 0.05; Figures 21 and 22D;

Table 8). While the 17°C treatment did not differ in palmitic fatty acid content from the other temperature treatments (Tukey's HSD, p-value > 0.05; Figure 21), the 20°C, 23°C, and 26°C treatments had significantly higher palmitic fatty acid content than the 14°C treatment (Tukey's HSD, p-value < 0.05; Figure 21). However, the 20°C, 23°C, and 26°C treatments did not differ amongst each other (Tukey's HSD, p-value > 0.05; Figure 21). The 26°C treatment had significantly higher saturated fatty acid content than the 14°C treatment (Tukey's HSD, p-value < 0.05; Figure 22D). The other treatments did not differ in saturated fatty acid content from the 14°C and 26°C treatments and each other (Tukey's HSD, p-value > 0.05; Figure 22D). The remaining fatty acids did not differ in content among the temperature treatments for juvenile monkeyface pricklebacks (ANOVA, p-value > 0.05; Table 8).

**Figure 21.**

*Bar plot depicting the palmitic fatty acid content in monkeyface prickleback fillets across the different temperature treatments (14°C = blue, 17°= light blue, 20°C = yellow, 23°C = light red, 26°C = red). Black dots indicate the mean, error bars show standard error, and letters are based on Tukey's multiple comparison of the mean. Bars that do not share the same letter are significantly different.*

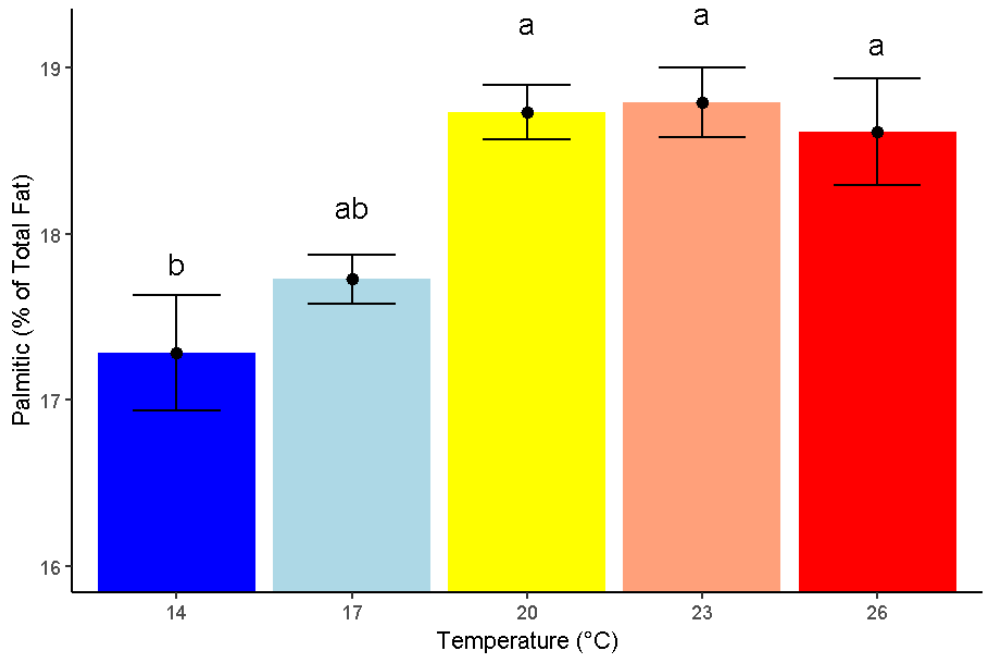
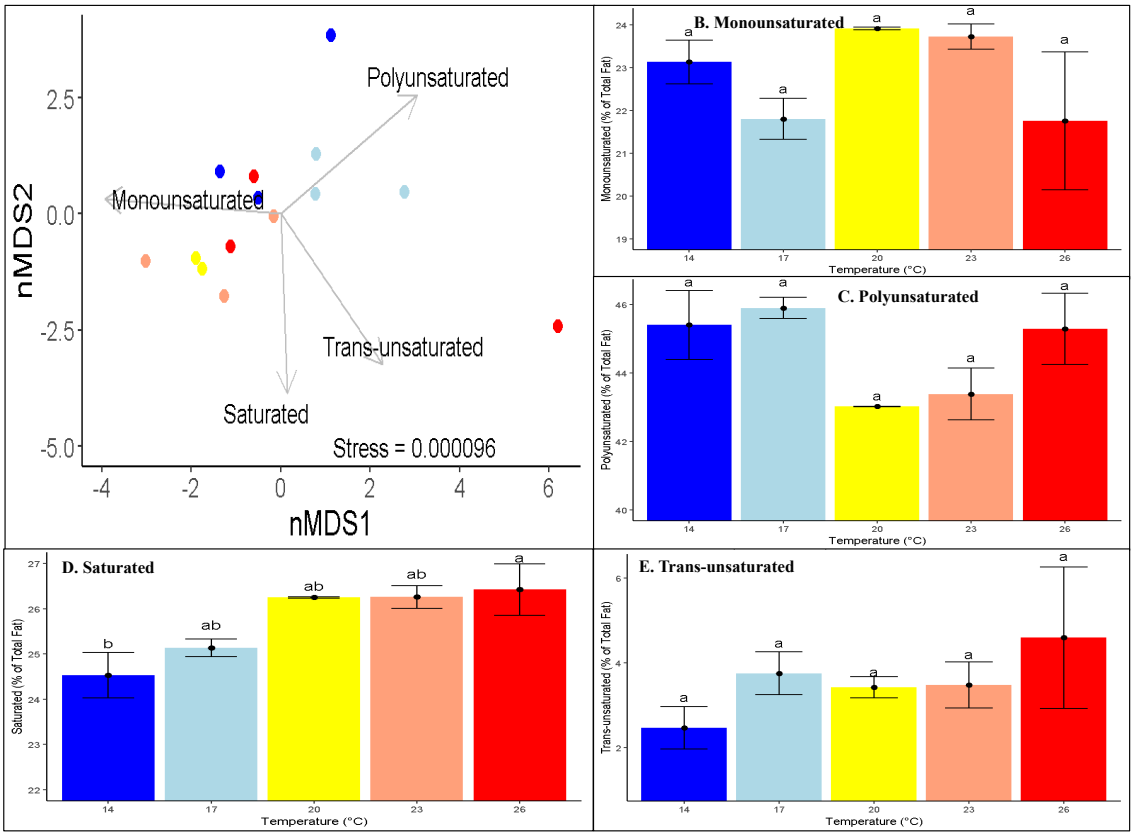


**Figure 22.**

*A non-metric multidimensional scaling (nMDS) and bar plots of monounsaturated, polyunsaturated, saturated, and trans-unsaturated fatty acids in fillets among temperature treatments. (A) An nMDS with arrows indicating strength (arrow length) and direction of fatty acids in relation to dietary treatments. Bar plots (B-E) depicting the fatty acid composition of monkeyface prickleback fillets across the different temperature treatments (14°C = blue, 17°= light blue, 20°C = yellow, 23°C = light red, 26°C = red). Shown are fatty acids of monounsaturated (B), polyunsaturated (C), saturated (D), and trans-unsaturated (E) content. Black dots indicate the mean, error bars show standard error, and the letters are based on Tukey's HSD post-hoc tests. Bars that do not share the same letter are significantly different.*




In a multivariate context, the permutational analysis of variance (PERMANOVA) indicated that there was no significant effect of the temperature treatments on the fatty acid composition of monkeyface prickleback fillets when grouping the fatty acids into monounsaturated, polyunsaturated, saturated, and trans-unsaturated fatty acids

(PERMANOVA, $F_{1,12} = 1.90$, p-value = 0.16, $R^2 = 0.14$). However, the non-metric multidimensional scaling (nMDS) test indicated a few interesting trends among temperature treatments (Figure 22A). Saturated and trans-unsaturated fatty acids tended to be elevated in fish exposed to higher temperatures (Figure 22A, 22D, and 22E), while polyunsaturated fatty acids had higher concentrations in fish exposed to colder temperatures (Figure 22A and 22C). Monounsaturated fatty acids tended to be highest in fish reared in intermediate temperatures (Figure 22A and 22B).

## Discussion

The temperature study was conducted to determine an optimal temperature to rear monkeyface pricklebacks for aquaculture. Standard metabolic rate (SMR) and feed intake increased with temperature; however, there was a peak in growth, condition, and critical thermal maximum (CTmax) in fish acclimated to intermediate temperatures. Since standard metabolic rate (SMR) increased rapidly at the warmest temperature, the increase in food consumption was insufficient to compensate for the exponentially increasing SMR, resulting in lowered growth and condition at the warmest acclimation temperature. The poor condition of fish reared in the highest temperature treatment likely led to their lowered CTmax. Maximum metabolic rate (MMR) unexpectedly remained the same despite changes in temperature. Most macronutrients and fatty acids also remained the same. Upon reviewing this study's results, a rearing temperature between 20–23°C is optimal for rearing monkeyface pricklebacks for aquaculture.

### Metabolism and Food Consumption

SMR increased with temperature in monkeyface pricklebacks (Figure 18A), which is expected for all ectotherms due to the $Q_{10}$ effect which states that biochemical reaction rates tend to double with a 10°C increase (Hochachka & Somero, 2002). $Q_{10}$ effects on SMR would predict an exponential increase in SMR as temperature increased, which was observed in this study (Figure 18A). These increases have been reported in numerous fish species, such as European sea bass (*Dicentrarchus labrax*), Atlantic cod (*Gadus morhua*), rock bream (*Oplegnathus fasciatus*), and the killifish *Fundulus heteroclitus*, further supporting the observed relationship between SMR and temperature (Schurmann & Steffensen, 1997;

Claireaux et al., 2000; Schulte, 2015; Prakoso & Chang, 2018; Stavrakidis-Zachou et al., 2022). Despite observed differences in fish condition, the SMR-temperature relationship reflected those reported in the literature, inferring condition did not influence SMR (Figures 17 & 18A).

Unlike SMR, juvenile monkeyface pricklebacks displayed a slightly upward parabolic curve between MMR and temperature, resulting in MMR remaining roughly constant despite temperature increases (Figure 18C). Typically, MMR is observed as a downward parabolic curve that initially increases, peaks, and then decreases with temperature (Figure 18C; Rummer et al., 2014; Schulte, 2015; Volkoff & Rønnestad, 2020). However, the association between MMR and temperature can take on different forms in other fish species, such as the exponentially increasing MMR with temperature of *F. heteroclitus* (Rummer et al., 2014; Schulte, 2015). However, *D. labrax*'s MMR also remained roughly the same with temperature since their cardiovascular output was incapable of increasing past 28°C, illustrating this relationship between MMR and temperature is found in nature (Stavrakidis-Zachou et al., 2022). It may be theorized that juvenile monkeyface prickleback's MMR does not differ as a function of temperature as they are also incapable of increasing their cardiovascular output (Stavrakidis-Zachou et al., 2022). Furthermore, the monkeyface prickleback is a sedentary species thermally adapted to its fluctuating environment. In effect, it may not need differing MMRs as it rarely moves despite temperature differences.

Aerobic scope (AS) was found to decrease as temperature increased in juvenile monkeyface pricklebacks (Figure 18E). AS is a proxy for aerobic performance and should appear like the fish performance curve with temperature, which has been observed in

multiple fish like brook, brown, rainbow, and lake trout (Farrell, 2009; Schulte, 2015; Volkoff & Rønnestad, 2020). However, the increase in SMR while MMR remained constant led to AS decreasing with temperature, which can be supported as it was also observed in *D. labrax* (Stavrakidis-Zachou et al., 2022; Figures 18A, 18B, & 18C). Since monkeyface prickleback's cardiovascular output may be incapable of increasing but its oxygen consumption increases with temperature, they have less oxygen available for aerobic activities at higher temperatures, reducing their fitness.

The limitations of the feed study's metabolic rate methods were accounted for in the temperature study. SMR and MMR were adjusted for background respiration as they were properly captured in the temperature study. Thus, the temperature study captured metabolic rates more accurately than the feed study. However, the potential influence of the systemic bacterial infection may have impacted metabolic rates. Both SMR and MMR measurements started 10 days after the final injection of ceftazidime. Despite the antibiotics likely taking effect, there is a possibility the infection raised the fish's metabolic rates to fight the infection, resulting in overestimations of SMR, MMR, and AS.

As temperature increased, monkeyface pricklebacks consumed more feed, which aligns with concomitant increases in SMR (Figure 16). This is supported by other fish increasing their food consumption with increasing temperature, as has been reported for species such as *D. labrax* and post-smolt Atlantic salmon (*Salmo salar*) (Handeland et al., 2008; Volkoff & Rønnestad, 2020; Stavrakidis-Zachou et al., 2022). Since temperature increases result in heightened metabolic rates, as observed in this study, more feed is required to satisfy the raised metabolic requirements (Volkoff & Rønnestad, 2020; Kuhn et al., 2023; Figure 18A).

Unlike most fish, however, monkeyface pricklebacks were not observed to have lower feed intakes outside their ideal range (Figure 16; Volkoff & Rønnestad, 2020). This typical reduction is due to high temperatures altering fish behavior, reducing their locomotion and sensory capabilities, leading to decreased feed intake (Volkoff & Rønnestad, 2020). Additionally, their appetite, digestion, and conversion efficiency decline at high temperatures, forcing the organism to halt growth to survive due to the high metabolic demand (Brett, 1979). Since monkeyface pricklebacks in the 26°C treatment still exhibited positive growth, it can be inferred that the metabolic demand at 26°C did not force the fish to reduce feed intake, supporting the observations that feed intake increased as temperature increased. However, feed intake methods may have influenced the total tank weight consumed per day data. Since the formulated pellets were quite unstable, satiation was determined when monkeyface pricklebacks had ceased feeding activities, and four or more pellets were left over. During the feeding, monkeyface pricklebacks were observed the entire time until fish no longer showed active feeding behavior. However, it should be noted that some of the experimental fish did not readily feed due to the presence of researchers. Thus, satiation may have been achieved for some individuals and not others, underestimating the total tank feed consumption. This may be true for the lower and intermediate (14°C, 17°C, & 20°C) temperature treatments as their metabolic rates might not have been high enough that missing a meal would be detrimental, but the higher temperatures (23°C & 26°C) needed to consume their daily rations to account for their heightened metabolism, regardless of being observed. It can be inferred that lower and intermediate temperatures would have higher feed

consumption, possibly resulting in a downward parabolic curve like other fish (Volkoff & Rønnestad, 2020).

**Survivorship, Growth, and Condition**

Juvenile monkeyface prickleback growth rates initially increased with temperature, peaked at 20°C, and then decreased past 20°C (Figures 15A & 15B). This downward parabolic curve observed in monkeyface pricklebacks is similar to the performance curve observed in multiple ectothermic fish as aerobic scope influences growth (Figure 1; Wang & Overgaard, 2007; Farrell, 2009; Volkoff & Rønnestad, 2020). It is also similar to the growth curves of brook trout, bull trout, and post-smolt *S. salar* (Handeland et al., 2008; Farrell, 2009). To grow, fish need to consume more calories than their basal metabolic demand. Since metabolism increases with temperature, fish need to increase their food consumption to maintain a caloric surplus to grow (Csik et al., 2023). Because digestion and enzyme activity increase with temperature, fish are able to efficiently convert food, leading to faster growth (Brett, 1979; Daniel et al., 2008; Volkoff & Rønnestad, 2020). This was observed in juvenile monkeyface pricklebacks as they consumed more feed to continue growing due to metabolic rates increasing with temperature. Beyond optimal temperatures, however, metabolic costs become too high, and feed consumption decreases, leading to reduced growth (Brett, 1979). This has been observed in *D. labrax*, with its appetite decreasing and growth ceasing while its SMR increased at 33°C (past its ideal growing temperature) (Stavrakidis-Zachou et al., 2022). Similarly, monkeyface pricklebacks reared at 26°C exhibited only slight growth rates likely because their metabolic demands were high and barely met by their caloric intake. Growth rates of the experimental fish were only reported for the first four weeks of the study

due to the systemic bacterial infection affecting all experimental fish. It is possible that monkeyface pricklebacks may have had higher growth rates if they were calculated for the study's entire duration. However, the observed growth curve was like other fish species, suggesting that the downward parabolic pattern would likely stay the same despite changes in growth rates.

Monkeyface pricklebacks were in better condition around the peak growth temperature (20°C) than at the extreme cold and warm temperatures (Figure 17). Specifically, Fulton's condition factor (FK) was higher at 20°C and 23°C treatments than the others. Because fish were fed the same feed across treatments, differences in condition are likely due to temperature differences as opposed to differences in diet. Since fish condition is influenced by environmental factors like temperature, individuals reared outside their optimal temperature are likely to have poor health due to decreased physiological performance (Saeed, 2013; Volkoff & Rønnestad, 2020). For instance, juvenile burbot (*Lota lota*) had an ideal growth rate at 15°C with the second-highest FK at 1.16, but its FK decreased to 1.02 as temperature increased to 21°C (Wolnicki et al., 2001). In effect, monkeyface pricklebacks reared near their optimal temperature should be in the best condition, supported with FK being high at the ideal growth temperature of 20°C (Figures 15A, 15B, & 17). Since food consumption, enzyme activity, and digestion increase with temperature, fish cultured close to their ideal temperatures likely have more capacity to store extra resources as their metabolic costs are not high enough to prevent resource storage while they are efficient at up taking resources (Brett, 1979; Daniel et al., 2008). Past ideal temperatures, however, SMR exceeds food consumption, likely leading to fish using their energy reserves to support the high

metabolic costs, resulting in poor condition (Brett, 1979). This was observed in the monkeyface pricklebacks of this experiment reared at 26°C, for they were in the poorest condition as a likely result of their metabolism exceeding their caloric intake. Similar to the growth rates, FK was only calculated for the first four weeks of the study due to the systemic bacterial infection and may be higher or lower if they were calculated for the study's entire duration.

Within the first four weeks, juvenile monkeyface pricklebacks had a high total survivorship, greater than 80% for all temperature treatments, except 17°C, which had a survivorship at ~79.13%. Seven of the observed mortalities can be attributed to the infection, while two of them were likely caused by prior fin injuries from containment before the experiment started. Causes for the five other mortalities were undetermined, but it can be inferred that the systemic bacterial infection was likely the cause. As previously mentioned, mortalities due to infection were not temperature-dependent. It can thus be surmised that the effects of temperature did not lead to these mortalities. Although the 20°C treatment had the highest survivorship in the first four weeks, it may have been due to a lower intensity or rate of infection. Nevertheless, juvenile monkeyface pricklebacks should be reared at 20°C to maintain high survivorship.

Based on the observed growth rates, 20°C is the optimal temperature to rear monkeyface pricklebacks to achieve maximum growth, with a specific growth rate (SGR) of 1.05% bodyweight gained per day (Figures 15A & 15B). Fish from the prior feed study that were reared at an average temperature of 12°C and were fed soy pellets had an SGR at 0.2% bodyweight gained per day (Figure 4A). Thus, rearing this species at warmer temperatures

can increase growth rates by 4–5 times over baseline values. Compared to European eels, monkeyface pricklebacks reared at 20°C had a higher mean SGR than European eels (0.8% bodyweight gained per day) acclimated to the same temperature (Sadler, 1979). In addition, the 20°C treatment had a high FK and the highest survivorship, making it the preferential temperature from those examined in this study for monkeyface prickleback growth, condition, and survivorship. While higher growth and better condition were achieved at 20°C, monkeyface pricklebacks would also require more feed than those reared at colder temperatures due to increased metabolic demands observed in this study. This would increase feed costs for the farmers, but the increased growth would likely result in monkeyface pricklebacks reaching a market size earlier, reducing the time and cost spent growing the species. For example, monkeyface pricklebacks reared at 12°C would have to be fed for 1500 days to grow by 300%, but fish reared at 20°C would only need to be fed for 286 days to reach 300% growth. Thus, monkeyface pricklebacks can be grown quickly and in good health at 20°C.

**Critical Thermal Maximum**

Monkeyface pricklebacks acclimated to 14°C had a mean critical thermal maximum (CTmax) of 30.36°C (Figure 19; Table 6). This is relatively close to the CTmax reported for the closely related prickleback, the high cockscomb, which exhibited a CTmax between 29.5–30°C when acclimated at 15°C (Barton, 1985). Both monkeyface pricklebacks and high cockscombs inhabit the intertidal zone, are from the same family Stichaeidae, and are considered eurythermal species, possibly explaining the similar CTmax between the species (Barton, 1985; Love, 2001).

The CTmax of monkeyface pricklebacks shifted warmer with increasing temperature, up to a point, as indicated by increases in acclimation response ratios (ARRs), supporting the hypothesis of CTmax increasing with warmer acclimation temperatures (Figure 19; Table 6). After peaking at 23°C, the CTmax decreased (Figure 19; Table 6). Similar to the growth rates, the CTmax of monkeyface pricklebacks exhibited a downward parabolic curve as temperatures increased, which is similar to the performance curve for ectothermic fish (Figures 12 & 19; Wang & Overgaard, 2007; Volkoff & Rønnestad, 2020). However, this pattern is not always observed, as CTmax has been reported to increase linearly as acclimation temperature increases in multiple fish species: channel catfish (*Ictalurus punctatus*), largemouth bass (*Micropterus salmoides*), rainbow trout (*Oncorhynchus mykiss*), the tropical guppy *Poecilla reticulata*, green chromide (*Etroplus suratensis*), *Therapon jarbua*, and the Qinling lenok (*Brachymystax lenok tsinlingensis*) (Currie et al., 1998; Chung, 2001; Rajaguru, 2002; Xia et al., 2017). Compared to the other treatments, the juvenile monkeyface pricklebacks in the 26°C treatment were in poor condition at the start of the CTmax experiments (Figure 17). Due to this, their CTmax may not have been accurately captured since it is recommended that experimental fish are in similar condition during measurement (Becker & Genoway, 1979). Furthermore, fish in the 26°C treatment already had difficulties maintaining themselves upright due to their poor condition near the end of the study, leading to CTmax being determined when they could not right themselves within a minute and were not moving. The additional requirement for CTmax quantification likely overestimated CTmax measurements for the 26°C treatment based on the loss of righting response method. Despite the overestimation, the 26°C treatment's mean CTmax was still

less than the other treatments, further highlighting the poor condition of these fish and their CTmax not being accurately captured.

Monkeyface pricklebacks can acclimate to temperature increases induced by climate change, as shown by ARRs at increasing temperatures (Table 6). This infers that monkeyface pricklebacks would still be capable of thriving in future warm conditions that would likely negatively affect stenothermal aquaculture species like coho salmon (*Oncorhynchus kisutch*) that have lower CTmaxes (Becker and Genoway, 1979). The European eel can also increase its CTmax from 33°C to 39°C at acclimation temperatures of 14°C and 29°C, respectively (Sadler, 1979). Since monkeyface pricklebacks and European eels can increase their CTmax, their ability to perform under culture conditions in a warming climate will likely not be affected. In addition, the CTmax results illustrate that monkeyface pricklebacks can be reared in a flow-through system without being impacted by acute temperature changes due to their relatively high CTmax, unlike stenothermal aquaculture species that would likely perish.

**Fillet Composition**

Macronutrient content of fillets did not generally differ among the temperature treatments, except for ash and calories (Figure 20; Table 8). Beyond optimal temperatures, calories decrease since energy digestibility decreases beyond ideal temperatures, which was observed in this study (Volkoff & Rønnestad, 2020). Ash is a measure of residue that includes undigestable materials and trace elements like phosphorus (Yin & Wang, 2024). Since absorbed phosphorus typically increases with temperature, it was expected that ash content would have been higher as temperatures increased, but this was not observed as ash content decreased with temperature (Chow & Schell, 1980; Figure 20C). This may be due to

the fish incorporating more trace elements around the ideal growing temperature to aid growth. Based on protein, macronutrients in fish fillets increase in content with temperature due to increased food consumption and utilization, as observed in juvenile meagre (*Argyrosomus regius*) (Kounna et al., 2021). However, protein, fat, moisture, and carbohydrates remained relatively the same in monkeyface prickleback fillets despite temperature increases (Figures 20A, 20B, 20D, & 20E). This may be explained by metabolism increasing with temperature and accounting for ingestion, digestion, and assimilation increases. At heightened temperatures, monkeyface pricklebacks were found to increase their feed intake to account for the increased SMR (Figures 16 & 18A). It can be inferred that the amount of food digested and assimilated increased accordingly, but it would be offset by the increase in SMR. Although ingestion, digestion, and assimilation rates increase with temperature in fish (Volkoff & Rønnestad, 2020), the increase in metabolism may prevent macronutrients from significantly differing in monkeyface prickleback fillets from different temperature treatments.

Similarly, fatty acid content did not differ among the temperature treatments besides palmitic and saturated fatty acids (Figures 21 & 22D; Table 8). Saturated fatty acids, like palmitic fatty acid, are often more abundant in fish tissues at warmer temperatures, as observed in this study (Figures 21 & 22D; Table 8), while unsaturated fatty acids are more abundant in fish at low temperatures to maintain membrane fluidity for protection against the cold, which has been observed in salmonids and *D. labrax* (Jobling & Bendiksen, 2003; Skalli et al., 2006; Halver, 1980). However, this was not observed in monkeyface pricklebacks as their unsaturated fatty acid content did not differ among temperature

treatments (Figures 22B, 22C, & 22E). Most of their fatty acids did not change in content despite temperature increases, unlike other ectotherms whose lipid contents are influenced by temperature (Jobling & Bendiksen, 2003). Temperature increases lead to increases in enzyme activity that impact the digestibility and metabolism of nutrients (e.g., protein and lipids) (Daniel et al., 2008; Volkoff & Rønnestad, 2020). Each fatty acid was expected to have a higher content at a specific temperature, as their respective enzymes likely perform better at different temperatures, but this was not observed. Because all temperature treatments were fed the same pellets and usually diet has a greater influence on fatty acid composition than temperature, monkeyface prickleback fillet fatty acid content did not differ among the temperature treatments due to diet alone (Table 8; Skalli et al., 2006).

Similar to the feed study, the statistical analysis of the macronutrient and fatty acid content in the fillets of juvenile monkeyface pricklebacks had uncertainties associated with it. Due to NP Analytical Laboratories' (NPAL) procedures requiring at least 10 g of sample material per test and fillets were pooled, only three samples per treatment were used for statistical analysis. In addition, the systemic bacterial infection likely impacted the digestion and assimilation of macronutrient and fatty acid content, leading to the lack of differences between temperature treatments. These call into question the reliability of the observed results.

Notwithstanding the experimental design limitations as discussed above, the results indicate that macronutrient and fatty acid content within the fillets of monkeyface pricklebacks does not generally differ between temperatures. Thus, monkeyface pricklebacks

can be reared at the ideal temperature for growth (20°C) without affecting changes in macronutrient and fatty acid content.

## Conclusions

Monkeyface pricklebacks are a potential alternative to freshwater eels for aquaculture. Upon conducting feed and temperature studies on monkeyface pricklebacks, an optimal diet and culture temperature were identified for efficiently rearing the species for commercial aquaculture. Based on biological and economic factors, the mixed/soy pellets and 20°C were the best performing diets and temperature for culturing this species because they produced the highest growth rates and fish in the best condition, which would likely reduce farm costs. However, leafy algae are more environmentally sustainable than soy, and they produced a higher quality macronutrient fillet in monkeyface pricklebacks (feed study). Farmers need to consider all factors before selecting a diet. Since the feed in the temperature study was a modified version of the original soy pellet tested, a combination of the two has already been examined, resulting in an SGR of 1.05% bodyweight gain per day that equates to 383% growth in a year; monkeyface pricklebacks are capable of reaching market-size quickly on a soy diet at 20°C. Combined with their thermal tolerance, herbivory, and lack of a commercial-scale fishery, monkeyface pricklebacks are a viable alternative to freshwater eels that have lower SGRs, require high levels of fishmeal and fish oil in their diets, and their populations are declining with no indications of recovery in the near future. The introduction of monkeyface prickleback aquaculture and products as an eel substitute to the seafood market may reduce freshwater eel demand and allow the species to recover while also generally increasing sustainability. Infections, parasites, and a territorial behavior, however, remain as issues for the culture of the monkeyface prickleback and require additional attention. In addition, an extended growth study at 20°C should be conducted to determine if

monkeyface pricklebacks can maintain the observed high SGR on the soy diet. It should also be conducted with whole algae to determine if it can have similar growth rates at 20°C. Moreover, achieving monkeyface prickleback reproduction in captivity is essential. In conclusion, the monkeyface prickleback is an intertidal, marine, and herbivorous fish capable of being efficiently raised in an aquaculture setting and may serve as a sustainable alternative to meet seafood demand for freshwater eel.

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